

Topic: Scenario of Underground Metal Mining in India



Uranium Corporation of India Limited

A government of India Enterprise

Department of Atomic Energy

ISO 9001:2015, 14001:2015 & IS-18001:2007 Company

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Introduction to UCIL



Uranium Corporation of India Limited was incorporated on 4th October 1967. It is a Public Sector Enterprise under the Department of Atomic Energy with a special standing at the forefront of Nuclear Power cycle. Fulfilling the requirement of uranium for the Pressurised Heavy Water Reactors, UCIL plays a very significant role in nuclear power generation of the country. UCIL is an ISO 9001:2015, 14001:2015 & IS 18001:2007 company and has adopted modern technologies for its mines and

process plants. The Company operates six underground mines (Bagjata, Jaduguda, Bhatin, Narwapahar, Turamdih and Mohuldih) and one open pit mine (Banduhurang) in the State of Jharkhand. Ore produced from these mines are processed in two process plants located at Jaduguda and Turamdih. UCIL is also operating one underground mine and process plant at Tummalapalle in Andhra Pradesh. The Company has taken up expansion of some of its operations in Jharkhand and started pre-project activities to set up new mines and plants in different parts of the country namely – Gogi in Karnatka, Lambapur in Telengana and KP M in Meghalaya.

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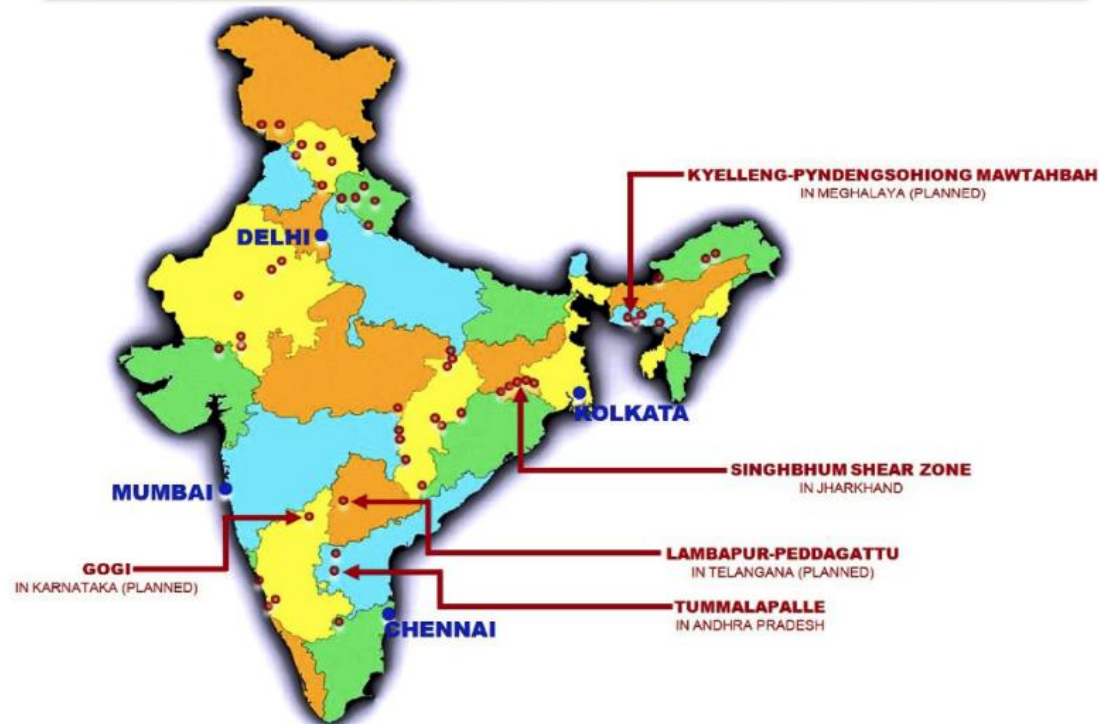
Opr. at jharkhand

Opr at other state

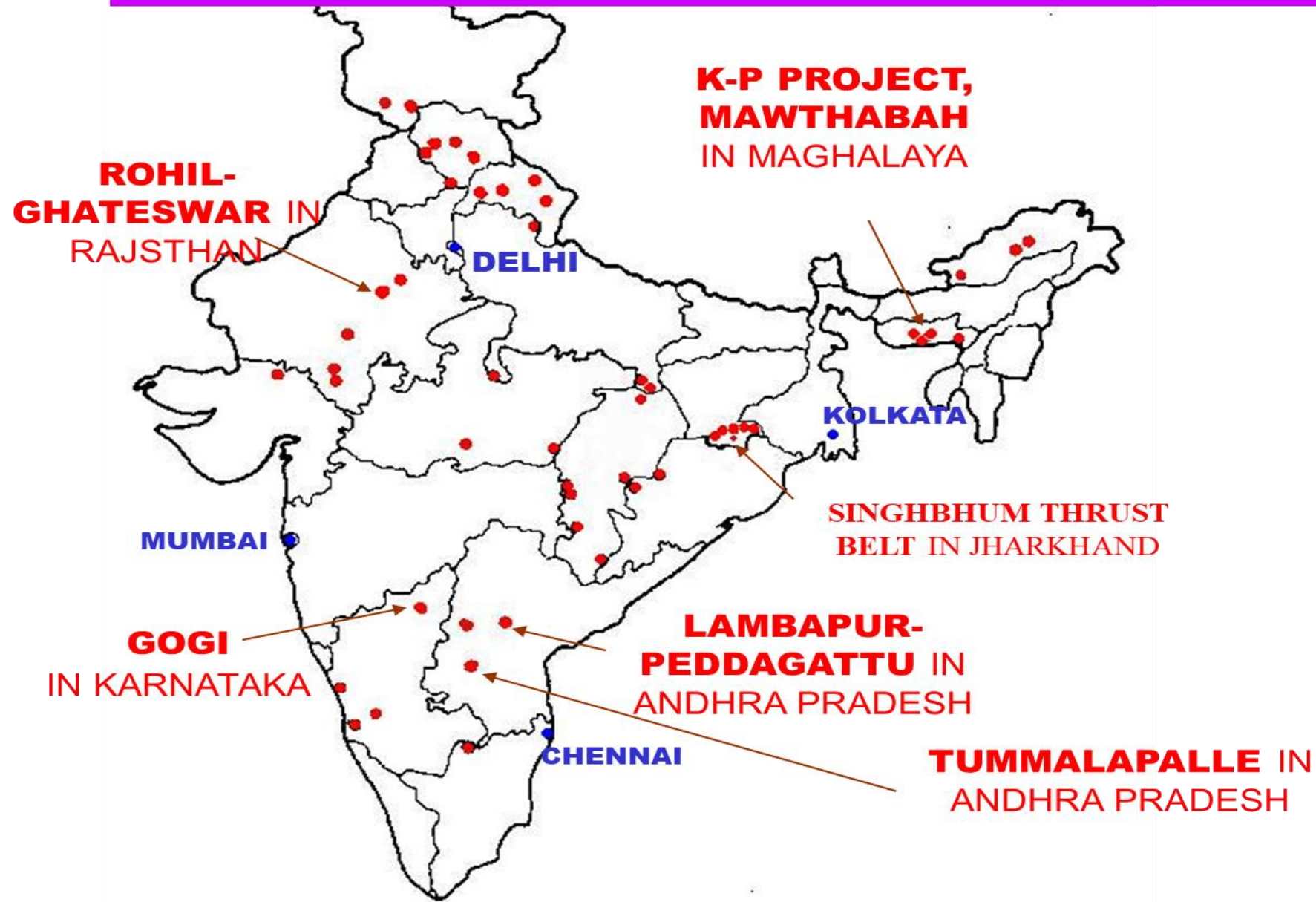
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Operation Of UCIL

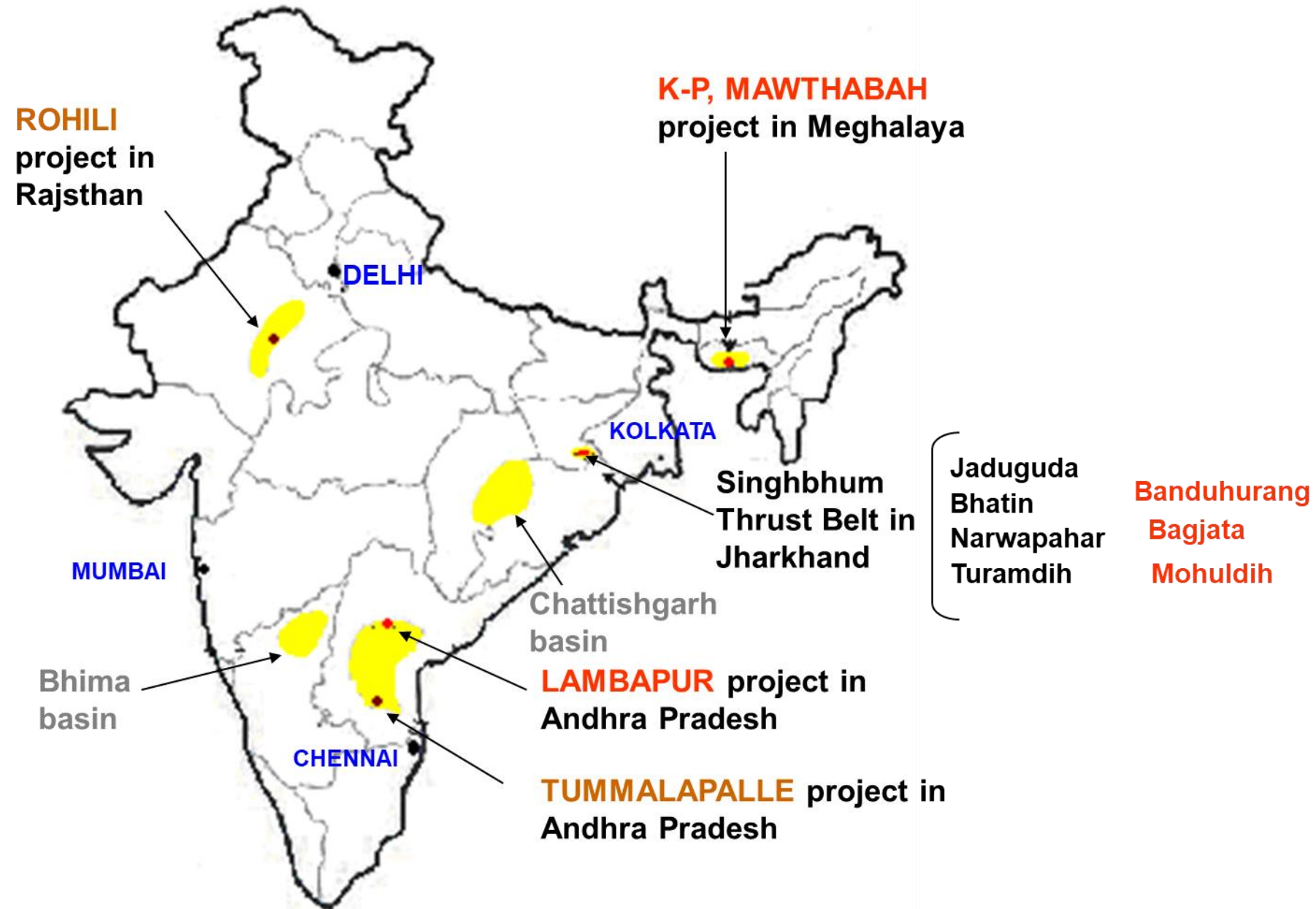
Uranium Occurrence and Production Centres in India



Uranium Occurrences in India

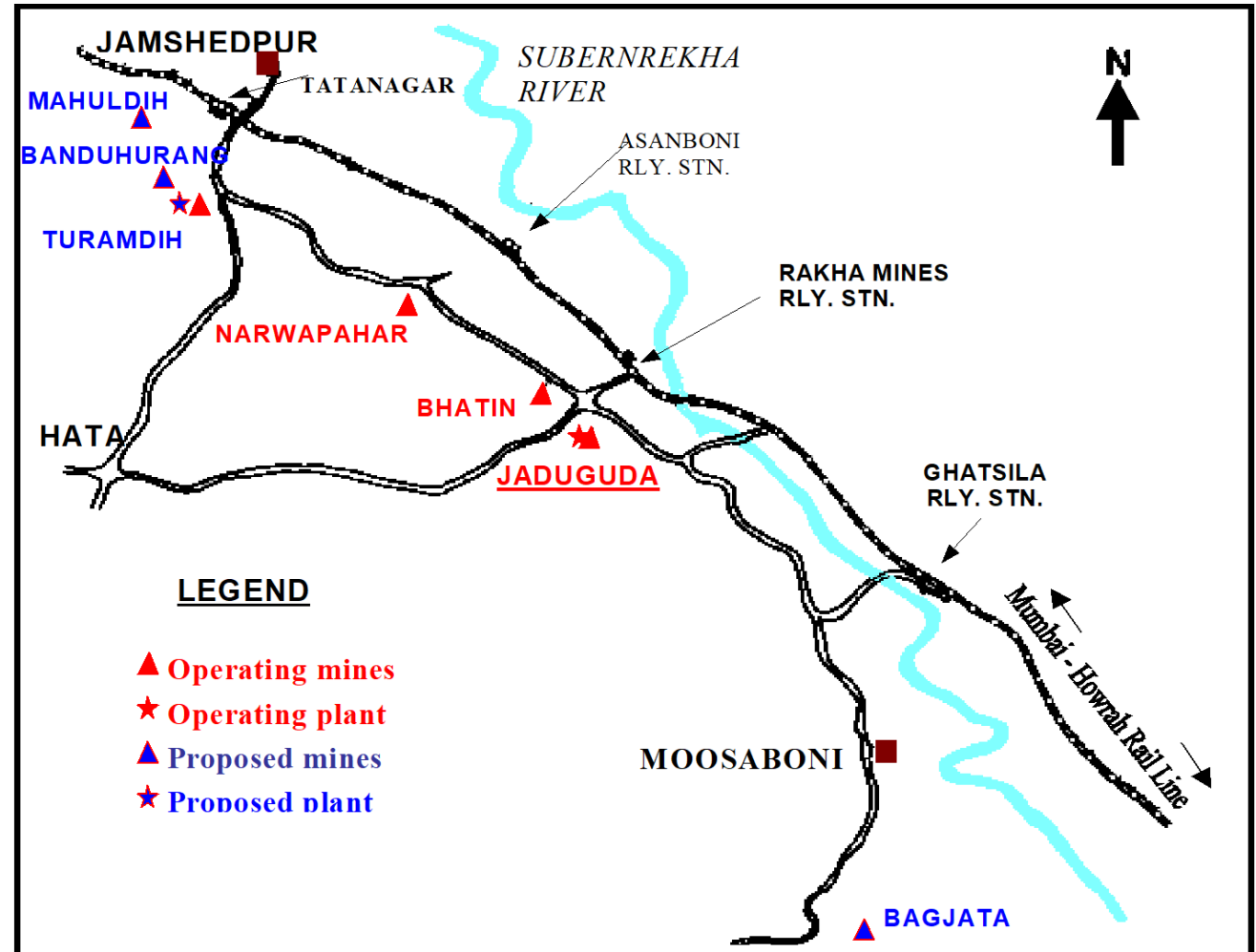


URANIUM MINING IN INDIA



Operation at Jharkhand

- » Jaduguda Mine
- » Bhatin Mine
- » Turamdih Mine
- » Bagjata Mine
- » Narwapahar Mine
- » Banduhurang Mine
- » Jaduguda Mill
- » Turamdih Mill
- » Mohuldi Mine



Bhatin Mine

Turamdih Mine

Bagjata Mine

Narwapahar Mine

Banduhurang Mine

Jaduguda Mill

Turamdih Mill

Mohuldih Mine

Jaduguda Mine



Uranium ore is found in the mineralized zone of Singhbhum Shear zone which is about 160 km in length and 1 to 10 km in width in the East & West Singhbhum Districts of Jharkhand. Rocks in the zone are highly folded and sheared. Uranium bearing minerals occur in very finely disseminated form. Ore lenses are hosted in metamorphic rocks of Proterozoic age.

Mining operations at Jaduguda began in the year 1967. Jaduguda Mine has the distinction of being the first Uranium Mine of

the country. The mine is accessed through a 5 metre diameter vertical shaft of depth of 640 metre. The shaft is lined with concrete. The commissioning of this shaft with tower mounted Friction Winders marked a technical milestone for India's mining industry. One of the winders is fitted with a cage and other with a skip. The cage accommodates 50 persons. The skip has a capacity of hoisting 5 tons of ore per trip. This shaft is also the main ventilation intake. The bottommost production level catering to this shaft is 555 metre level. Subsequently, a similar auxiliary vertical shaft with latest friction winders was sunk from 555 metre level to 905 meters to cater to ore hoisting requirement up to 880 metre level. The mining method followed for extraction of ore is Horizontal Cut and Fill with deslimed mill tailings as back fill. Demarcation of contact line for uranium ore and waste is carried out by using Geiger Muller counters and grade control of transported ore is performed by Scintillating Probes. Ore hoisted to surface is transferred to the adjacent process plant by conveyor. Jaduguda Mine has created a large skill base for the mining industry in general and uranium mining in particular.

URANIUM MINE AT JADUGUDA



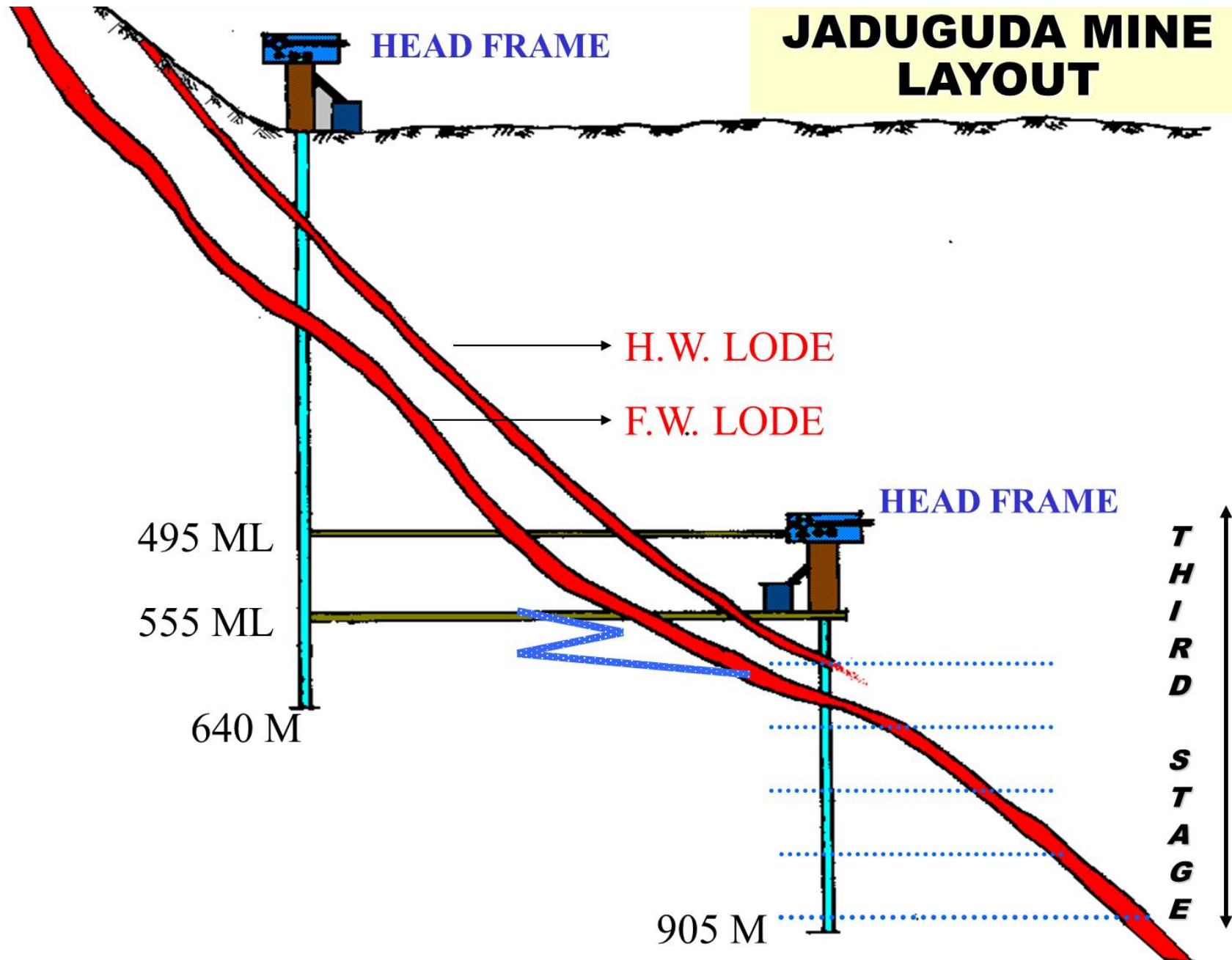
Jaduguda is the first uranium mine in the country commissioned in 1968.

In the **FIRST STAGE**, it was developed upto 315m and deepened upto 640m in the **SECOND STAGE**. Today it is 905m deep after **THIRD STAGE** of deepening.

The entry into the mine is through a 640m deep vertical shaft. An underground vertical shaft from 555m depth provides entry upto the depth of 905m.

The success story of Jaduguda mine symbolises the beginning of a mammoth nuclear programme for the country.

JADUGUDA MINE LAYOUT



Jaduguda Mine

Turamdih Mine

Bagjata Mine

Narwapahar Mine

Banduhurang Mine

Jaduguda Mill

Turamdih Mill

Mohuldih Mine

Bhatin Mine



Bhatin Mine is a small underground mine located 3 km away from Jaduguda. It shares most of the infrastructure of Jaduguda Mine. The access to the ore body is through adits. Deeper levels are accessed through winzes with man winding facility. The method of mining is horizontal cut and fill similar to that of Jaduguda mine. Mining of this small deposit illustrates UCIL's commitment to optimally utilise the country's scarce Uranium resources.

URANIUM MINE AT BHATIN



Bhatin is a small underground uranium mine commissioned in 1987.

Entry to the mine is through a horizontal passage - called adit. An underground incline provides entry into the deeper levels.

The present mining operation is carried out at a depth of 185m.

Bhatin mine will continue upto a depth of 250m.

Jaduguda Mine

Bhatin Mine

Turamdih Mine

Bagjata Mine

Banduhurang Mine

Jaduguda Mill

Turamdih Mill

Mohuldih Mine

Narwapahar Mine Of UCIL



Narwapahar Mine is the first fully mechanized mine operating since April 1995. This is a trackless mine with a decline access to underground and ramp accesses to the stopes. This permits use of large diesel powered underground equipment resulting in high productivity, eliminating fatigue of workmen and providing a good working environment. Diesel traction and Electro-hydraulic Drill Jumbos are used for drilling. The capability to drill long and parallel holes has improved drilling

productivity. The mucking of blasted ore and waste is carried out by diesel powered Load – Haul – Dump Loaders and transported by Low – Profile – Dump – Trucks. To be able to utilize the high productive capacities of these machines, adequate back-up in terms of service vehicles such as passenger carriers, explosive vans, service and carriage trucks have been deployed. These make Narwapahar one of the most modern mines in India.

Narwapahar Mine: Salient features

- **Narwapahar Mine is the First Fully mechanized underground Metal Mine Operating Since April 1995**
- **This is a trackless mine with a decline access to underground and ramp accesses to the stopes**
- **This permits use of large diesel powered (tyre mounted) underground equipment resulting in high productivity, eliminating fatigue of workmen and providing a good working environment**
- **Diesel traction and electro-hydraulic drill jumbos are used for drilling in level development, Decline, ramps, and in stopes. The capability to drill long and parallel holes has improved drilling productivity**
- **The mucking of blasted ore and waste is carried out by diesel powered Load-Haul-Dump (LHD) Loaders and transported by Low-Profile-Dump-Truck (LPDT) to the near by ore pass**
- **To be able to utilize the high productive capacities of these machines, adequate back-up in-terms of service vehicles such as passenger carriers, explosive vans, service and carriage trucks have been deployed**

Narwapahar Mine: Salient features

- **Mine is equipped with a vertical shaft of 355 m depth which serves for man-material winding and mineral hoisting from deeper levels.**
- **Mine is equipped with ground mounted koepe winding system for both man-material and mineral hoisting**
- **Jack hammer is used for raise drilling**
- **Alimak Raise climber is used for long Raise (End Raise) development**
- **Mine is developed up to 8th level (320 mL) and proposed to work up to 460 m**
- **Boundary ventilation system (BVS) is adopted in the mines to improve the ventilation efficiency**
- **Three ventilation shafts are there in the mine namely East, West and KND ventilation shaft**
- **The principal mining method/stoping method is Horizontal-cut-and-Fill with Room and Post Pillar**
- **Mill Tailing and Waste rock generated during mine development is used as back-filling material**
- **Mined ore is processed at Jadugoda Mill**
- **These make Narwapahar Mine one of the most modern mines in the country**

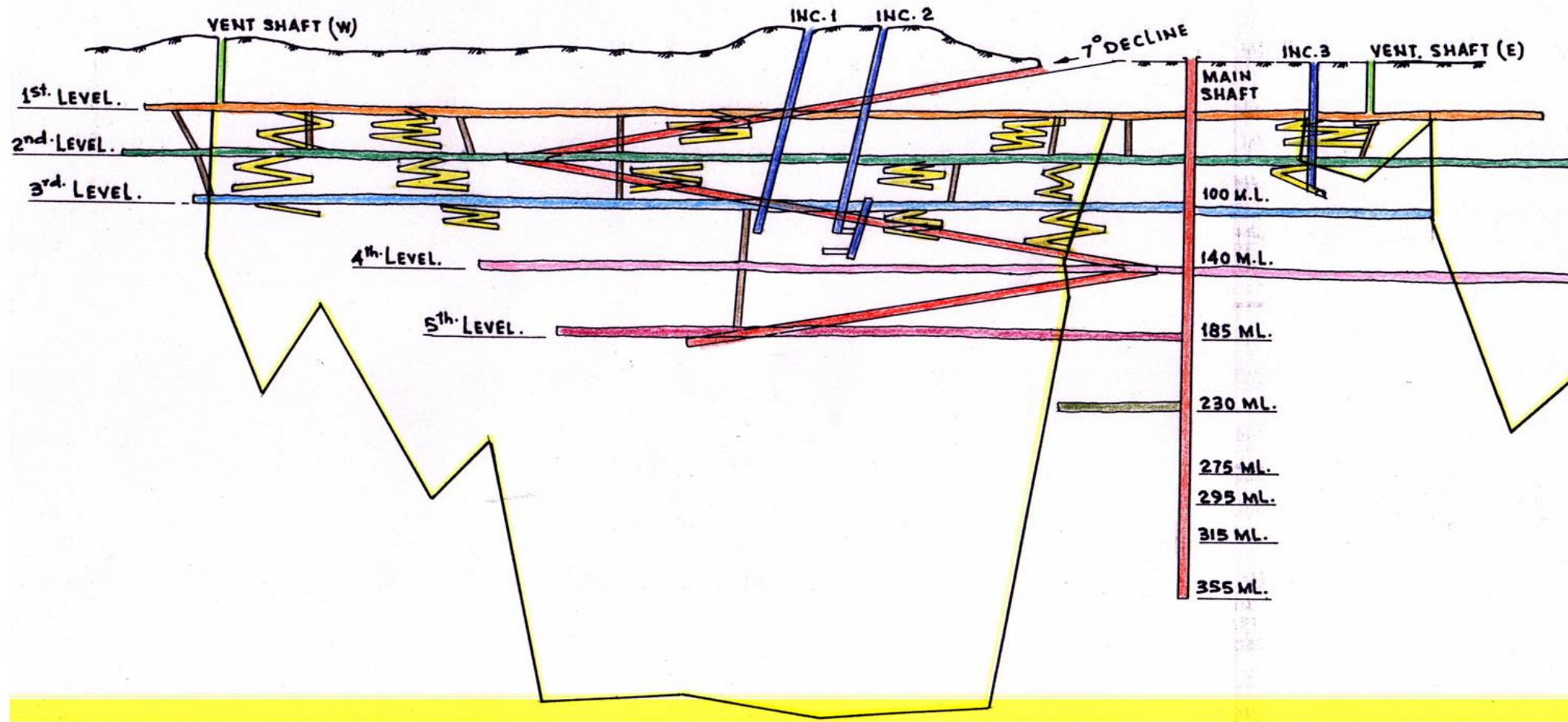
MINE ENTRY



7° ACCESS DECLINE



VERTICAL SHAFT



LAYOUT OF NARWAPAHAR MINE

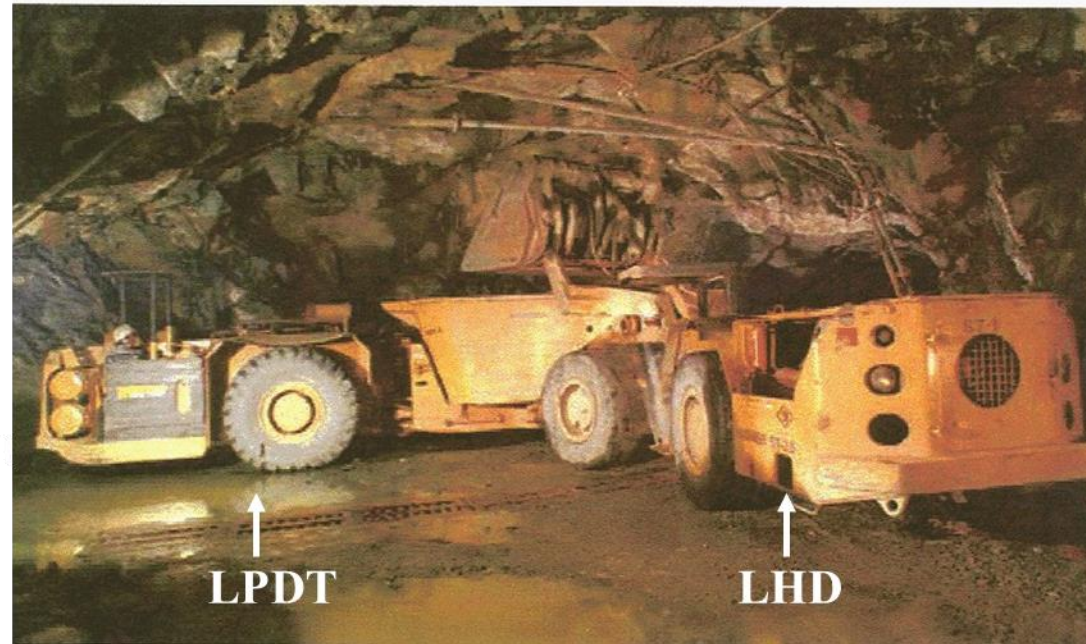
LONGITUDINAL SECTION

DRILLING BY JUMBO



- **MORE ENERGY EFFICIENT**
- **LESS DRILLER FATIGUE**
- **MORE PRODUCTIVITY**
(METERAGE/MAN)
- **DRILLING QUALITY IS BETTER**

TRACKLESS HAULING



- MANUAL STRAIN IS MINIMISED
- LESS PERSONS REQUIRED IN U/G
- MORE FLEXIBLE

MT426

H281

Jaduguda Mine

Bhatin Mine

Bagjata Mine

Narwapahar Mine

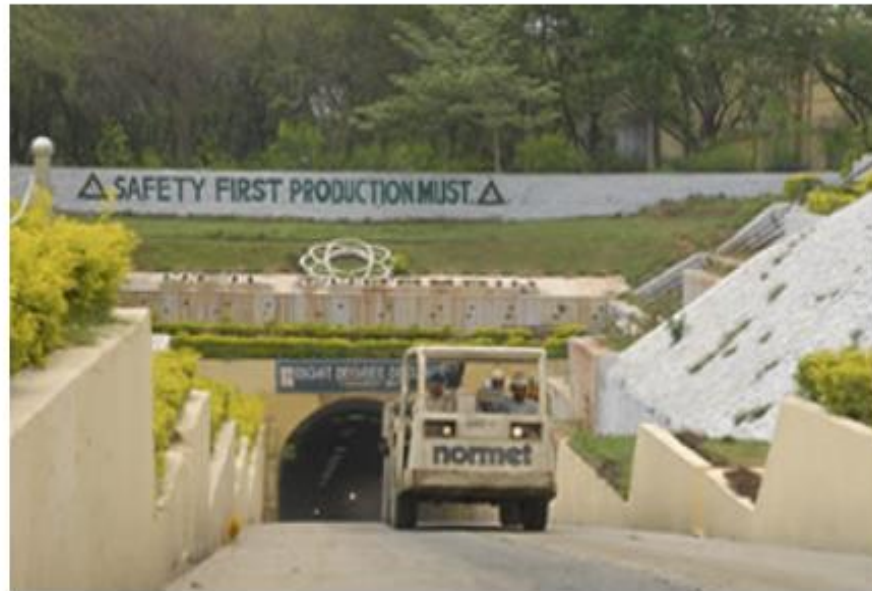
Banduhurang Mine

Jaduguda Mill

Turamdih Mill

Mohuldih Mine

Turamdih Mine



for cage and skip. The mine employs modern trackless mining system. The principal method of mining is Horizontal Cut-and-Fill with Post and Pillar.

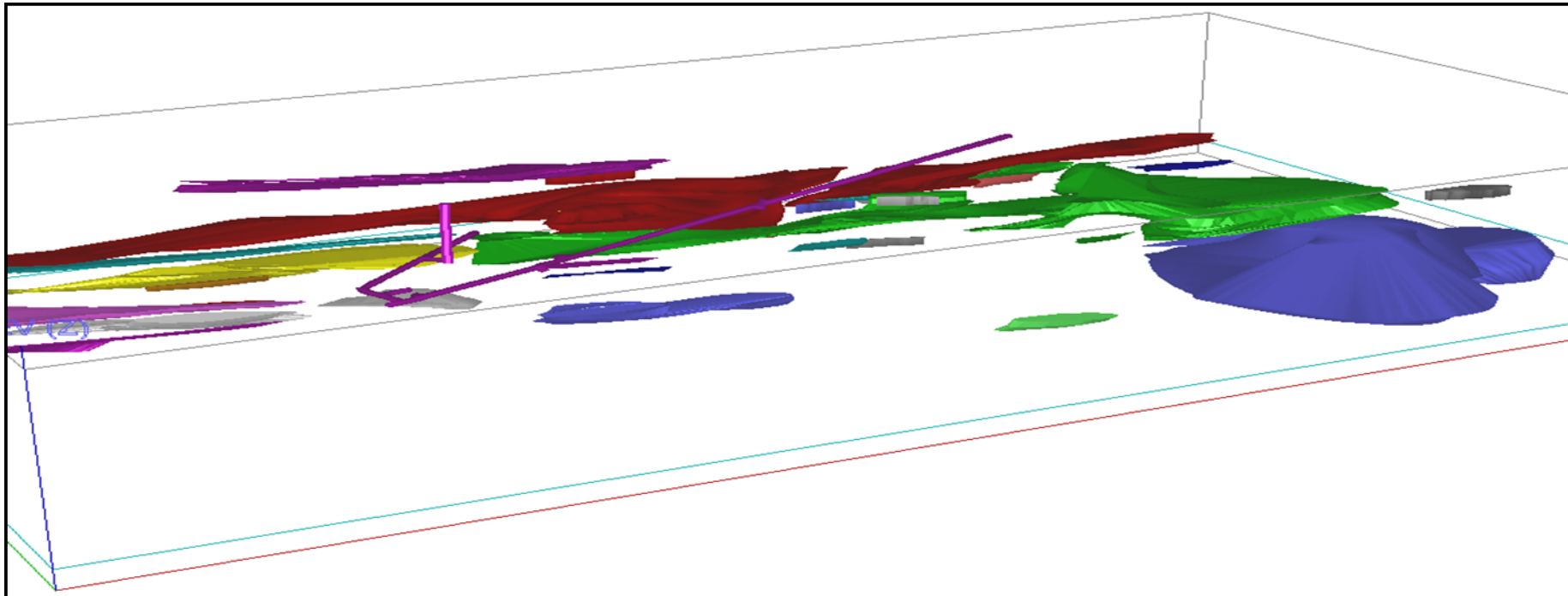
Turamdih uranium deposit is located about 24km west of Jaduguda. This mine was commissioned in 2003. The entry into the mine is through an 8 degree decline which provides facilities for using trackless mining equipment like passenger carriers, drill jumbos, low-profile dump trucks etc. At a depth of 70m, the orebody has been accessed from the Decline by cross-cuts. A 260 metre vertical shaft with a double Drum Winder caters to hoisting of ore from deeper levels. The shaft has a change over facility

Turamdih Mine : Salient Features

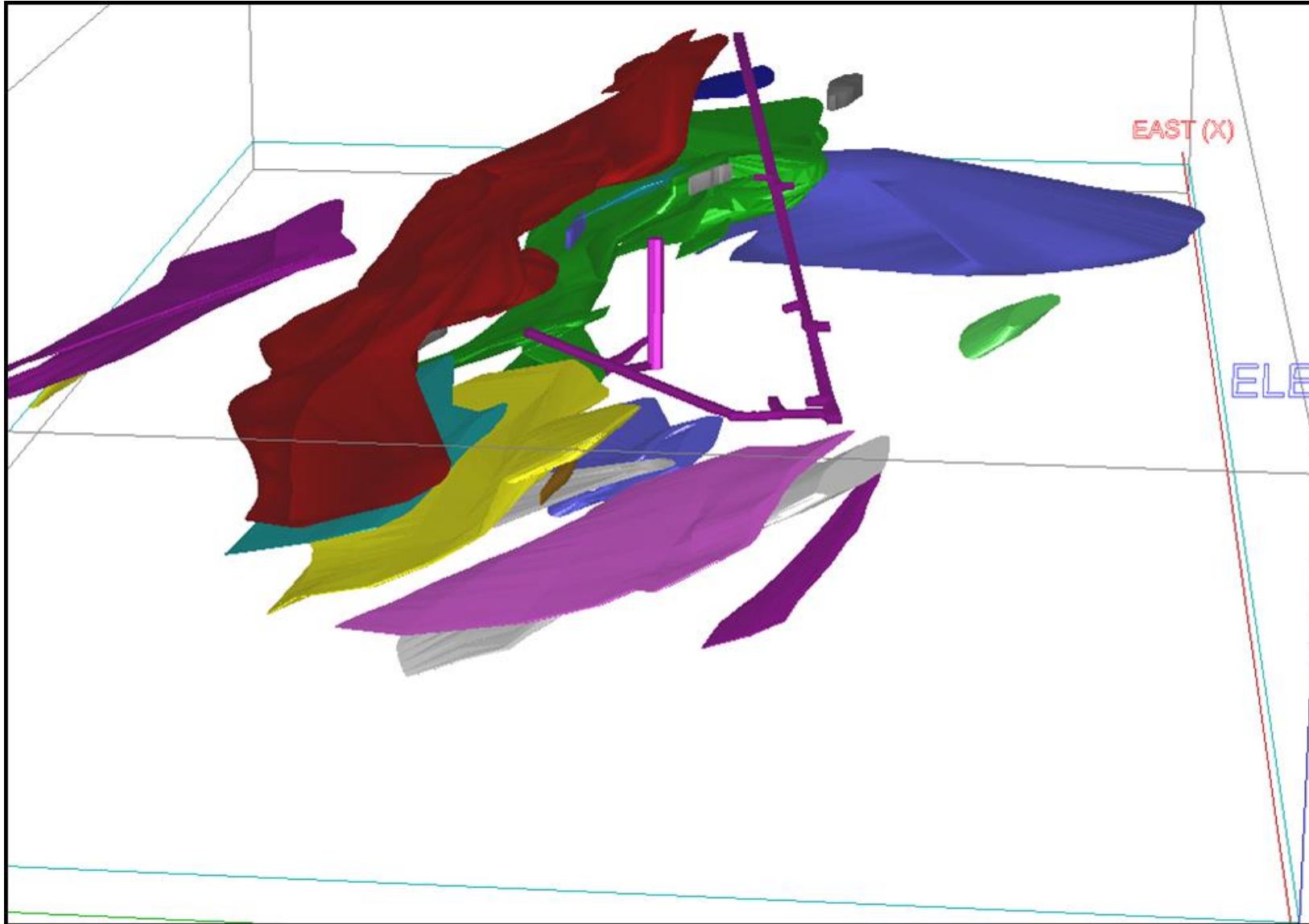
- **Turamdih uranium deposit/mine is located about 24 KM west of Jaduguda**
- **This mine was commissioned in the year 2003**
- **The Entry into the mine is through an 8 degree Decline which provides facilities for using trackless mining equipment like passenger carrier, Drill Jumbo, LHD & LPDT etc.**
- **At a depth of 70 meter, the ore body has been accessed from Decline by cross-cuts**
- **A 260 meter vertical shaft with a Double Drum Winder caters to hoisting of ore from deeper levels**
- **The shaft has a change over facility from cage and skip**
- **The Mine employs modern trackless mining system**
- **The principal mining method/stoping method is Horizontal-cut-and-Fill with Room and Post Pillar**
- **Mill Tailing and Waste rock generated during mine development is used as back-filling material**
- **Mined ore is processed at Turamdih Mill**



**3-D VIEW SHOWING ALL THE LODES ALONGWITH MINE-
DEVELOPMENTS (VIEWIING ANGLE : 29° HORZ & 110° VERT)**



**ZOOMED 3-D VIEW SHOWING ALL THE LODES ALONGWITH
MINE-DEVELOPMENTS
(VIEWIING ANGLE : 84° HORZ & 105° VERT)**



Jaduguda Mine

Bhatin Mine

Turamdih Mine

Narwapahar Mine

Banduhurang Mine

Jaduguda Mill

Turamdih Mill

Mohuldih Mine

Bagjata Mine



materials from the deeper levels. The mine employs modern trackless mining system. The principal method of mining is Horizontal Cut-and-Fill with Post and Pillar. The ore of Bagjata mine is sent to Jaduguda by road for processing.

Bagjata mine is located in an extremely remote area about 25 km east of Jaduguda. Construction of an all weather road and a bridge over river Sankh Nalla by UCIL during the course of opening up of the mine had a tremendous impact in the development of the surrounding localities. The mine was commissioned in 2008. The principal entry into the mine is through a 7 degree decline. A vertical shaft has been sunk to a depth of 375 metre to meet the requirements of ore hoisting and transportation of men and

Bagjata Mine : Salient Features

- **Bagjata uranium deposit/mine is located about 25 KM east of Jaduguda Mine**
- **This mine was commissioned in the year 2008**
- **The Principal Entry into the mine is through a 7 degree Decline which provides facilities for using trackless mining equipment like passenger carrier, Drill Jumbo, LHD & LPDT etc.**
- **A vertical shaft has been sunk to a depth of 375 meter to meet the requirements of ore hoisting and transportation of men and materials from deeper level.**
- **At a depth of 70 meter, the ore body has been accessed from Decline by cross-cuts**
- **A 260 meter vertical shaft with a Double Drum Winder caters to hoisting of ore from deeper levels**
- **The Mine employs modern trackless mining system**
- **The principal mining method/stopping method is Horizontal-cut-and-Fill with Room and Post Pillar**
- **Mill Tailing and Waste rock generated during mine development is used as back-filling material**
- **Mined ore is transported (via road) at Jadugoda Mill for its processing**



Jaduguda Mine

Bhatin Mine

Turamdih Mine

Bagjata Mine

Narwapahar Mine

Banduhurang Mine

Jaduguda Mill

Turamdih Mill

Mohuldih Mine



Mohuldih uranium deposit Mine in Gamharia block of Seraikella-Kharsawan district in state of Jharkhand has been developed as a modern underground mine by Uranium Corporation of India Ltd. The mine was commissioned on 17th April 2012.

Presently, the principal entry into Mohuldih mine is through an 80 decline of 5m width and 4.5 m height which facilitates the use of high productive trackless mining equipment like low profile dump truck, diesel traction & electric- hydraulic drill jumbo etc. A vertical

shaft is under sinking. The mine adopts modern trackless mining system. The mining method for extraction of ore blocks is horizontal cut-and-fill with post pillar. Ore produced from the mine is fed to Turamdih mill.

Mohuldih Mine : Salient Features

- **This mine was commissioned in the year 2012**
- **The Principal Entry into the mine is through a 8 degree Decline (Size 5m*4.5m) which provides facilities for using trackless mining equipment like passenger carrier, Drill Jumbo, LHD & LPDT etc.**
- **A vertical shaft is under sinking**
- **The Mine employs modern trackless mining system**
- **The principal mining method/stoping method is Horizontal-cut-and-Fill with Room and Post Pillar**
- **Mill Tailing and Waste rock generated during mine development is used as back-filling material**
- **Mined ore is transported (via road) to Jadugoda Mill for its processing**



Jaduguda Mine

Bhatin Mine

Turamdih Mine

Bagjata Mine

Narwapahar Mine

Banduhurang Mine

Turamdih Mill

Mohuldih Mine

Jaduguda Mill



Ore from Jaduguda, Bhatin, Bagjata and Narwapahar mines is processed in the processing plant (Mill) at Jaduguda. It is located adjacent to Jaduguda mine. Uranium is extracted from ore by hydro-metallurgical process. After three stages of crushing, and two stages of wet grinding, the slurry thus obtained is pumped to leaching pachucas for dissolution of Uranium. The leached slurry is filtered to obtain liquor containing uranium which is purified and concentrated by ion exchange method. Uranium is then

precipitated from this concentrated liquor as magnesium Di-Uranate, generally known as “YELLOW CAKE”. This is thickened, washed, filtered and dried in the spray dryer and finally packed in drums and then sent to Nuclear Fuel Complex at Hyderabad for further processing into UO₂ pellets..

The plant has undergone several modifications adopting technologies to maximize the re-use of water, high recovery of the product and minimum discharge of effluents. It has several automated process control mechanism and online monitoring systems.

Jaduguda Mine

Bhatin Mine

Turamdih Mine

Bagjata Mine

Narwapahar Mine

Banduhurang Mine

Jaduguda Mill

Mohuldih Mine

narwapahar

Turamdih Mill



A new plant at Turamdih has been set up to treat ore produced from Turamdih and Banduhurang mines. The flowsheet of this plant is similar to that of Jaduguda. However, taking account of developments in hydrometallurgy / processing technology worldwide, some efficient equipment like apron feeders, particle size monitors, horizontal belt filter, pressure filter etc have been incorporated in this plant. A very high degree of instrumentation has been introduced that includes PLC based control

system. The mill caters to processing of ore produced from Turamdih, Banduhurang and Mohuldih mines.

Operation at Other States

- » KPM Project
- » Tummalapalle uranium project
- » Lambapur uranium project
- **Gogi Uranium Project**
- **Rohil Uranium Project**

Tummalapalle Uranium Project

Lambapur Uranium Project

KPM Uranium Project

Kyelleng-Pyndengsohiong, Mawtahbah uranium project, Meghalaya -The sandstone hosted orebody at Killung and Rangam in West Khasi Hills district of Meghalaya is the first of its kind to be discovered in the country. UCIL has planned to construct open pit mines at this site and a processing plant at Mawthabab. Infrastructure development and some welfare activities have already been taken up. Site activities are expected to start soon.

KPM Uranium Project

Tummalapalle Uranium Project

Lambapur Uranium Project

A large uranium resource has been established at Lambapur-Peddagattu region in the Nalgonda district of Telengana. UCIL proposes to set up one open pit and three underground mines. A central plant about 48 km from the mine site based on acid leaching has been planned.

KPM Uranium Project

Lambapur Uranium Project

Tummalapalle Uranium Project



based on alkali leaching under controlled temperature and pressure technology has been put under trial run..

Tummalapalle is part of a large tabular, strata bound and flatly dipping uranium reserve occurring in carbonate host rock in YSR district of Andhra Pradesh. The lodes as well as the parting in between are narrow. A highly mechanised underground mine has been opened up using state of the art technology through three numbers of 9-degree Declines. The central Decline has a belt conveyor meant for transport of ore and the other two are meant for movement of men and machineries. A processing plant

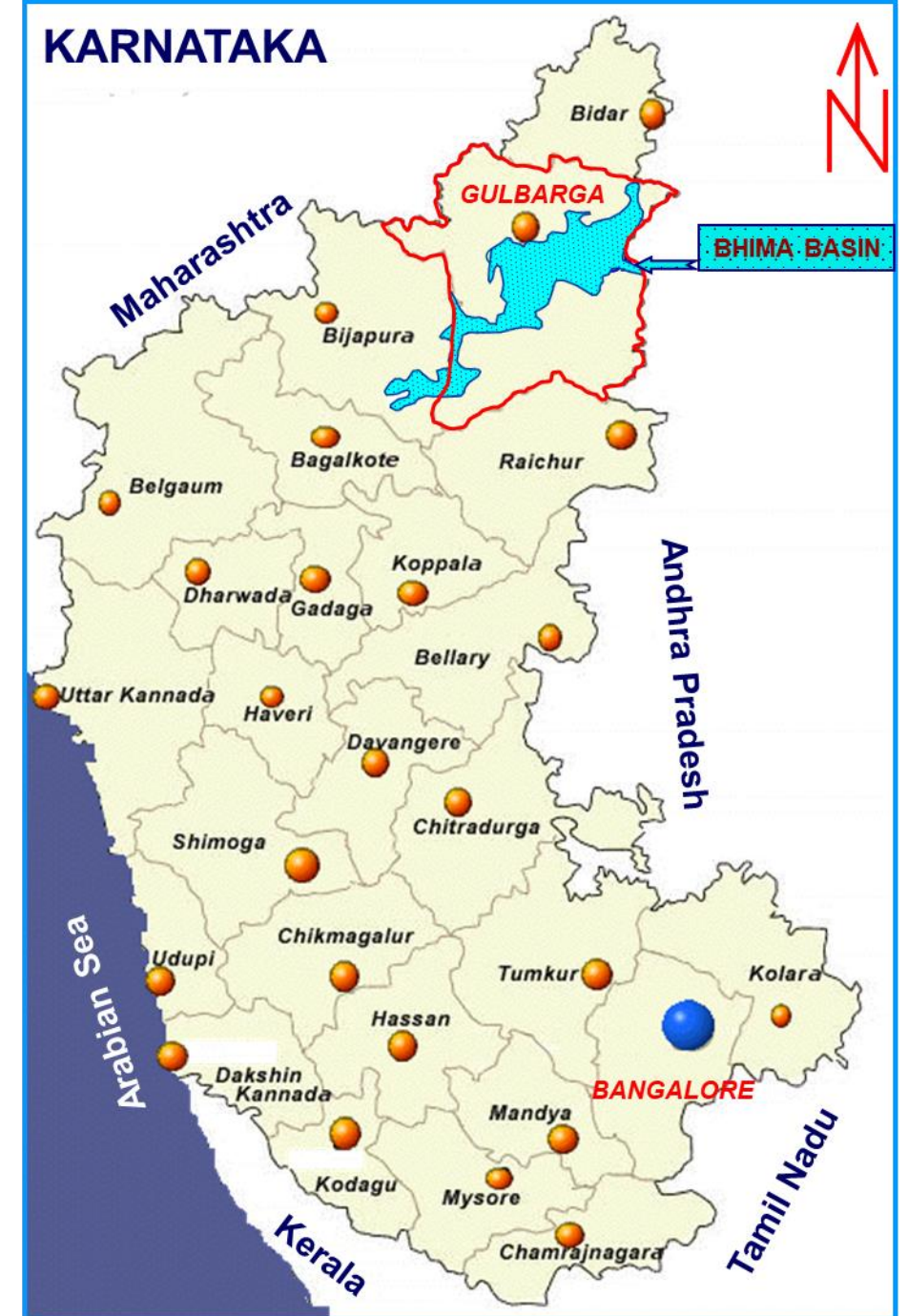
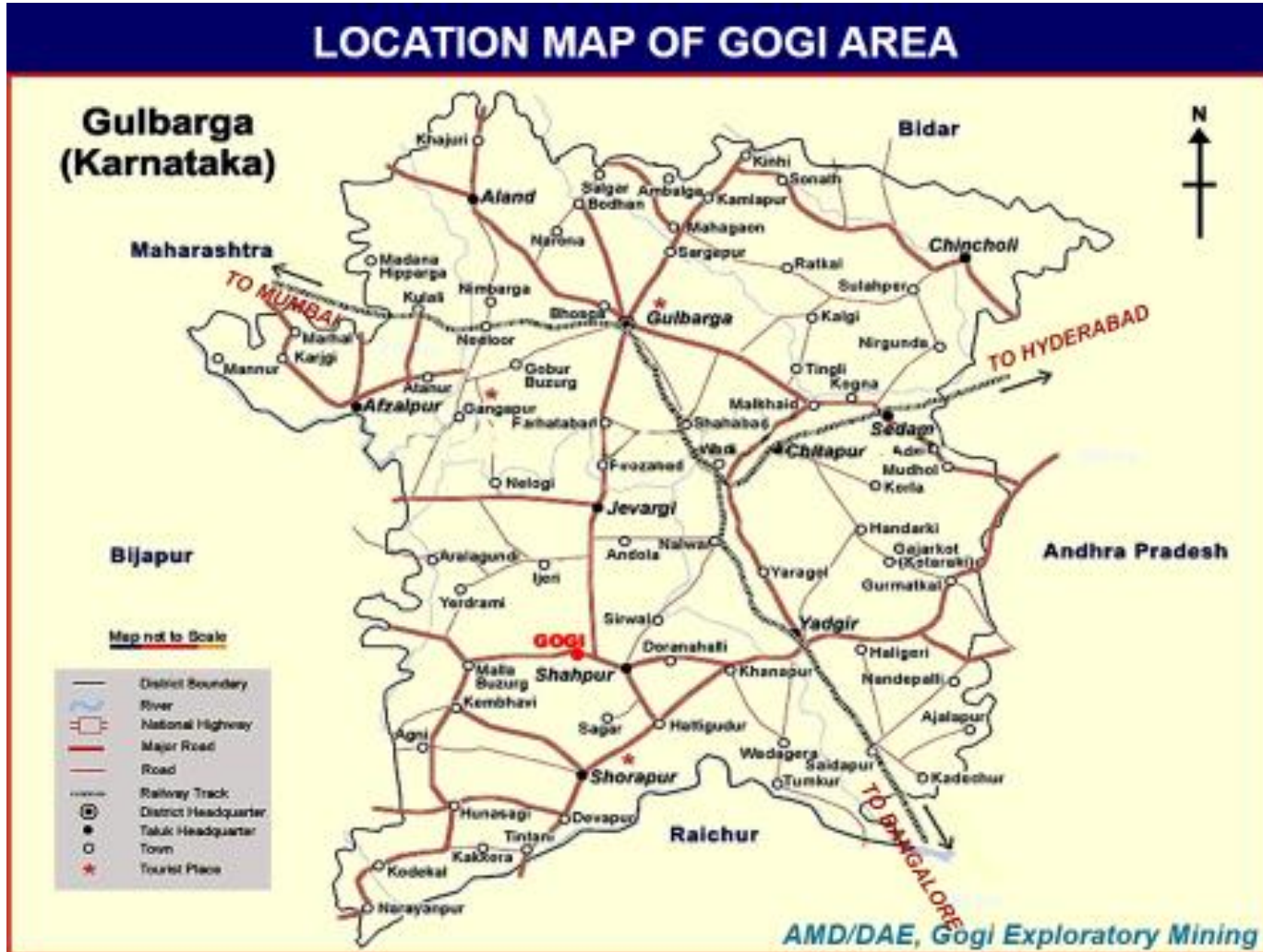
Tummalapalle Mine : Salient Features

- **This mine was commissioned in the year 2012**
- **Tummalapalle is part of a large tabular strata bound and flatly dipping uranium reserve located in YSR district of Andhra Pradesh**
- **The lodes as well as parting in between are narrow**
- **A highly mechanized underground mine has been opened up using of state of the art technology through three numbers of 9-degree declines**
- **The central decline has a belt conveyor meant for transport of ore and other two meant for movement of men and machineries**
- **Mined ore is transported to Tummalapalle Mill for its processing**



Gogi Uranium Project

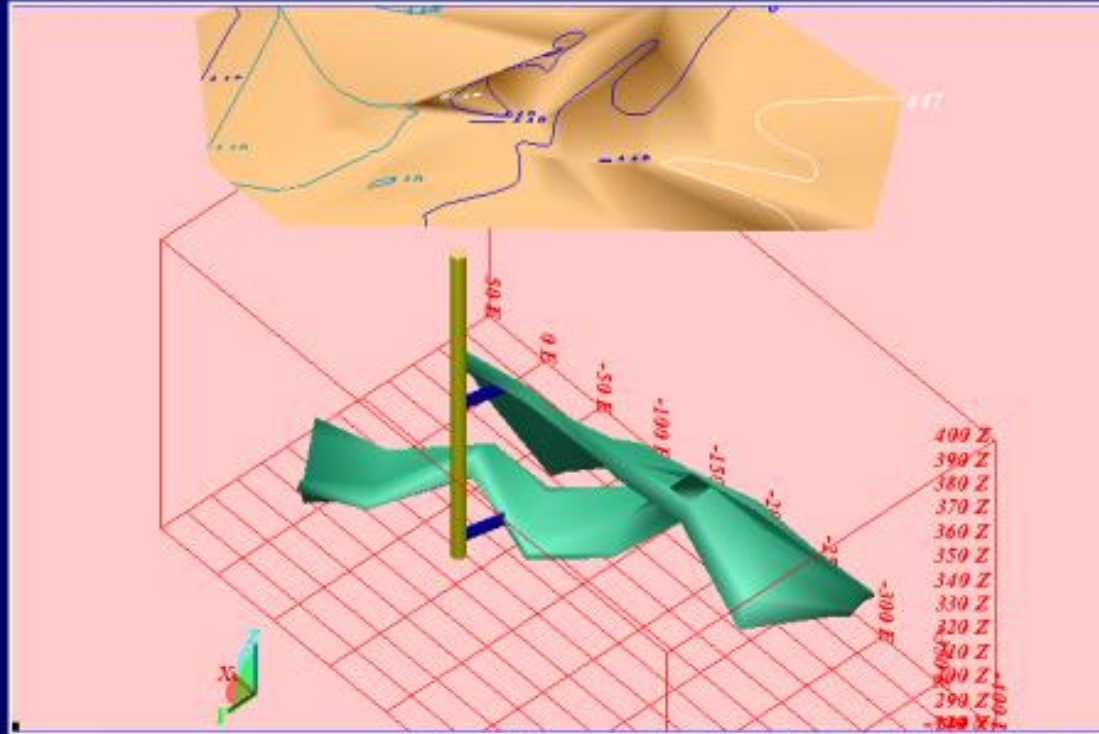
- **Exploratory Mining is completed**
- **Mining yet to be started**



Gogi Uranium Project : Salient Features

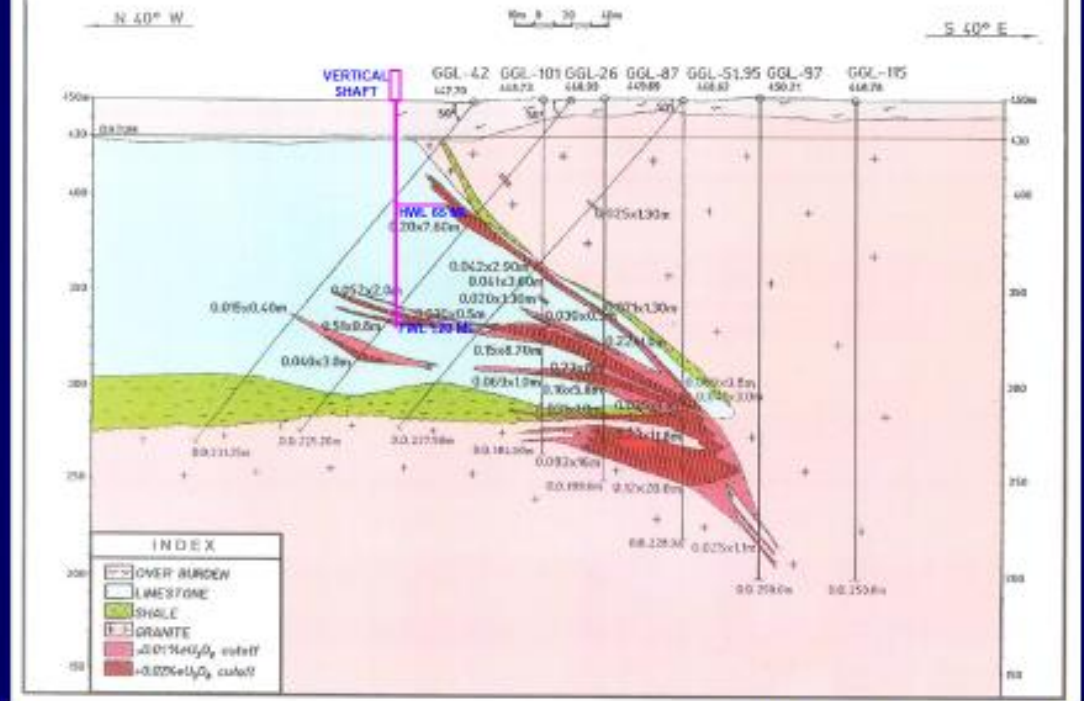
- **Exploratory Mining is completed**
- **Mining yet to be started**

3D MODEL SHOWING LIMESTONE OREBODY & EXPLORATORY MINE SHAFT



AMD/DAE, Gogi Exploratory Mining

MINE ENTRY AND LEVELS FOR PROPOSED EXPLORATORY MINE AT GOGI GULBARGA, KARNATAKA.



AMD/DAE, Gogi Exploratory Mining

Gogi Uranium Project

- **Exploratory Mining is completed**
- **Mining yet to be started**

DETAILS OF ORE LODES

Lode	Depth from surface (m)		Between RLs(m)	Strike Length (m)
	From	To		
Eastern Hangwall	38	176	410 - 272	142
Central Hangwall	63	145	385 - 303	107
Western Hangwall	60	133	388 - 315	174
Footwall	110	175	338 - 273	529
Granite	158	233	290 - 215	270

AMD/DAE, Gogi Exploratory Mining

SUMMARY OF ORE RESERVE

Host Rock	METHOD OF TRIANGLES		
	Ore tonnage (million tonnes)	U ₃ O ₈ (tones)	Av. Grade % eU ₃ O ₈
0.02% e U ₃ O ₈ grade & 1.2 m cut-off thickness			
Limestone Reserves	1.306	2413	0.185
Granite Reserves	0.512	664	0.129
Total	1.815	3077	0.170

AMD/DAE, Gogi Exploratory Mining

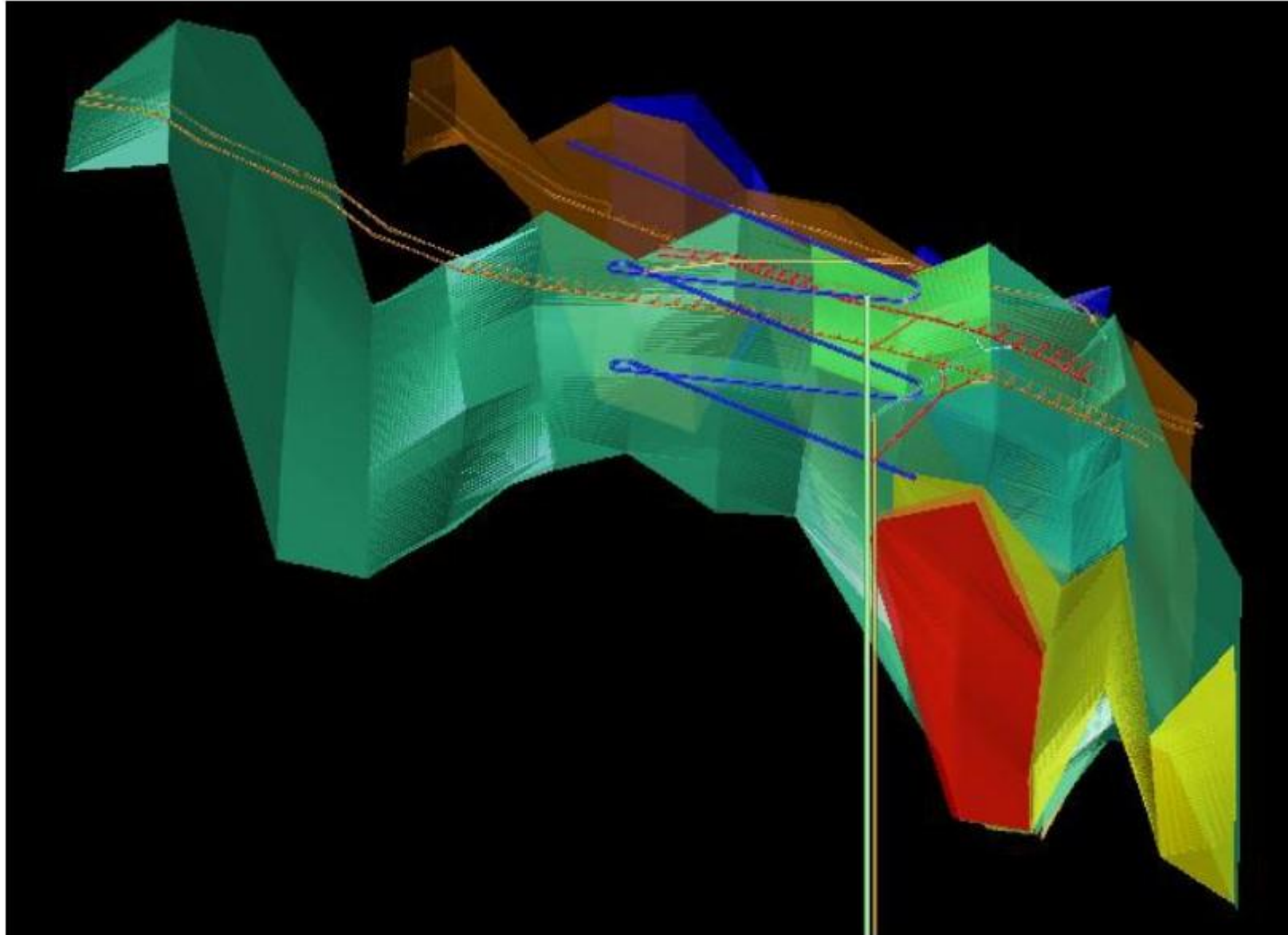
Rohil Uranium Project: Salient Features

- It is under commissioning
- Rohil Uranium deposit is located in Sikar district (27° 33' 25" N Latitude and 75° 29' 25" E Longitude) of Rajasthan
- The Rohil deposit shall be mined by underground mechanized method of mining
- The proposed capacity for the mine shall be 0.75 MTPA.
- The proposed method of mining shall be longitudinal sublevel stoping mainly as the deposit is favourable for sublevel method.
- the life of the mine for the underground operations have been worked out to be 16 years including initial years of partial production
- There are two modes of entries proposed for the underground mine: Vertical shaft and Decline
- A decline (at 7 degree gradient) and a vertical shaft upto a depth of 1000 m is envisaged to be the most suitable mode of entries into the mine considering orebody configuration.
- Decline may be developed parallel to the orebody with dimension of 4.5m X 5.5m and a shaft of 6m dia may be sunk with 60 cm RCC/PCC lining upto 1000 metres.

Rohil Uranium Project: Salient Features

- The shaft with provisions, men, material and ore hoisting will be equipped with suitable winder(s)
- With the available information the strata have been considered to be more or less competent
- The vertical interval between consecutive underground levels is proposed as 50m with a crown pillar of 10m.
- All the development and stoping operations shall be mechanized.
- For development jumbo drills/jack hammer and shall be used for drilling of face.
- LHD & dumper shall be used for loading and transporting of waste rock.
- For production drilling drifters (57 mm hole dia) and ITH drills (165 mm dia) shall be used and for mucking LHD in combination of LPDT will be deployed.
- It is proposed that mine will be ventilated through 2 Nos. of main mechanical ventilators (exhaust fans of 250 KW, 156 m³/sec capacity each) installed at two ventilation shafts (3.6m dia) at exhaust ends. It is proposed to allow fresh air to enter through decline and main vertical shaft.
- during stoping operations, rock bolts should be the primary means of support.
- Mined Ore will be processed at acid leaching based processing plant of 0.75 MTPA capacity which is proposed near the mine site

Rohil Uranium Project: Salient Features



3D model showing schematic development of upper levels



HINDUSTAN COPPER LIMITED

(A Govt. of India Enterprise)

The Copper Miner to the Nation



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Company Profile

Hindustan Copper Limited (HCL), a public sector undertaking, incorporated on 9th November 1967, is under the administrative control of the Ministry of Mines, Government of India . It has the distinction of being the nation's only vertically integrated copper producing company as it manufactures copper right from the stage of mining to beneficiation, smelting, refining and casting of refined copper metal into downstream saleable products.

The Company has the facilities for production & marketing of copper concentrate, copper cathodes, copper wire bar, continuous cast copper rod and by-products, such as anode slime (containing gold, silver, etc.), copper sulphate and sulphuric acid. Presently, company is focusing on mining & beneficiation operation and is primarily selling copper concentrate as the main product. In the concluded financial year 2020-21, the Company has earned a net profit (PAT) of Rs 109.98 crore against a sales turnover of Rs 1760.84 crore (excluding GST) HCL's mines and plants are spread across five operating Units, one each in the States of Rajasthan, Madhya Pradesh, Jharkhand, Maharashtra and Gujarat as named below:

- Malanjkhand Copper Project (MCP) at Malanjkhand, Madhya Pradesh
- Khetri Copper Complex (KCC) at Khetrinagar, Rajasthan
- Indian Copper Complex (ICC) at Ghatsila, Jharkhand
- Talaja Copper Project (TCP) at Talaja, Maharashtra
- Gujarat Copper Project (GCP) at Jhagadia, Gujarat

November 1967	Incorporated to take over from National Mineral Development Corporation Ltd.
March 1972	M/s Indian Copper Corporation Limited, Private Sector Company, located at Ghatsila, Jharkhand having facilities of Smelter and Refinery was Nationalized and made part of HCL.
February 1975	Fully integrated Copper complex from mining to refining came on stream at Khetri (capacity 31,000 tonnes of refined copper)
November 1982	The largest hard rock open pit mine in the country came into stream at Malanjkhand in Madhya Pradesh with capacity 2 million tonnes ore
December 1989	Continuous Cast Wire Rod plant of South Wire Technology of capacity 60,000 MT was commissioned at Talaja in Maharashtra.
June 2015	Secondary Copper Smelter and Refinery Unit at Jhagadia, Gujarat (50,000 TPA of Copper Cathodes Conforming to LME-A grade) was acquired

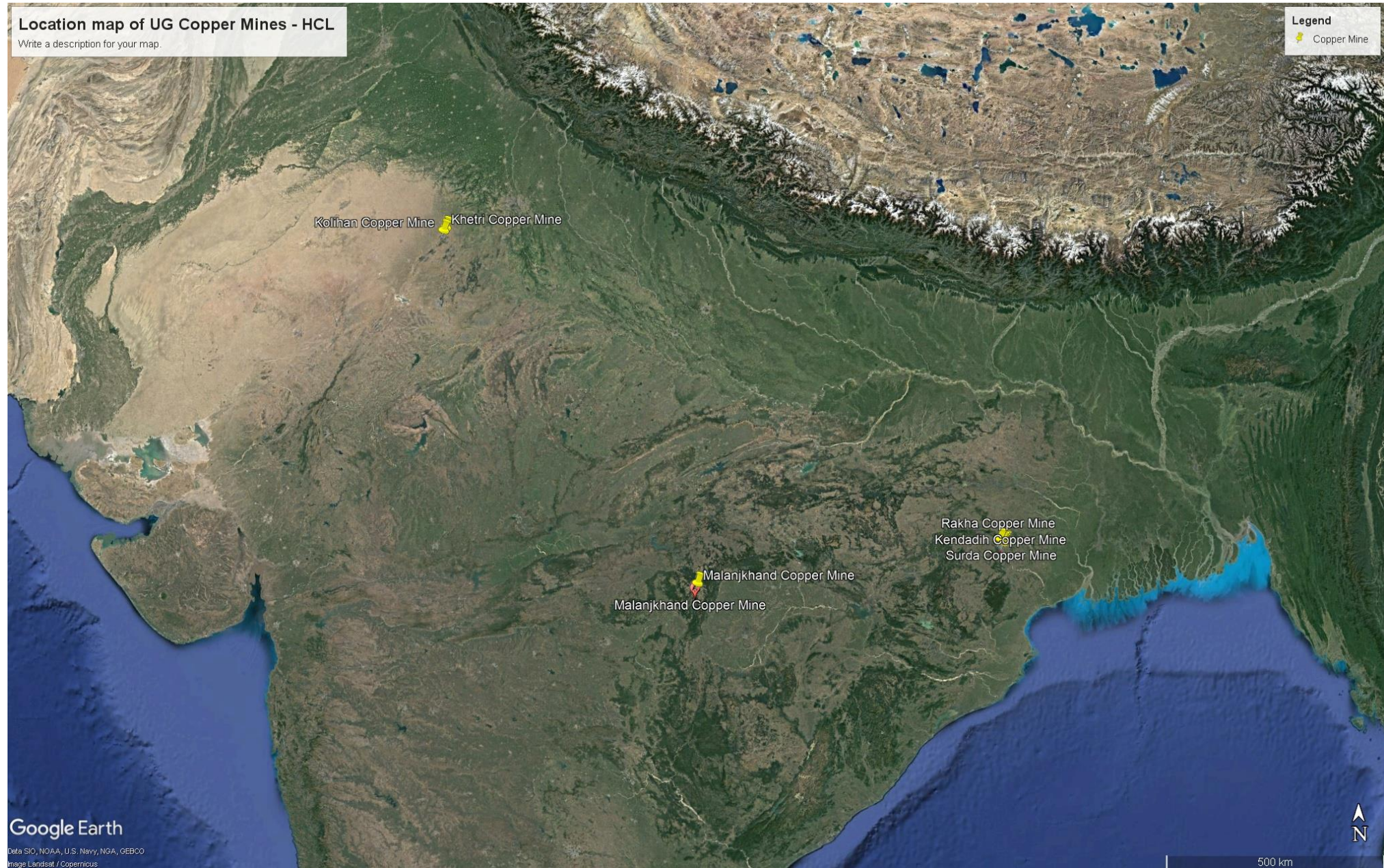
Hindustan Copper Limited (HCL), a public sector undertaking, incorporated on 9th November 1967, is under the administrative control of the Ministry of Mines, Government of India . It has the distinction of being the nation’s only vertically integrated copper producing company as it manufactures copper right from the stage of mining to beneficiation, smelting, refining and casting of refined copper metal into downstream saleable products.

The Company has the facilities for production & marketing of copper concentrate, copper cathodes, copper wire bar, continuous cast copper rod and by-products, such as anode slime (containing gold, silver, etc.), copper sulphate and sulphuric acid. Presently, company is focusing on mining & beneficiation operation and is primarily selling copper concentrate as the main product. In the concluded financial year 2020-21, the Company has earned a net profit (PAT) of Rs 109.98 crore against a sales turnover of Rs 1760.84 crore (excluding GST) HCL’s mines and plants are spread across five operating Units, one each in the States of Rajasthan, Madhya Pradesh, Jharkhand, Maharashtra and Gujarat as named below:

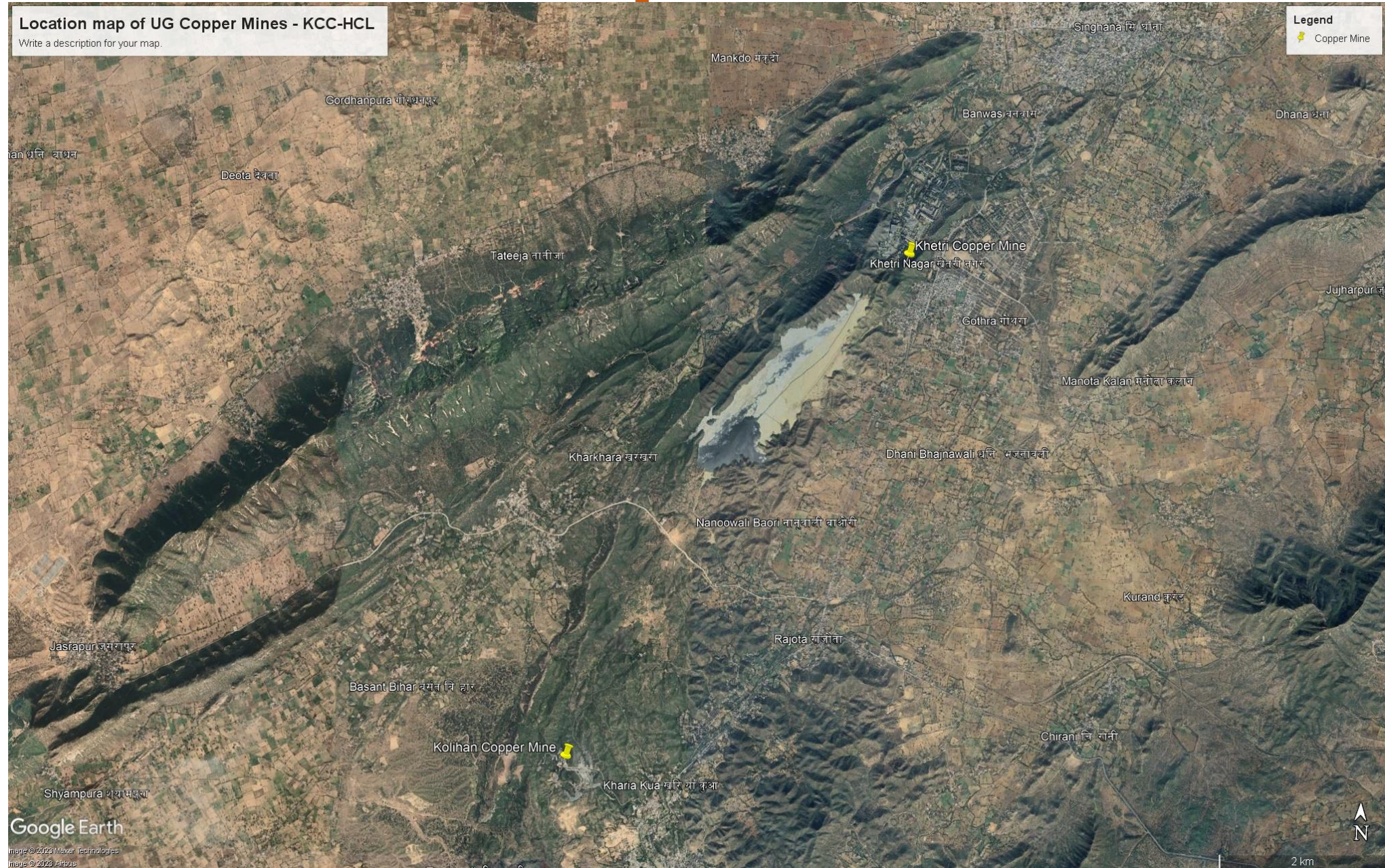
- Malanjkhand Copper Project (MCP) at Malanjkhand, Madhya Pradesh
- Khetri Copper Complex (KCC) at Khetrinagar, Rajasthan
- Indian Copper Complex (ICC) at Ghatsila, Jharkhand
- Taloja Copper Project (TCP) at Taloja, Maharashtra
- Gujarat Copper Project (GCP) at Jhagadia, Gujarat

November 1967	Incorporated to take over from National Mineral Development Corporation Ltd.
March 1972	M/s Indian Copper Corporation Limited, Private Sector Company, located at Ghatsila, Jharkhand having facilities of Smelter and Refinery was Nationalized and made part of HCL.
February 1975	Fully integrated Copper complex from mining to refining came on stream at Khetri (capacity 31,000 tonnes of refined copper)
November 1982	The largest hard rock open pit mine in the country came into stream at Malanjkhand in Madhya Pradesh with capacity 2 million tonnes ore
December 1989	Continuous Cast Wire Rod plant of South Wire Technology of capacity 60,000 MT was commissioned at Taloja in Maharashtra.
June 2015	Secondary Copper Smelter and Refinery Unit at Jhagadia, Gujarat (50,000 TPA of Copper Cathodes Conforming to LME-A grade) was acquired

Location map of HCL mines



Location map of HCL mines



HCL U/G mines at KCC

1. Khetri Copper Mine

2. Kolihan Copper Mine

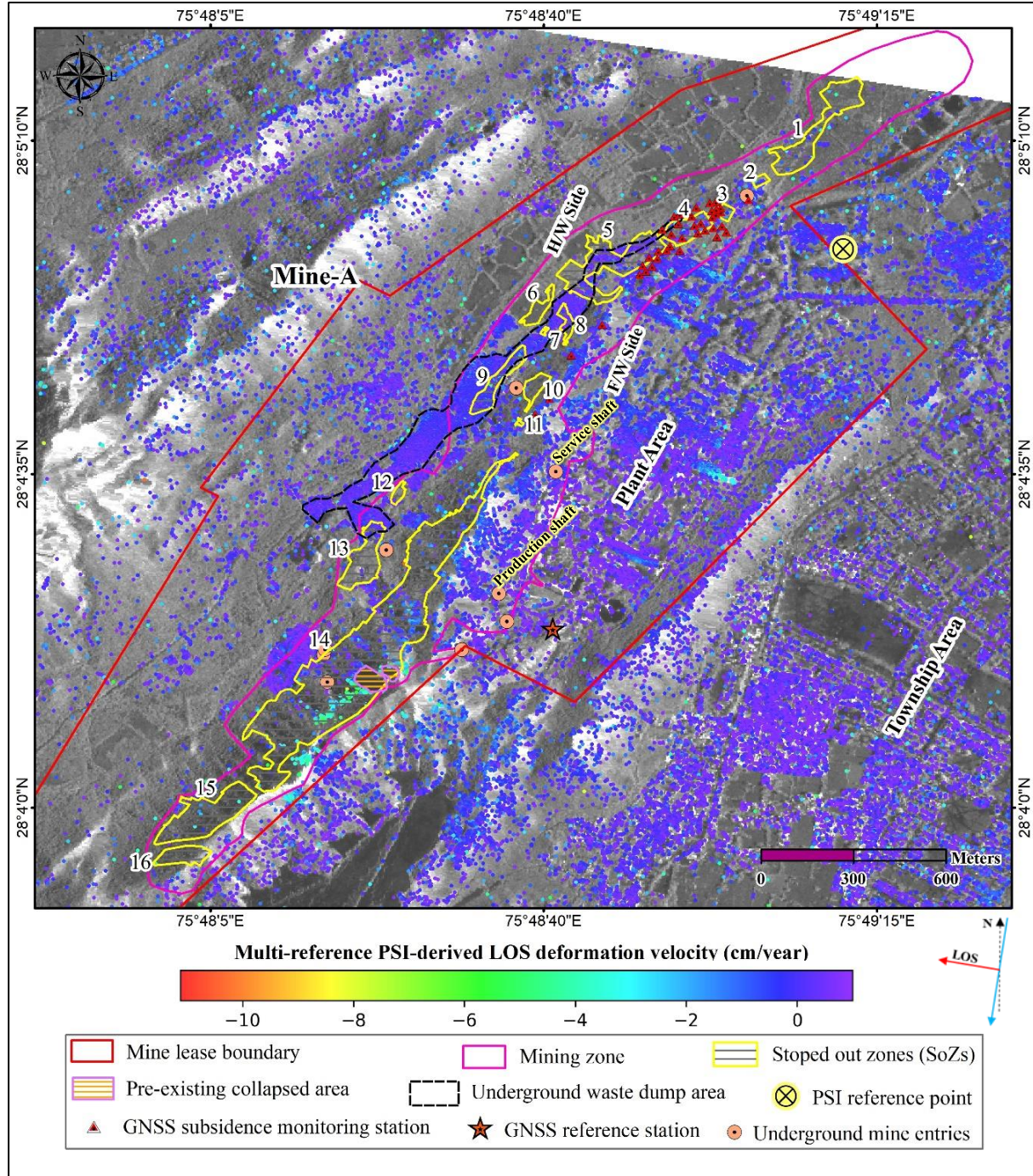
1. Khetri Copper Mine

- Khetri Mine has been developed over a strike length of 3425 m and has eight levels at a vertical interval of 60 m viz. 421, 350, 300, 240, 180, 120, 60 and 0 mRLs.
- The numbers indicate the height of the respective level above MSL.
- The upper five levels viz. 421, 350, 300, 240 and 180 mRL are exhausted, and the topmost three levels are isolated from the rest of the mine.
- Production is going on in three levels, namely 120, 60, and 0 mRL, and mine development is also going on in these three levels 120, 60 & 0 mRL. 2nd stage mine development below 0 mRL is under progress.
- Sub-level open stoping method has been adopted as the standard method of extraction of ore. The blast hole stoping method, a variation of the sub-level open stoping method, has also been adopted later on.
- Khetri Copper Mine is divided into two Blocks namely Khetri and Banwas
- In Khetri Block track bounded mining system adopted whereas in Banwas block trackless mining system adopted.
- Upper two levels 421 and 350 mRLs mined through Adit

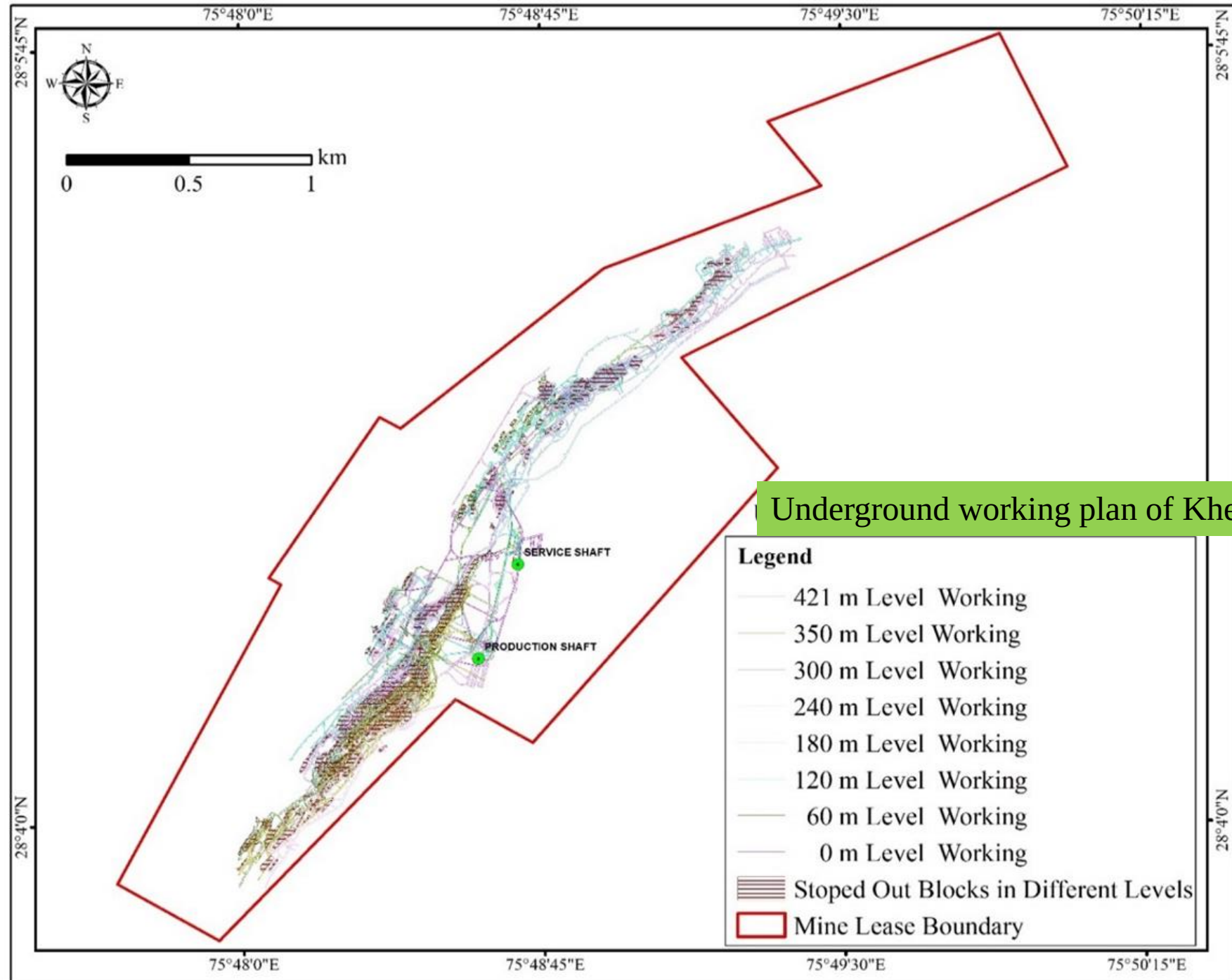
1. Khetri Copper Mine cont.....

- **Mode of Entries:**

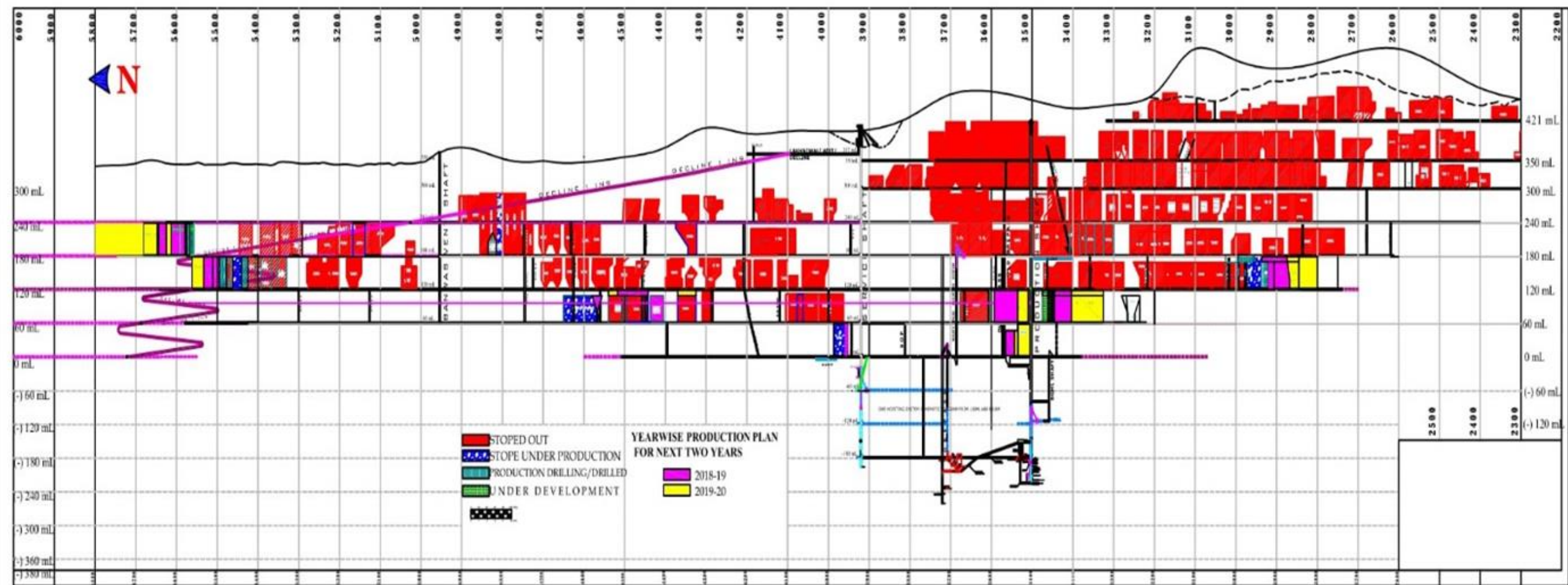
- ❑ **Service Shaft:** it is equipped with double drum winder having double deck cage facility. It is rectangular in shape (6.11m x 4.93m). All the levels from 300 ML to 0 ML are connected with this shaft. The shaft is utilized for man-winding, waste rock hoisting and materials lowering and raising for all levels below. It is also used as main intake for mine ventilation. Service shaft is a vertical shaft.
- ❑ **Production Shaft,** For hoisting of ore from the mine another entry was made. This is a vertical 5.5 m diameter production shaft having depth of 477 meters. This is also concrete lined. This shaft is equipped with 2870 kW Koepe winder. It has two skips each of 14 tonnes capacity for hoisting the ore from the mine. The skips move in shaft on rope guides (4 ropes each skip). Production shaft is a vertical shaft
- ❑ **Banwas Ventilation shaft:** The shaft is from surface (RL 362.64) to 66 m RL and connected to 240 m RL, 180 m RL, 120 m RL and 60 m RL. This is a vertical shaft having 4.5 m diameter and is used as return airway.
- ❑ **Shaft No. 3 (Incline Shaft):** An incline shaft from surface (374.07 m RL) to 186 m RL of 4 m x 3 m size is used for pumping out the water from 180 m RL sump. This is also used as a 2nd out let from 180 m RL of the mine.
- ❑ **Decline:** The mine is being extended in the north direction (Banwas Block) and a decline of 5.0 m x 3.5 m dimension is being prepared having 1 in 9 gradient. This decline is being used for men, material supply, ventilation, waste handling and use of trackless equipment in Banwas section of the mine.



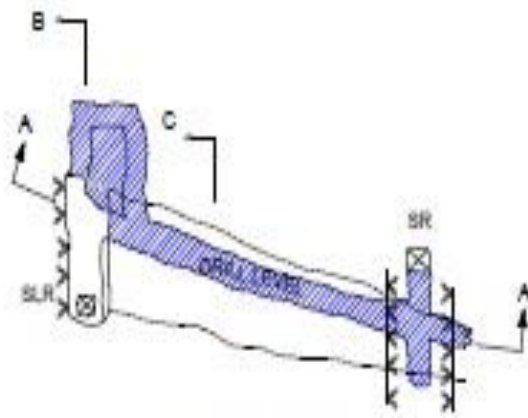
Mine Entries at Khetri Mine



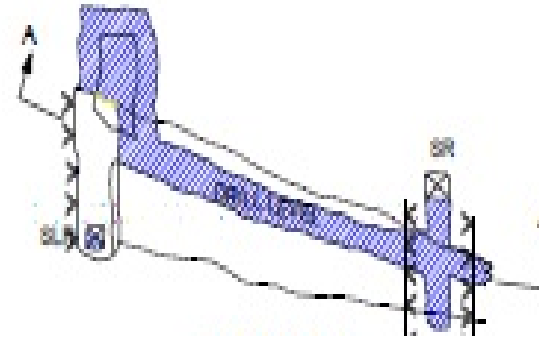
Underground working plan of Khetri Mine



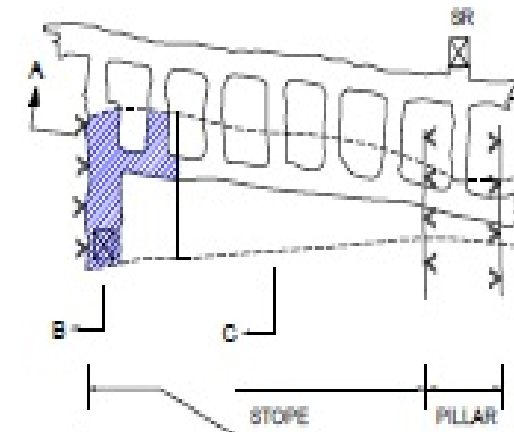
Longitudinal section of underground workings of Khetri Mine



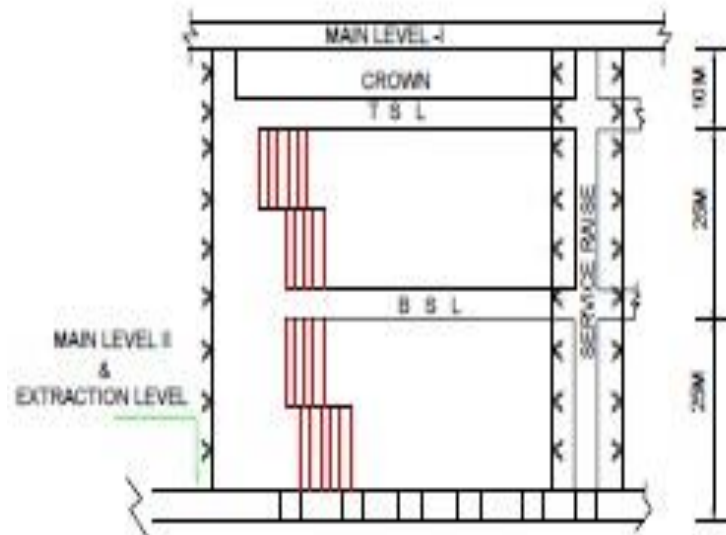
Top sublevel Plan



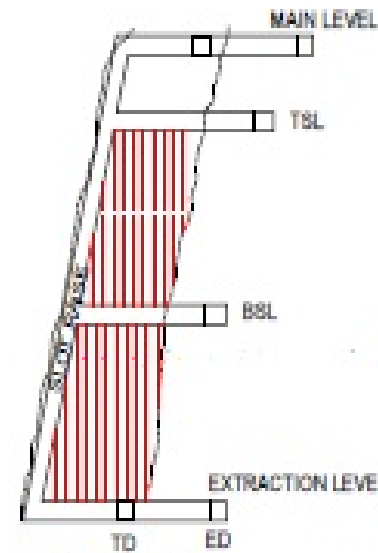
Bottom sublevel Plan View



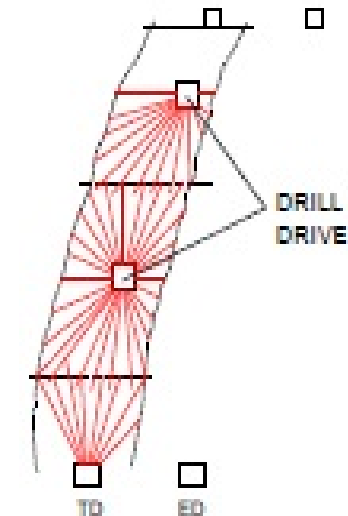
Extraction Level Plan View



Section-AA (Longitudinal Section)

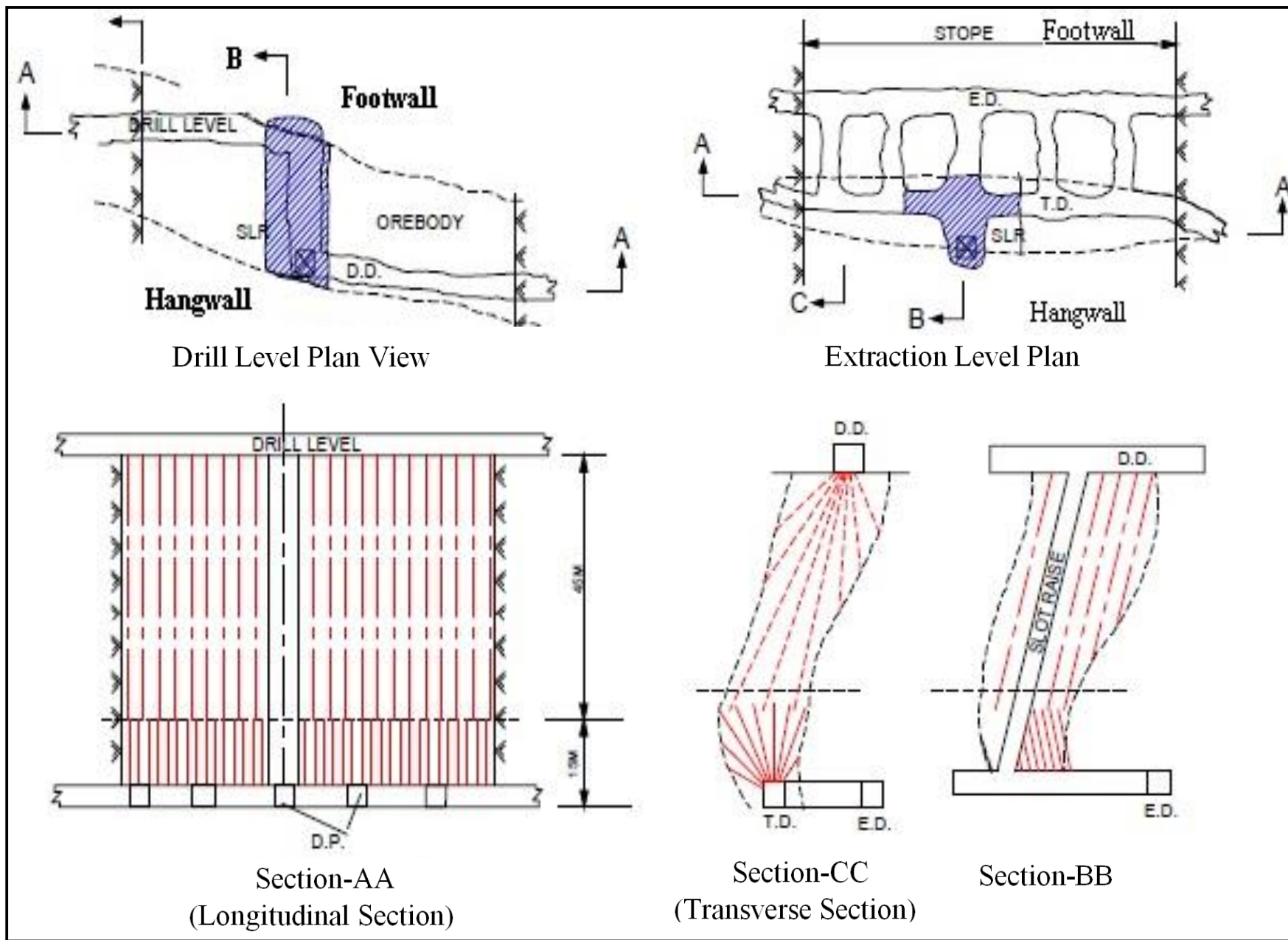


Section-BB



Section-CC (Transverse Section)

A schematic drawing of longitudinal sublevel stoping in Khetri Mine



A schematic drawing of longitudinal blast hole stoping in Khetri Mine

HCL U/G mines at KCC

1. Khetri Copper Mine

2. Kolihan Copper Mine

2. Kolihan Copper Mine

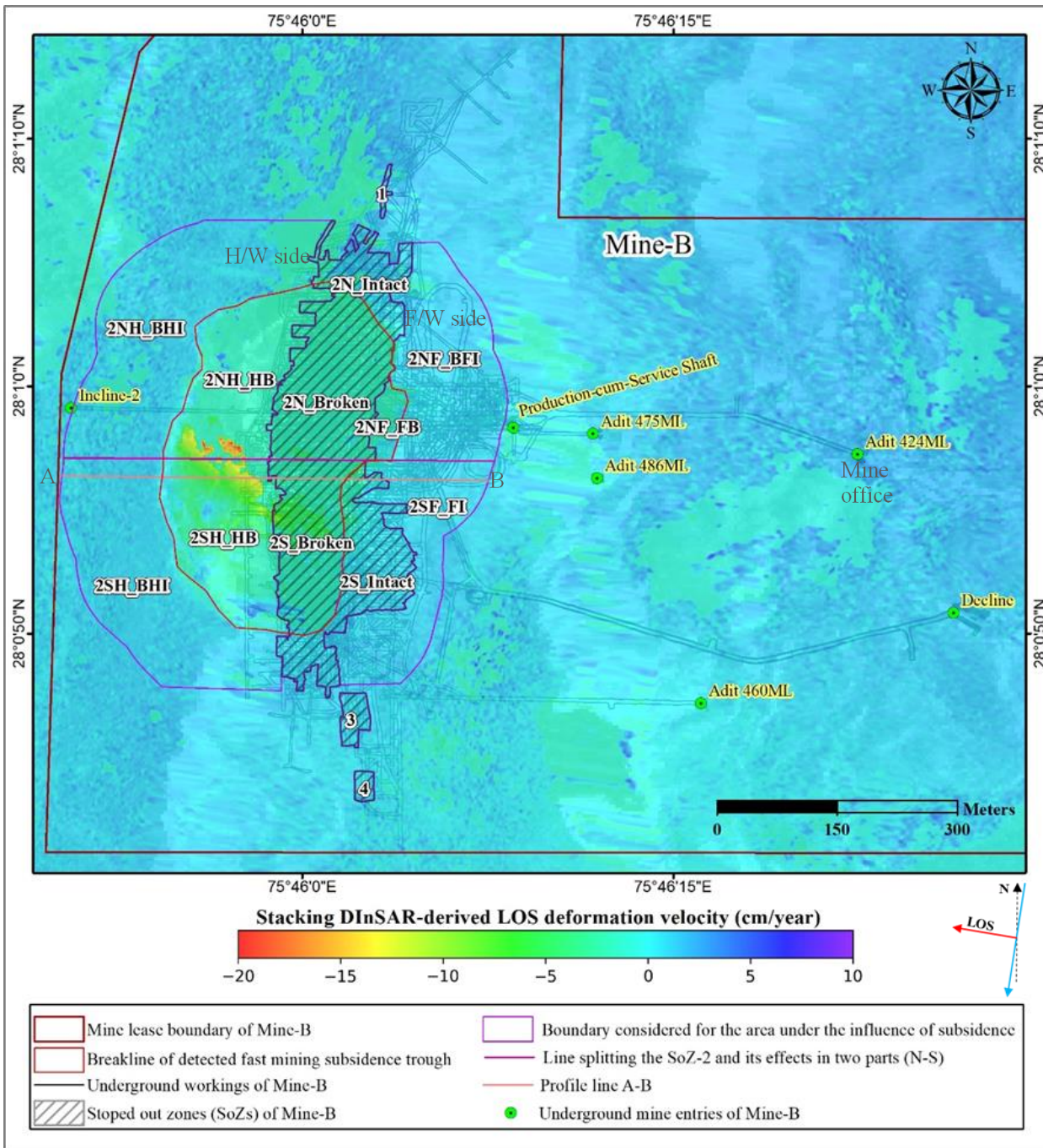
- Kolihan Mine has been developed over a strike length of 700 m and has eight levels at a vertical interval of 60 m viz. 424, 364, 306, 246, 184, 124, 64 and 0 mRLs.
- The numbers indicate the height of the respective level above MSL.
- The upper four levels viz. 424, 364, 306, and 246 mRL are exhausted, and topmost three levels are isolated from the rest of the mine.
- Production is going on in two levels, namely 184, 124 mRL, and mine development at four levels 184, 124, 64 & 0 mRL.
- Stope development in strike extension at 306 mRL and 246 mRL is also being planned to increase the stope availability.
- Blast hole stoping method has been adopted as the standard method of extraction of ore.
- Trackless mining system adopted in the mine

HCL U/G mines at KCC

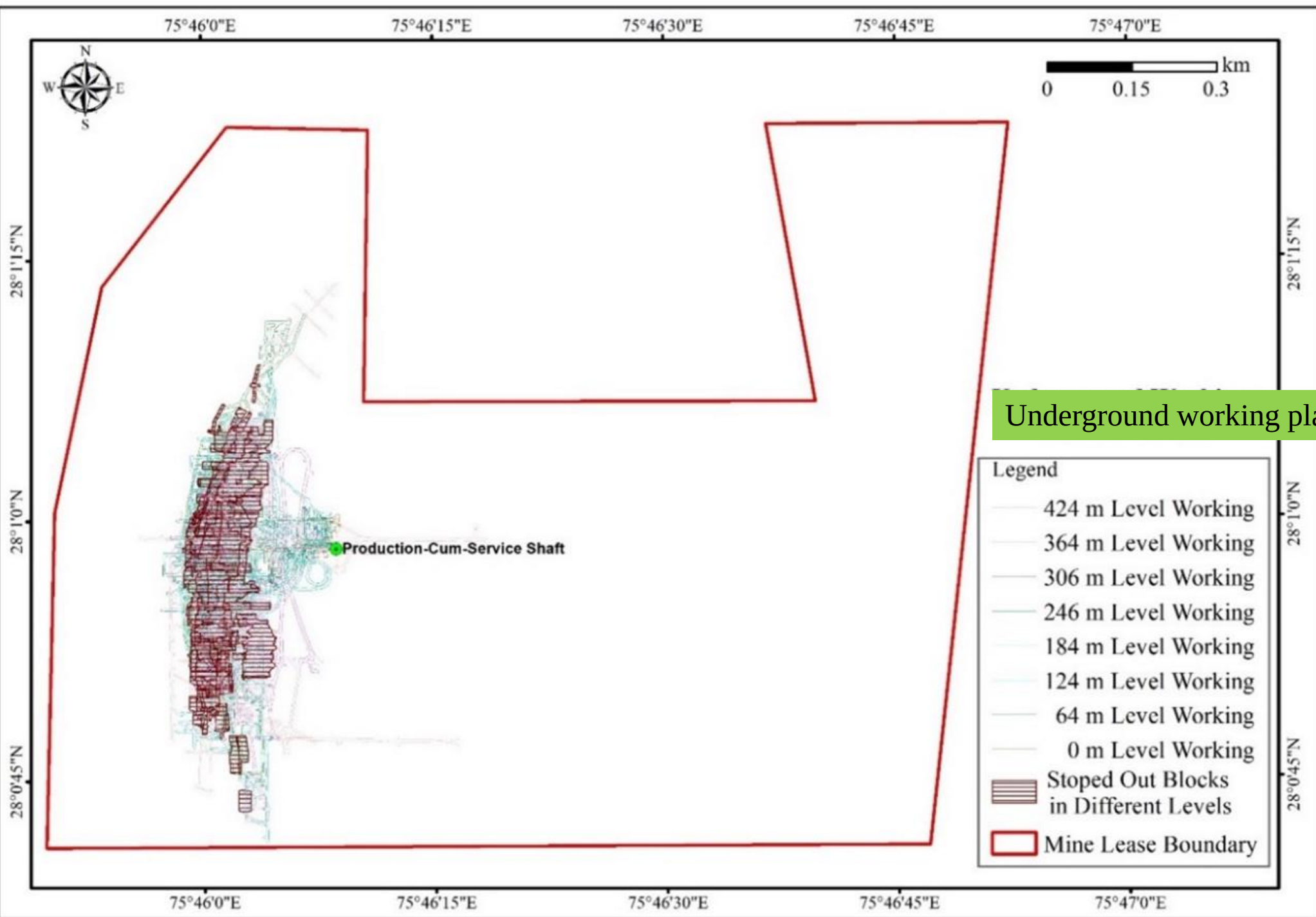
2. Kolihan Copper Mine contd...

❖ Modes of Entry:

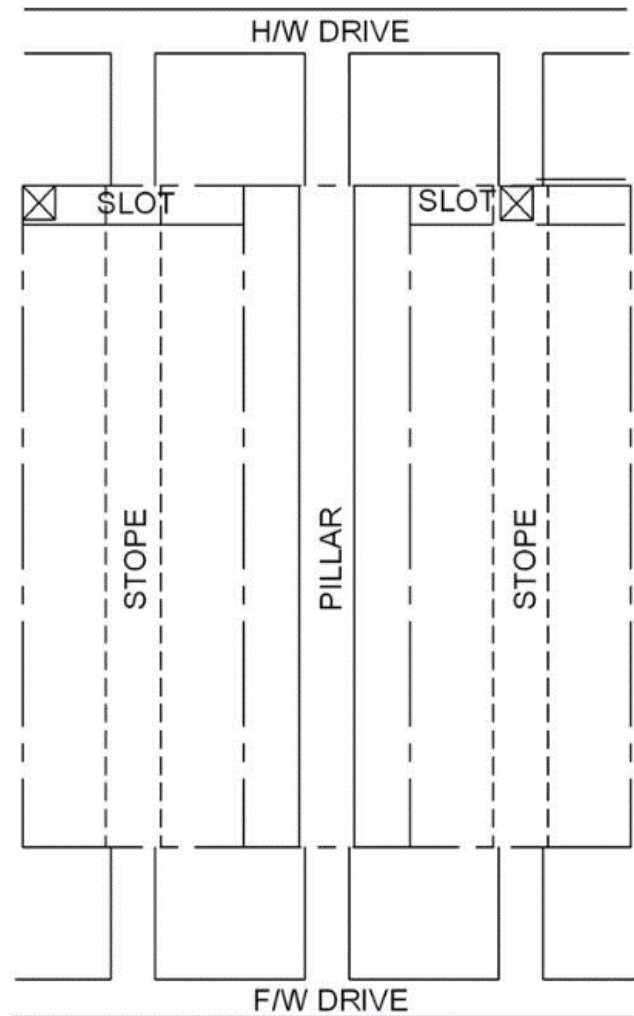
- ❑ **424 ML Adit:** 5.0 m x 3.5 m Adit is developed at 424 m RL up to underground service cum production shaft. This also serves as entry for intake air.
- ❑ **Service cum production shaft:** 5.9 m diameter service cum production shaft is available from 424 m RL to (-) 102 m RL. This shaft is having two compartments, one for cage and counter weight for men, material and waste hoisting and other compartment is for skip hoisting using 6 t skips. This also serves as entry for intake air
- ❑ **Decline:** 5.0 m x 3.5 m decline 1 in 9 gradient have been developed from surface to 103 mRL and below 103 mRL up to 26 mRL it is developed in 1 in 7 gradient. Work is in progress for extending this decline up to 0 mRL. This decline is used for men, material supply, ventilation, waste handling and use of trackless equipment. This decline also serves the purpose of return air way from 326 mRL to surface.
- ❑ **460 mRL Ventilation Adit:** 5.0 m x 3.0 m adit has been developed for ventilation exhaust from the mine.
- ❑ **475 mRL Adit:** 4m x 3 m adit for approach to koepe winder tower at 475 mRL in underground. This also serves as entry for intake air.
- ❑ **Inclined Shaft No. 2:** For mining of 424 mRL



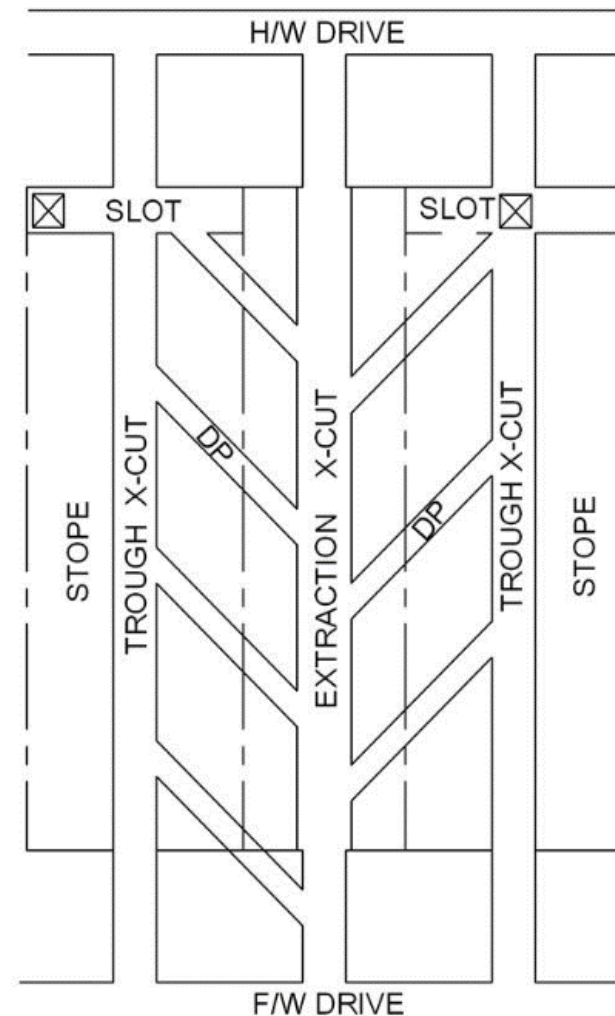
Mine Entries at Kolihan Copper Mine



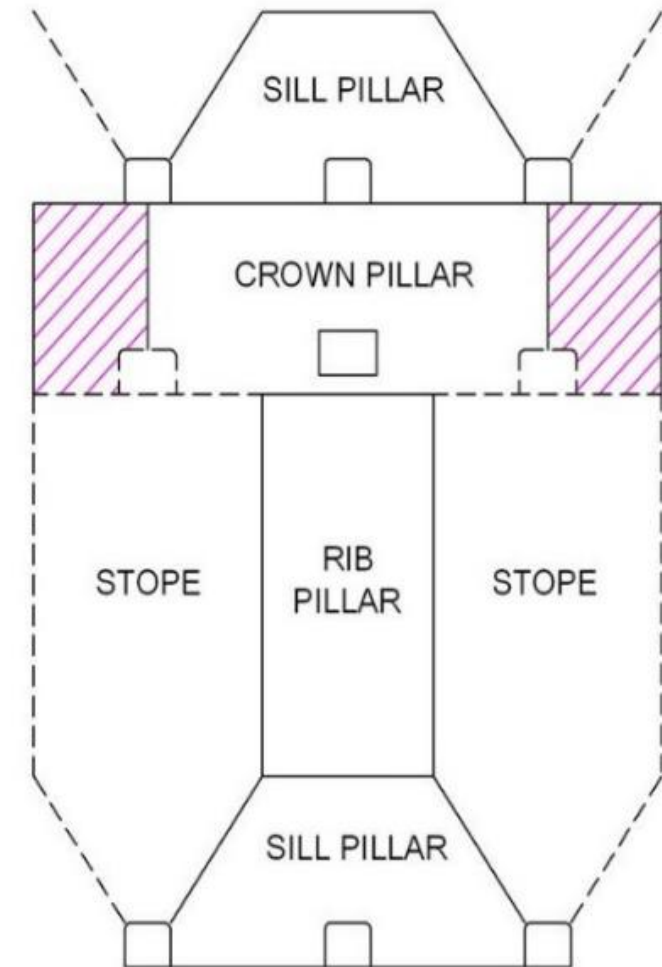
Underground working plan of Kolihan Mine



Drill Level Plan View



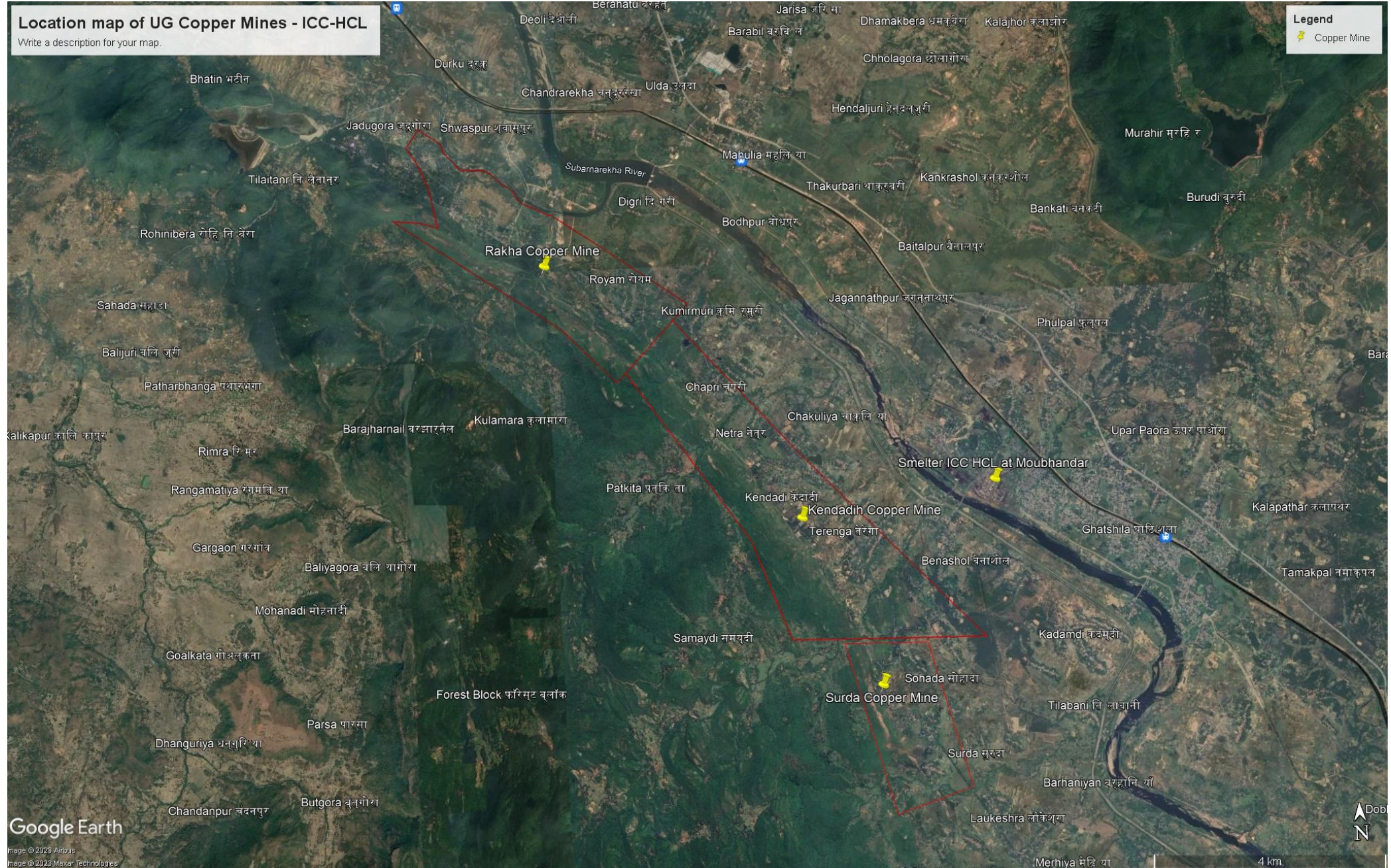
Extraction Level Plan



Longitudinal Section

A schematic drawing of transverse blast hole stoping in Kolihan Mine

Location map of HCL mines



HCL mines at ICC

1. **Surda copper u/g Mine**
2. **Kendadih copper u/g Mine**
3. **Rakha copper u/g Mine**



1. **Surda copper u/g Mine – Salient Features**

- The Surda Mine is located between latitude N22°32'43.119" - N22°34'17.401" and longitude 86°25'31.849" - E86°26'45.097" in Ghatsila Sub-Division, East Singhbhum district of Jharkhand state in the famous Singhbhum Copper Belt. It is situated 10 km away from Ghatsila Railway Station, S.E. Railway and 45 km from Tatanagar.
- The mine was opened in 1930's at a modest scale with available information of ore reserves but production and exploration was suspended later on. After independence, the mine was re-opened in 1956. In the year 2003, mine production was suspended again due to high production cost etc. Dewatering, equipment maintenance and other essential activities kept continuing. Production activities re-started again from 14.04.2007.
- The mine has been developed over a strike length of 2.2 km and to a depth of 474m. i.e. 13th level.
- The present production capacity of the mine is **3,17,238 tonnes**

1. Surda copper u/g Mine – Salient Features contd....

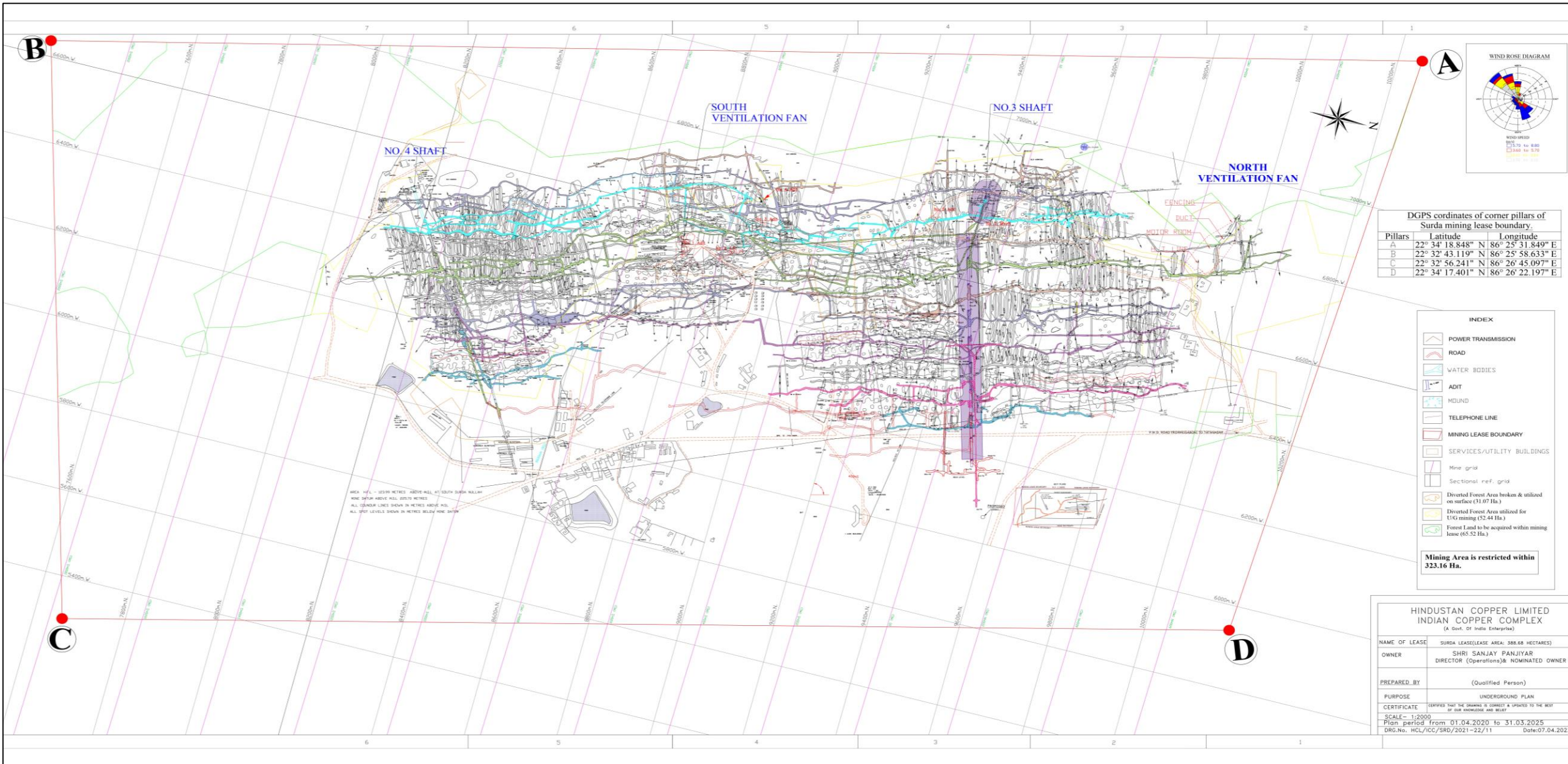
❑ The access to the underground mine is through:

- No. 3 vertical shaft from northern end sunk up to 5th level (i.e. about 176 m below from the surface) at central part of the mine .
- No. 4 incline shaft from southern side approx. 1200m south of No.3 shaft at 40 degree inclination to the horizontal is sunk up to 10th level i.e. 360 m below surface.
- At the close proximity of the vertical shaft No. 3 there is sub incline shaft from 5th level to 13th level (i.e.474 m below surface).
- The sub-incline is at an angle of 31 degree– 35 degree to horizontal.
- The proposed vertical shaft of 6 m diameter shall be an additional mine entry after commissioning. Shaft will be sunk up to 18th level (i.e. 600m below surface).

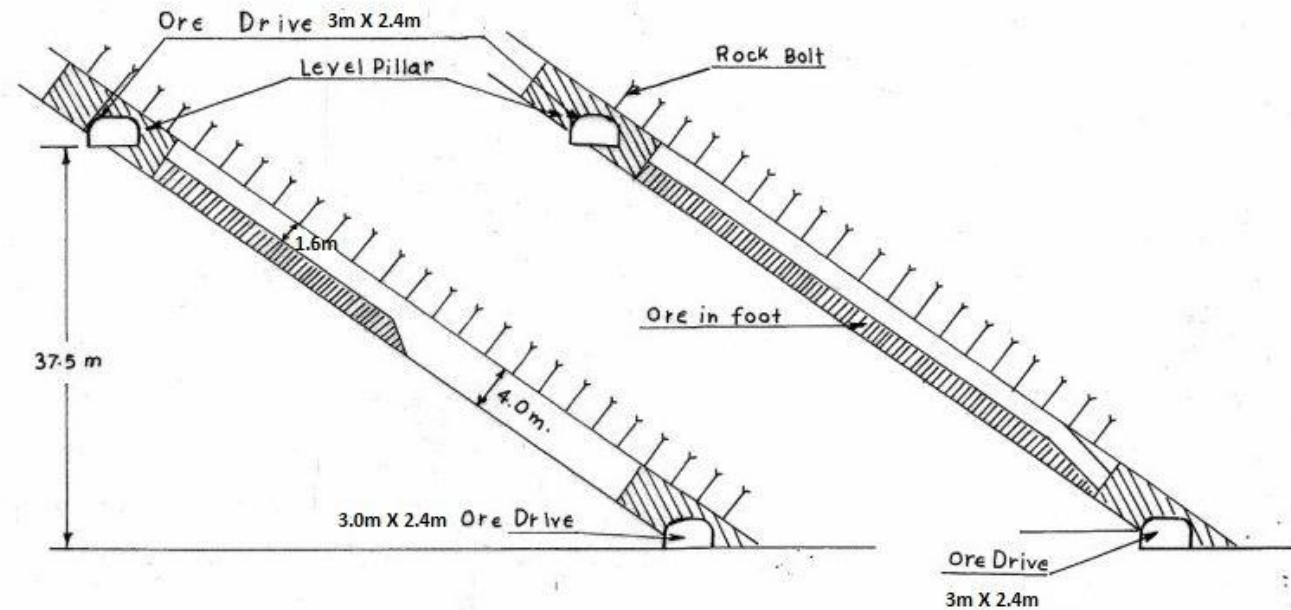
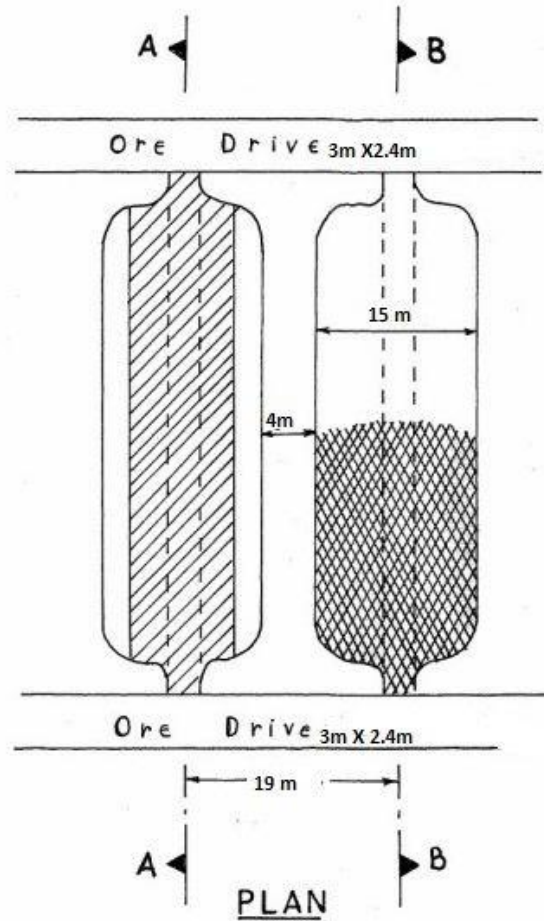
❑ Mining method followed are:

- Room & pillar with back filling methods for ore body width – 1.5 m to 4 m.
- Cut &fill method for ore body width – 4 m to 6 m.
- HCF with Post pillar methods for ore body width - 6 m and above

Underground Plan of Surda Mines

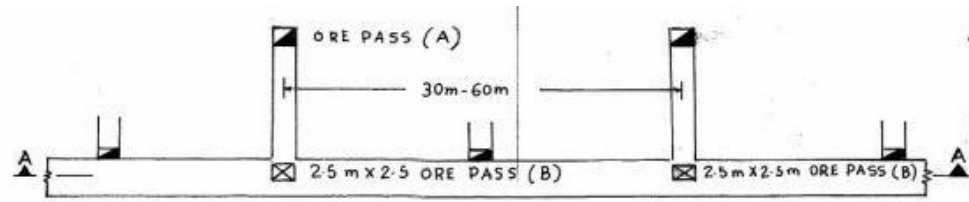


LAY-OUT OF ROOM AND PILLAR METHOD.



SEQUENCE OF OPERATIONS.

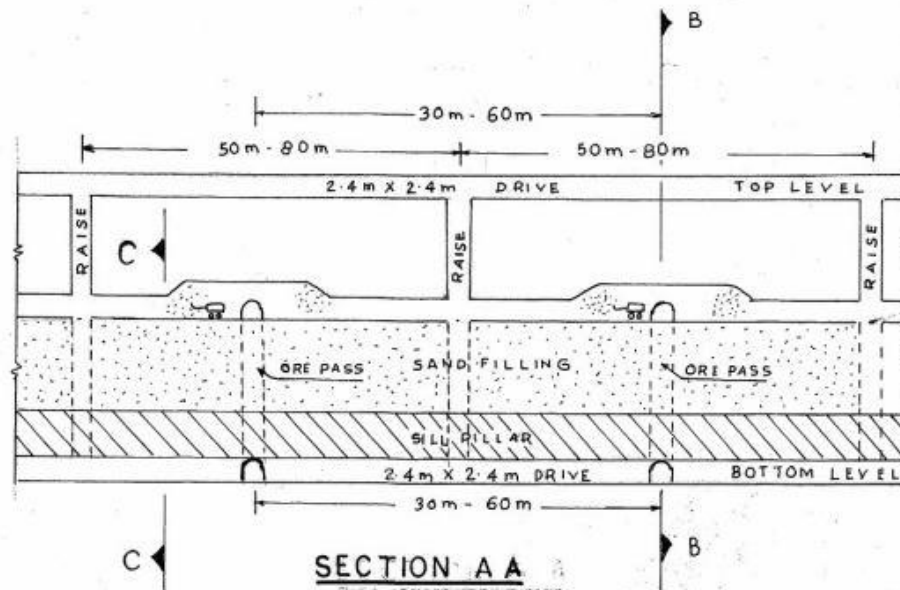
1. RAISES DRIVEN BETWEEN LEVELS AT 19m CENTRES ALONG HANGING WALL CONTACT OF 2.5m X 1.6m
2. RAISE IS WIDEN TO 15m BY BREASTING, HANG WALL BEING ROCK BOLTED AT 1.2m X 1.2m PATTERN.
3. ORE IN THE FOOT WALL STRIPPED.



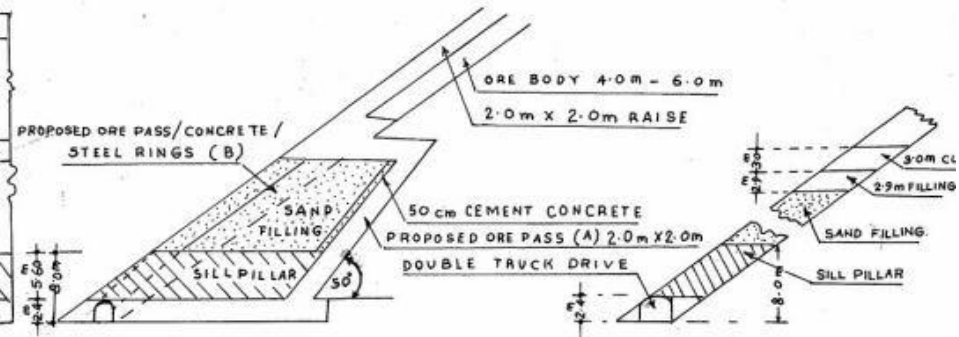
SILL PLAN

1. RAISE CONNECTION BETWEEN TWO COSECUTIVE LEVEL WILL BE SPACED AT 50m-80m INTERVAL, THESE RAISES WILL BE DRIVEN ALONG ORE BODY.
2. ORE PASS RAISES WILL BE SPACED AT 30m-60m INTERVAL, THESE RAISES WILL BE MAINTAINED AT 50° INCLINATION EITHER IN THE FOOT OF THE ORE BODY (A) OR WITHIN THE ORE BODY WITH THE HELP OF STEEL/ CONCRETE RINGS (B).
3. DEVELOPMENT AT SILL HORIZON WILL COMMENCE FROM THE ORE RAISES.
4. SILL LEVEL WILL BE STRIPPED TO A HEIGHT OF 5.9m.
5. TAILINGS WILL BE FILLED LEAVING AN OPEN HEIGHT OF ABOUT 2.9m FOR THE MOVEMENT OF MEN AND EQUIPMENT.
6. SUBSEQUENT CUTS WILL BE OF 3.0m EACH AND FILLING WILL BE CYCLIC.
7. HANGING WALL WILL BE SUPPORTED WITH ROCK BOLTS AT 1.5m X 1.5m INTERVAL.

LAY-OUT OF CUT AND FILL STOPE.



SECTION AA

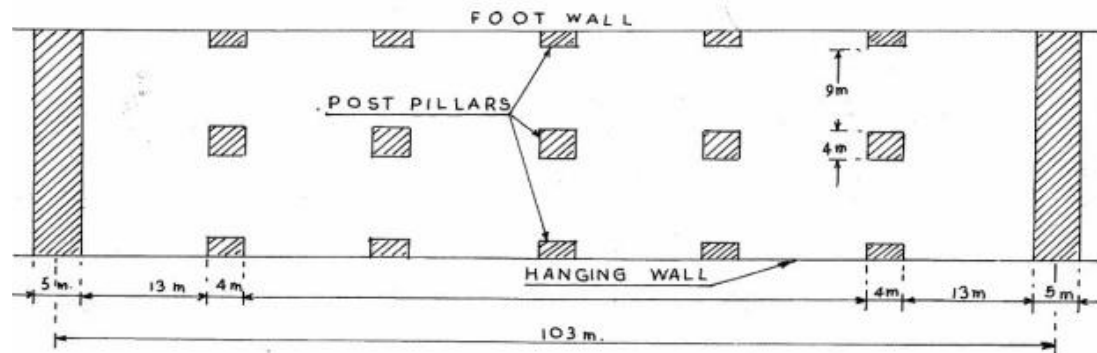


SECTION BB

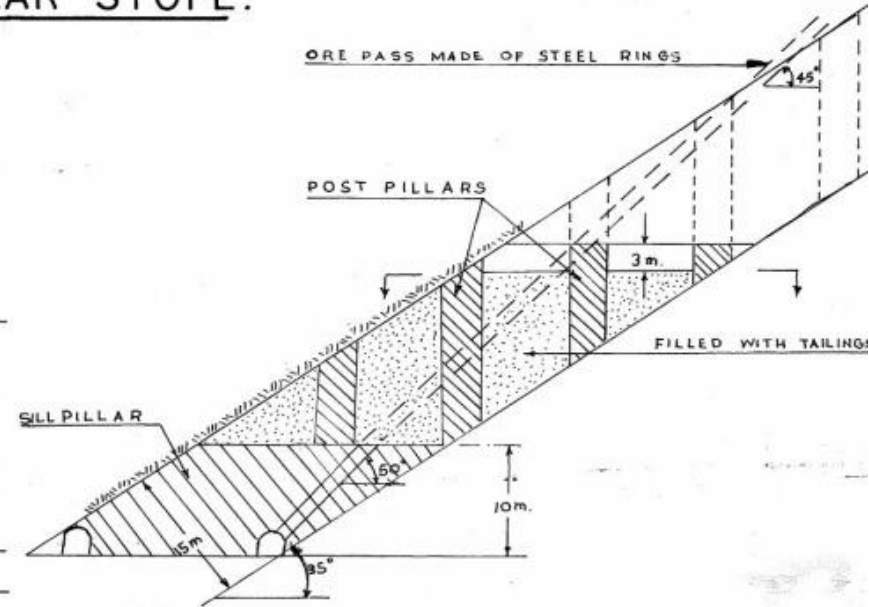
SECTION CC

LAY- OUT OF A POST PILAR STOPE.

SCALE :- 1cm. = 5m.



PLAN OF THE STOPE FLOOR.



SECTION

Sequence of operations

1. Ore passes at 30m. interval inclined at 50°
2. One service Raise in the centre of the stope along F/W. contact, two ventilation Raises at two extremities of the stope along H/W. contact.
3. Development of sill horizon through ventilation, services & ore pass raises.
4. Stripping of sill level to a height of 5m. leaving 4m. x 4m. pillars at spacing of 13m. along strike and 9m. across, back to be rock bolted 1.5m. x 1.5m. pattern.
5. Fill with Tailing leaving an open height of 2.5m.
6. Then 2nd cut 2.5m.
7. For operational convenience a service raise is cut at the bottom of the stope.

HCL mines at ICC

1. **Surda copper u/g Mine**
2. **Kendadih copper u/g Mine**
3. **Rakha copper u/g Mine**

2. **Kendadih copper u/g Mine – Salient Features**

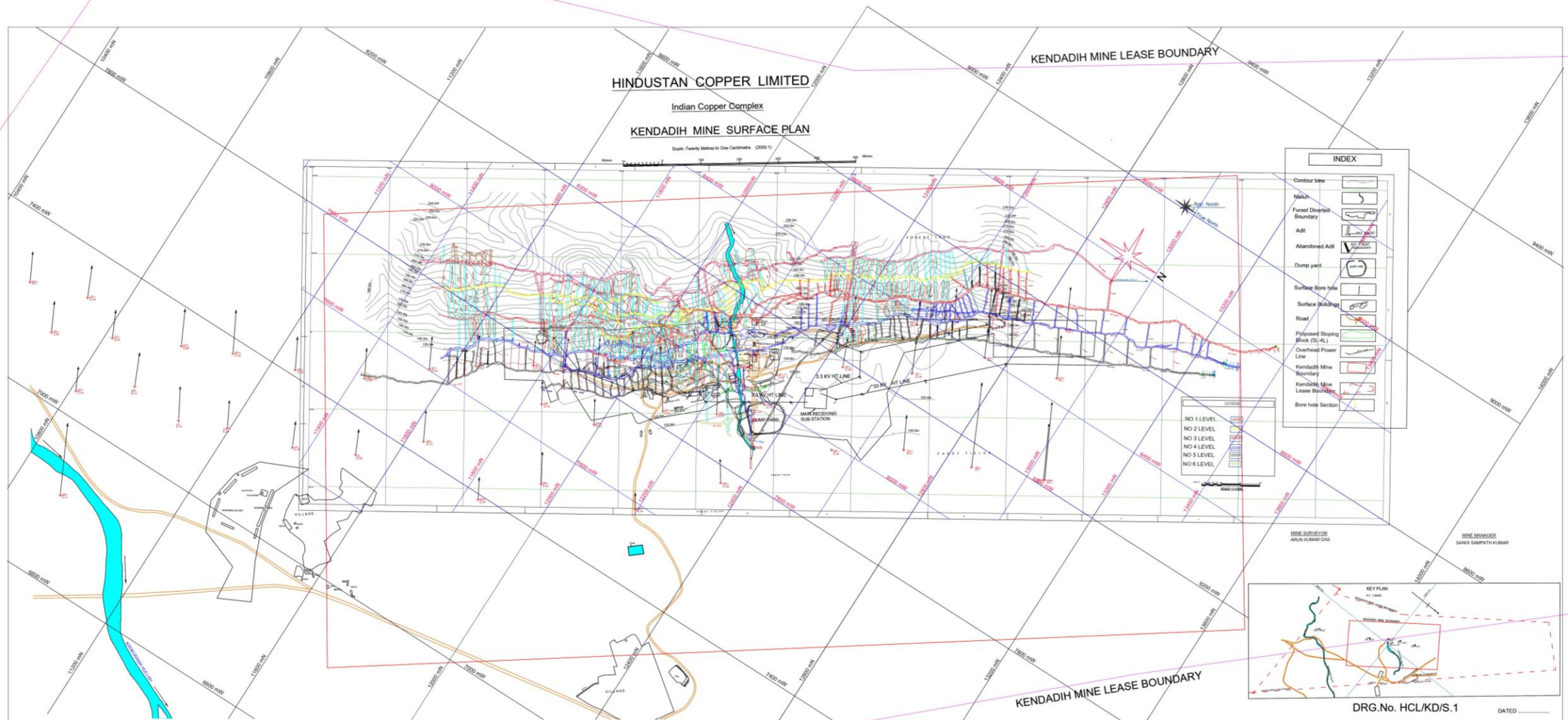
- Kendadih underground copper mine falls on Survey of India Toposheet- 73 J/6 and is located at Ghatsila subdivision of east Singhbhum district of Jharkhand state in the Singhbhum copper belt. It is at distance of about 15 km from Ghatsila Rly. Stn. and around 240 km from Kolkata and 50 km from Tatanagar.
- The Kendadih mining lease was executed on 03.06.1973
- The mine was under temporary discontinuance w.e.f 02.01.2000. Notice of reopening of mines has been submitted in Form-C on 20.07.2012
- The Kendadih mines has been accorded Environmental Clearance for production of 0.45 MTPA copper ore by the MoEFCC
- As the width of the ore body is 1.5 to 4.0m hence the most suitable method of mining is Room & Pillar method
- In case the width of the ore body is more other methods may be adopted, e.g. Cut &fill method for ore body width – 4 m to 6 m; HCF with Post pillar methods for ore body width - 6 m and above



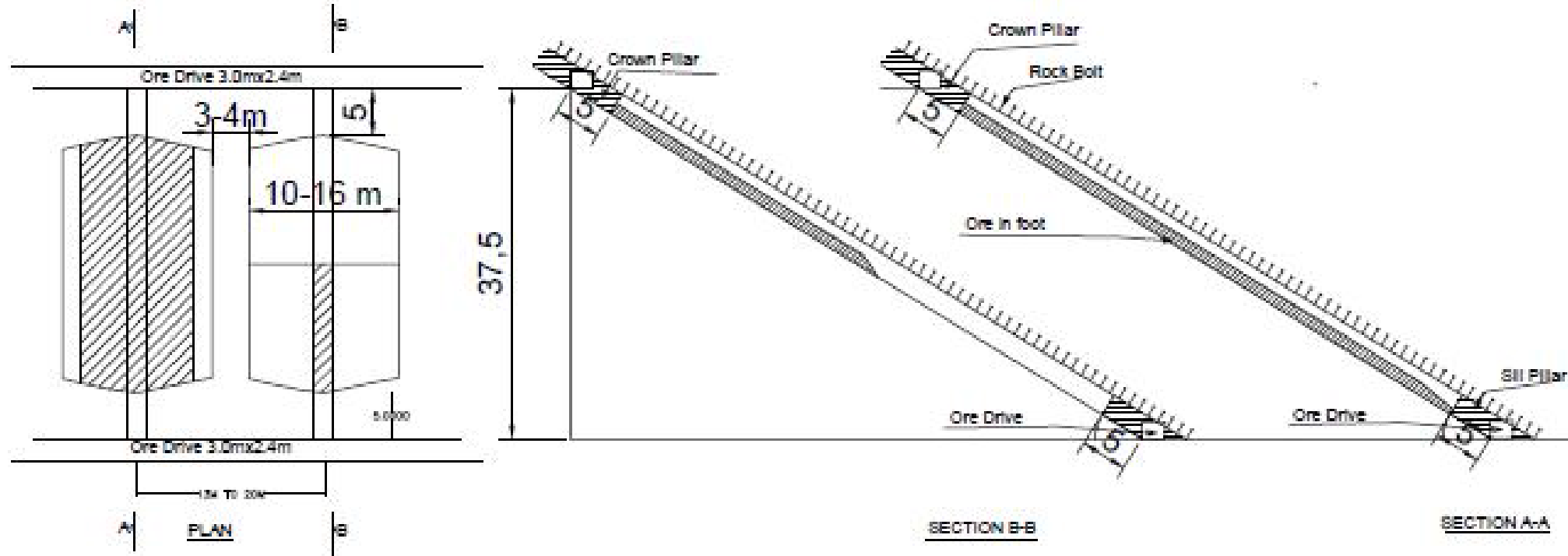
Principal existing mode of entries at Kendadih Mine are given in table below.

Entry	Inclination	Dimension Width X Height (metre)	Vertical depth (metre)	From Level	To Level	Remarks
550N Shaft	32 ⁰	5 x 2.2	154.15	Surface	5 th	
No. 1 Adit	Horizontal	3.0 x 2.1	--	Surface	1 st	Closed
No. 3 Adit	Horizontal	3.5 x 2.4	--	Surface	1 st	
480N Winze	25 ⁰ – 36 ⁰	3.0 x 2.8	187.2	1 st	6 th	
780N Winze	35 ⁰ - 37 ⁰	4.3 x 2.2	142.0	1 st	5 th	
Ventilation Shaft/adit						
No. 2 Adit	Horizontal	4.0 x 3.0	--	Surface	1 st	
No. 4 Adit	Horizontal	4.0 x 2.2	--	Surface	1 st	

Surface Plan of Kendadih Mines



LAY-OUT OF ROOM AND PILLAR STOPING METHOD

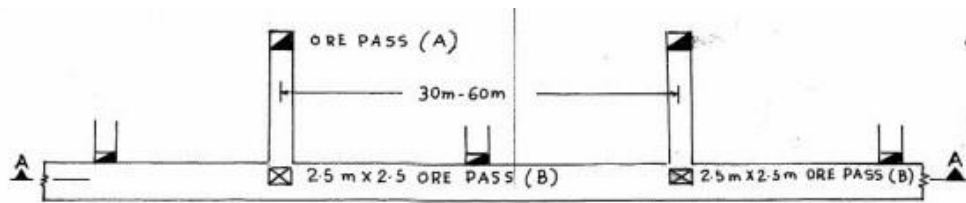


SEQUENCE OF OPERATION :-

1. RAISES DRIVES BETWEEN LEVELS AT 13M TO 20M CENTRES ALONG HANGING WALL CONTACT
2. RAISES IS WIDEN TO 10M TO 16 M BY BREASTING,HANGWALL BEING ROCK BOLTED AT 1.2MX1.2M PATTERN
3. ORE IN FOOT WALL STRIPPED
4. MINED OUT AREA FILLED WITH SANDMILL TAILINGS IN LOWER LEVEL.

SURVEYOR
HOLLAND SURVEY DEPT.
Dated :-

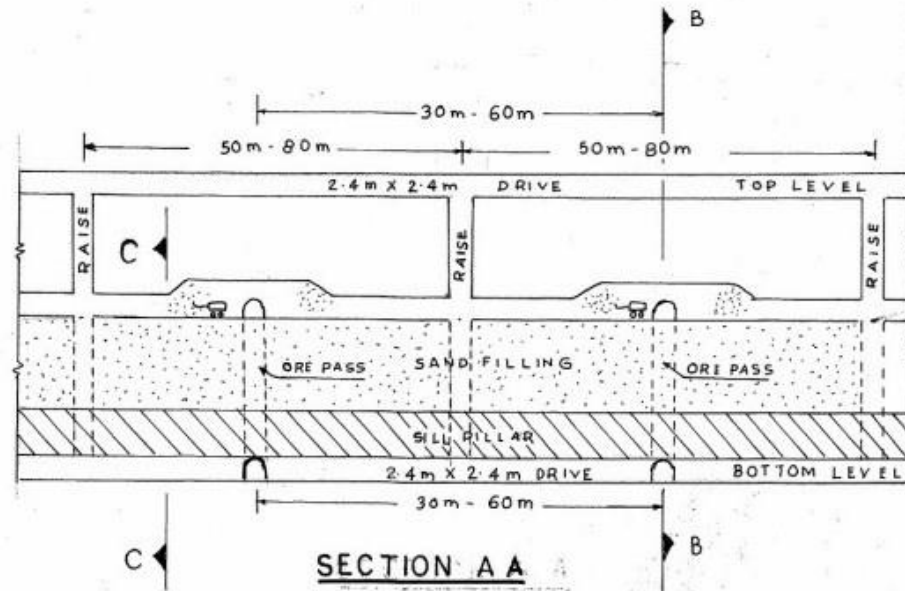
MANAGER



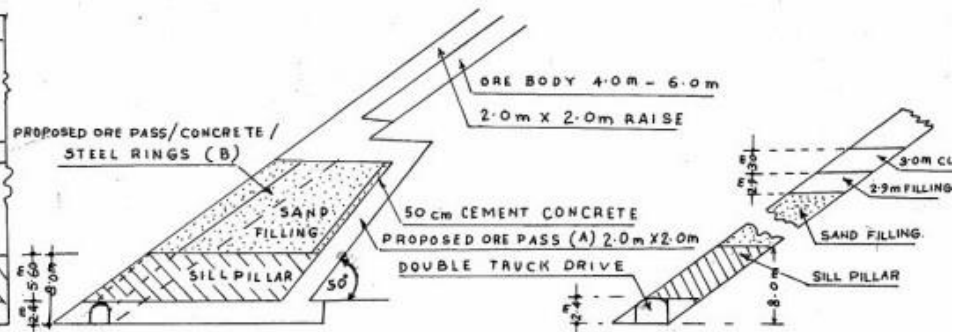
SILL PLAN

1. RAISE CONNECTION BETWEEN TWO COSECUTIVE LEVEL WILL BE SPACED AT 50m-80m INTERVAL, THESE RAISES WILL BE DRIVEN ALONG ORE BODY.
2. ORE PASS RAISES WILL BE SPACED AT 30m-60m INTERVAL, THESE RAISES WILL BE MAINTAINED AT 50° INCLINATION EITHER IN THE FOOT OF THE ORE BODY (A) OR WITHIN THE ORE BODY WITH THE HELP OF STEEL/ CONCRETE RINGS (B).
3. DEVELOPMENT AT SILL HORIZON WILL COMMENCE FROM THE ORE RAISES.
4. SILL LEVEL WILL BE STRIPPED TO A HEIGHT OF 5.9m.
5. TAILINGS WILL BE FILLED LEAVING AN OPEN HEIGHT OF ABOUT 2.9m FOR THE MOVEMENT OF MEN AND EQUIPMENT.
6. SUBSEQUENT CUTS WILL BE OF 3.0m EACH AND FILLING WILL BE CYCLIC.
7. HANGING WALL WILL BE SUPPORTED WITH ROCK BOLTS AT 1.5m X 1.5m INTERVAL.

LAY-OUT OF CUT AND FILL STOPE.



SECTION A A

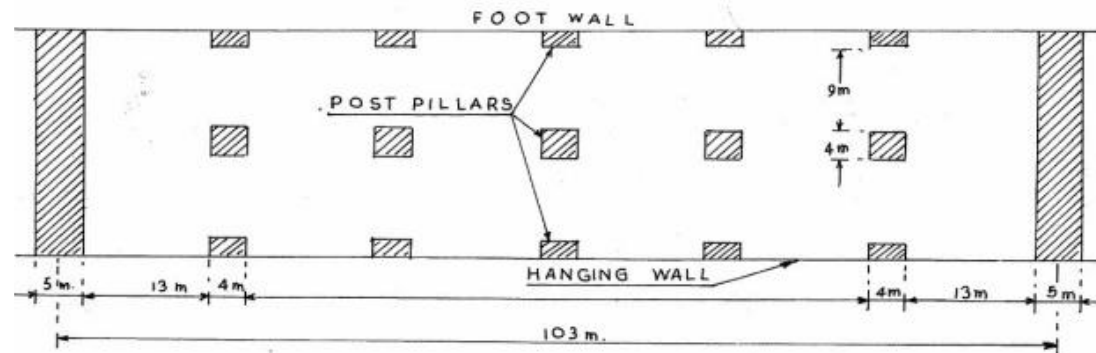


SECTION B B

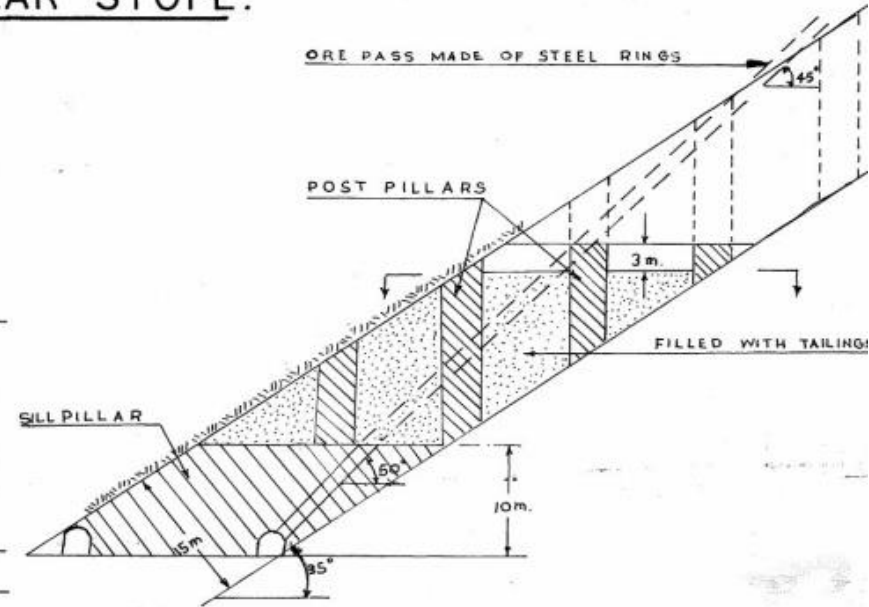
SECTION C C

LAY- OUT OF A POST PILAR STOPE.

SCALE :- 1cm. = 5m.



PLAN OF THE STOPE FLOOR.

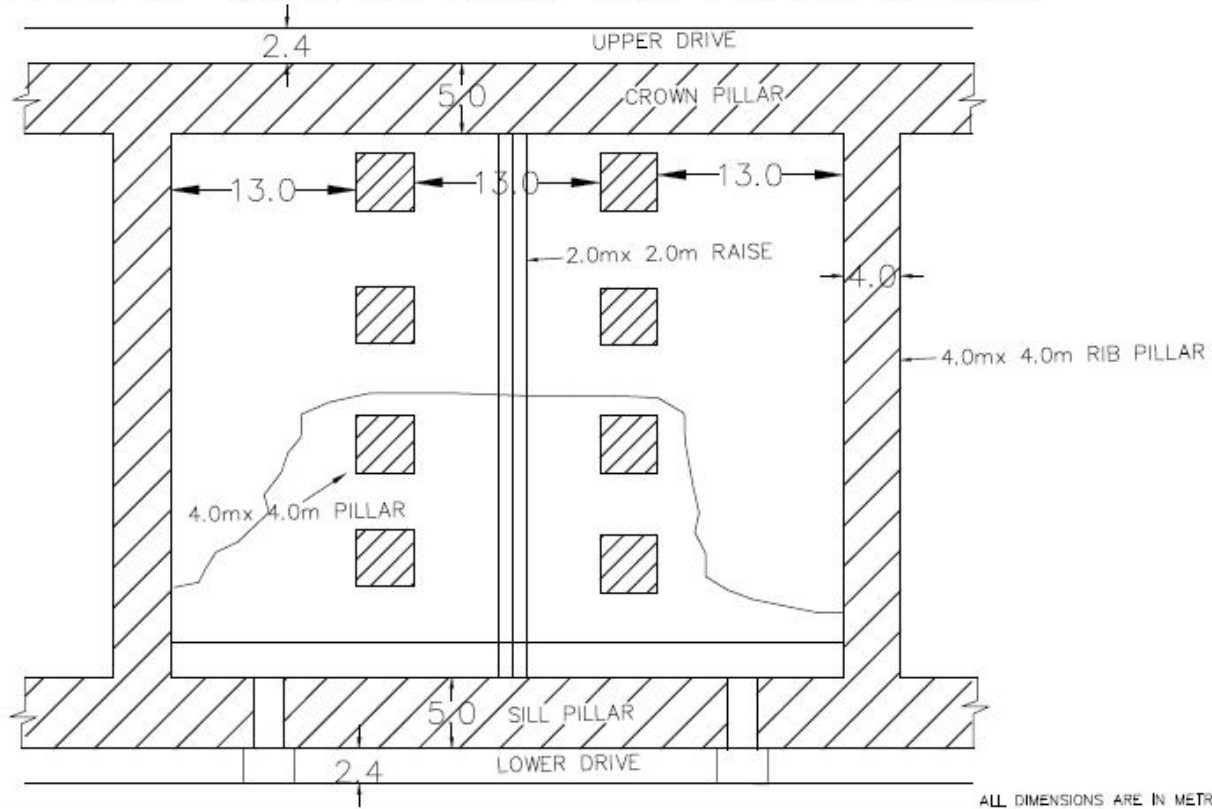


SECTION

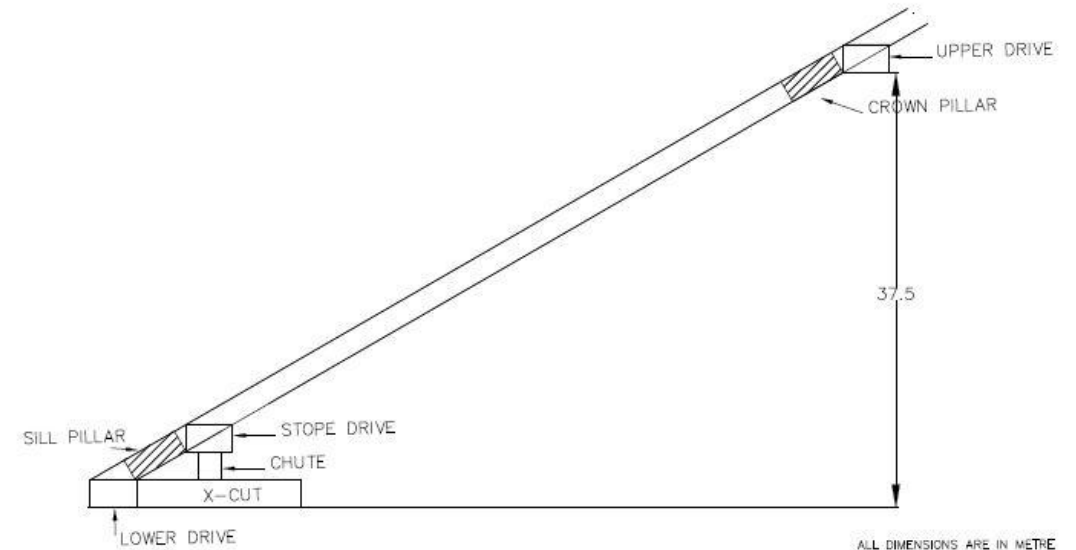
Sequence of operations

1. Ore passes at 30m. interval inclined at 50°
2. One service Raise in the centre of the stope along F/W. contact, two ventilation Raises at two extremities of the stope along H/W. contact.
3. Development of sill horizon through ventilation, services & ore pass raises.
4. Stripping of sill level to a height of 5m. leaving 4m. x 4m. pillars at spacing of 13m. along strike and 9m. across, back to be rock bolted 1.5m. x 1.5m. pattern.
5. Fill with Tailing leaving an open height of 2.5m.
6. Then 2nd cut 2.5m.
7. For operational convenience a service raise is cut at the base of the stope.

PLAN OF MODIFIED ROOM AND PILLAR METHOD



SECTION OF MODIFIED ROOM AND PILLAR METHOD



HCL mines at ICC

1. **Surda copper u/g Mine**
2. **Kendadih copper u/g Mine**
3. **Rakha copper u/g Mine**



3. Rakha copper u/g Mine – Salient Features

- Rakha lease is located in SOI Toposheet No. 73 J/6 between latitudes 22o 36' 52.652"N and 22o39' 19.497" N and longitudes 86o 21' 10.752" E and 86o 24' 07.556" E.
- Rakha Mining Lease of M/s Hindustan Copper Limited was initially granted for 20 years from 29.08.1971 to 28.08.1991.
- The mine workings have been discontinued w.e.f. 07.07.2001 due to low LME copper price.
- It is planned to augment the production capacity to 3.0 MTPA of ore by reopening and expansion of closed Rakha block and Chapri block respectively.
- Rakha lease will consists of two mine. Rakha mine and Chapri Sideshwar. The combined capacity of both the mine is 3 MTPA for which mine plan have been approved.
- Since the capacity is increasing, apart from existing entries, two vertical shaft and two declines have been proposed to achieve the target of 3 MTPA.

Existing mine entries:**Rakha Main Block:**

Sl.	Entry	Inclination	Dimension W x H (metre)	Vertical Depth (m)	Extent	
					From	To
1	No. 4 Shaft	Surface-1L=53 ⁰ , 1L-3L=43 ⁰ ,3L-7L=40 ⁰	4.8 x 2.2	207	Surface	7L
2	No. 5 Shaft	40 ⁰	4.5 X 2.2	200	Surface	7L
3	Ventilation Shaft	Surface-110 mRL – Vertical, 110mRL-5L- 35 ⁰	3.0 X 3.0	153	Surface	5L
4	Comp. Winze 4 shaft	32 ⁰ to 42 ⁰	3.75 X 2.75	206	3L	Below 8L
5	Comp. Winze 5 shaft	40 ⁰ e	2.5 X 2.5	153	5L	9L

Rakha NW Block:

Sl	Entry	Inclination	Vertical Depth (m)	Extent	
				From	To
1	Main inclined Shaft	400	231	Surface (+147 mRL)	9L (-84.08 mRL)
2	No. 6 Shaft	Vertical	123	Surface	3L
3	No 7 shaft	Vertical	32	Surface	2L

Chapri Block:

Sl	Entry	Inclination	Vertical Depth (metre)	Extent	
				From	To
1	Adit 1	Horizontal	0	136.89 mRL	
2	Adit 2	Horizontal	0	148.59 mRL	
3	Adit 3	Horizontal	0	147.00 mRL	
4	Eastern Shaft	Vertical	55	180 mRL	125 mRL

The mining methods at the time of suspension of work were Room & pillar, cut & fill and post pillar

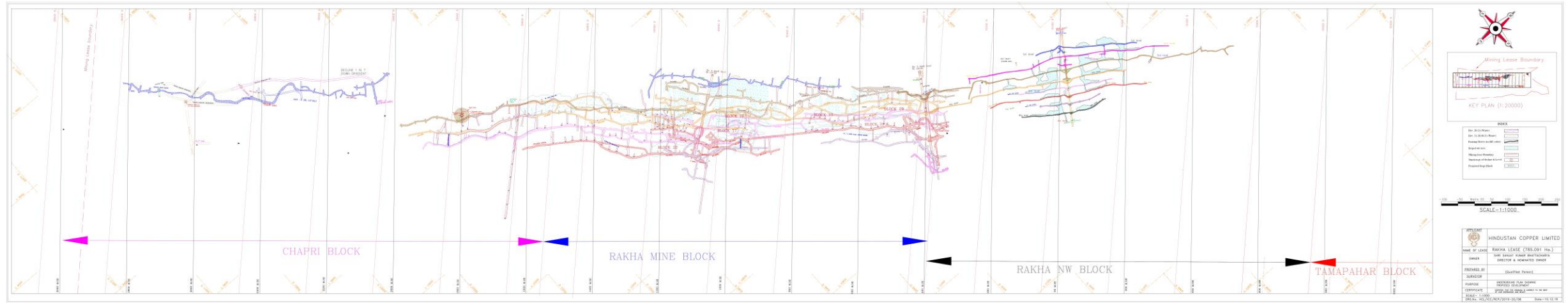
methods:-

Room & pillar methods for ore body width – 1.5 m to 4 m.

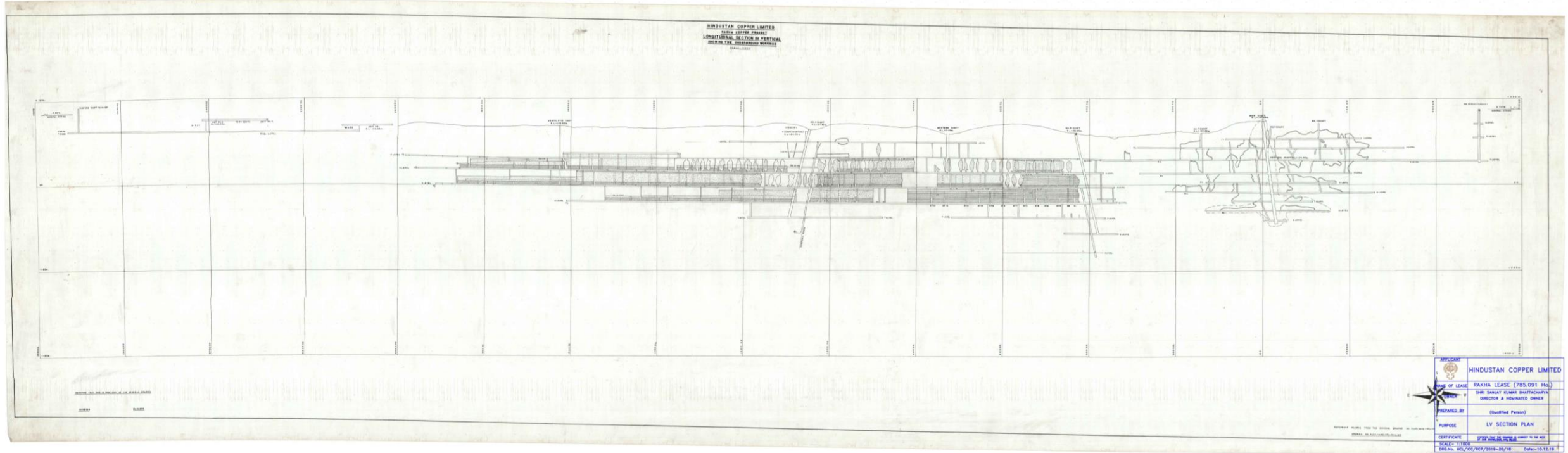
Cut & fill method for ore body width – 4 m to 6 m.

Post pillar methods for ore body width - 6 m and above.

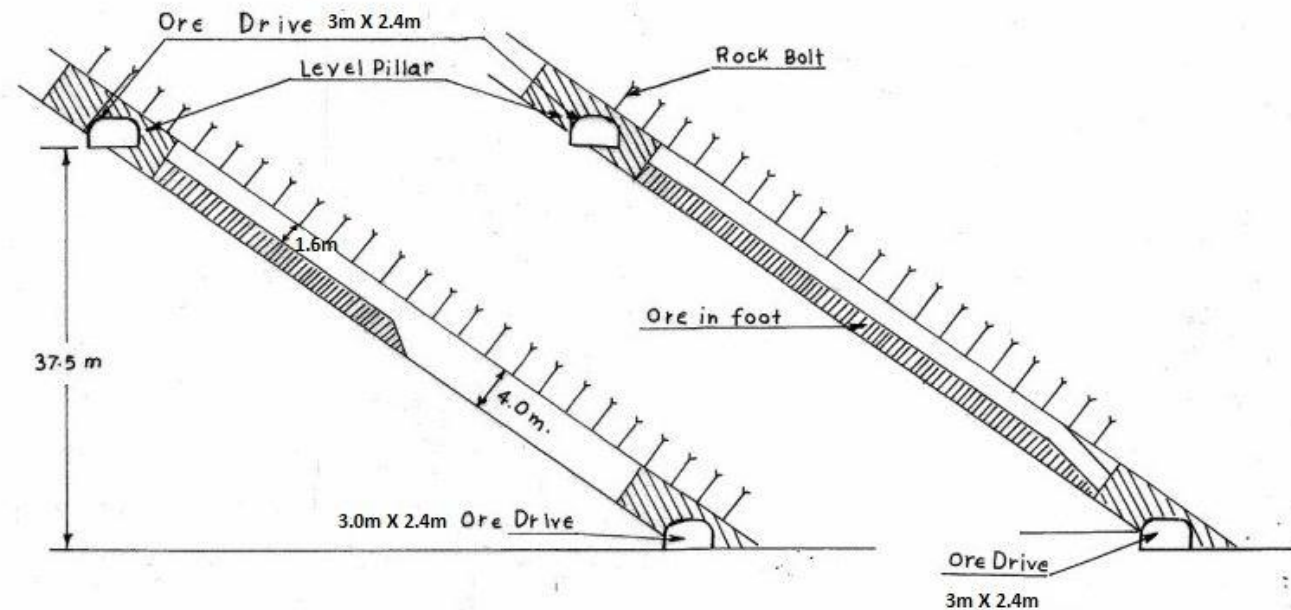
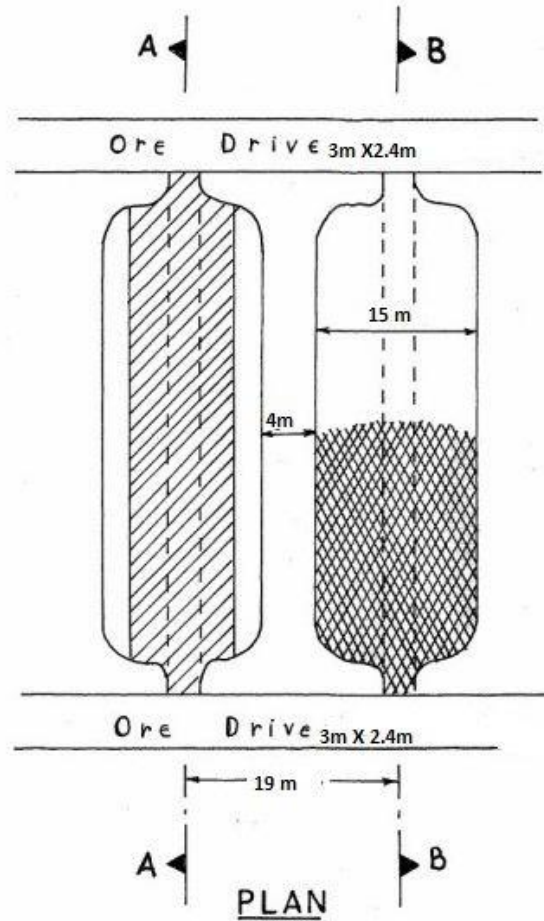
Underground Plan of Rakha and Chapri Mines



LVS of Rakha and Chapri Mines



LAY-OUT OF ROOM AND PILLAR METHOD.

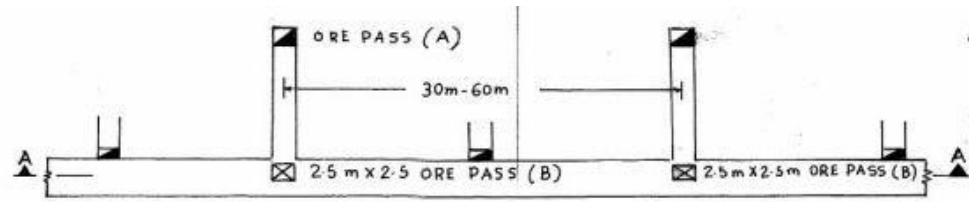


SECTION BB

SECTION AA

SEQUENCE OF OPERATIONS.

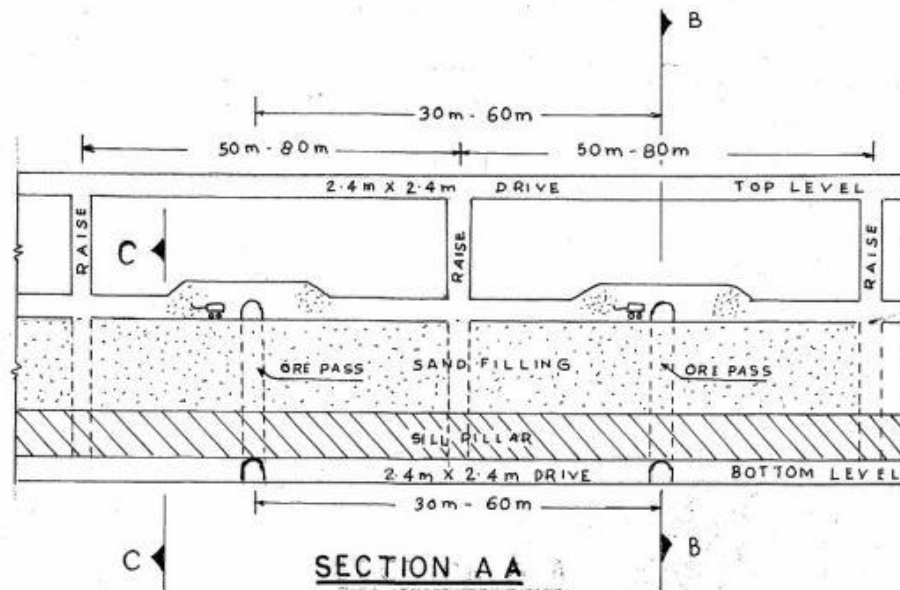
1. RAISES DRIVEN BETWEEN LEVELS AT 19m CENTRES ALONG HANGING WALL CONTACT OF 2.5m X 1.6m
2. RAISE IS WIDEN TO 15m BY BREASTING, HANG WALL BEING ROCK BOLTED AT 1.2m X 1.2m PATTERN.
3. ORE IN THE FOOT WALL STRIPPED.



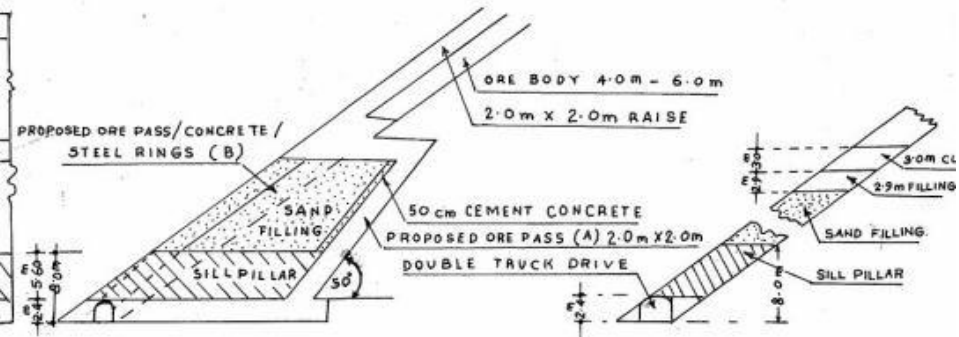
SILL PLAN

1. RAISE CONNECTION BETWEEN TWO COSECUTIVE LEVEL WILL BE SPACED AT 50m-80m INTERVAL, THESE RAISES WILL BE DRIVEN ALONG ORE BODY.
2. ORE PASS RAISES WILL BE SPACED AT 30m-60m INTERVAL, THESE RAISES WILL BE MAINTAINED AT 50° INCLINATION EITHER IN THE FOOT OF THE ORE BODY (A) OR WITHIN THE ORE BODY WITH THE HELP OF STEEL/ CONCRETE RINGS (B).
3. DEVELOPMENT AT SILL HORIZON WILL COMMENCE FROM THE ORE RAISES.
4. SILL LEVEL WILL BE STRIPPED TO A HEIGHT OF 5.9m.
5. TAILINGS WILL BE FILLED LEAVING AN OPEN HEIGHT OF ABOUT 2.9m FOR THE MOVEMENT OF MEN AND EQUIPMENT.
6. SUBSEQUENT CUTS WILL BE OF 3.0m EACH AND FILLING WILL BE CYCLIC.
7. HANGING WALL WILL BE SUPPORTED WITH ROCK BOLTS AT 1.5m X 1.5m INTERVAL.

LAY-OUT OF CUT AND FILL STOPE.



SECTION AA

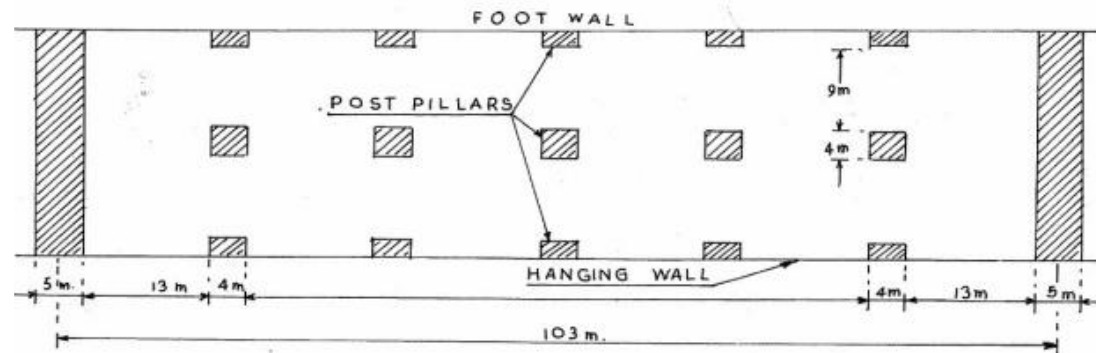


SECTION BB

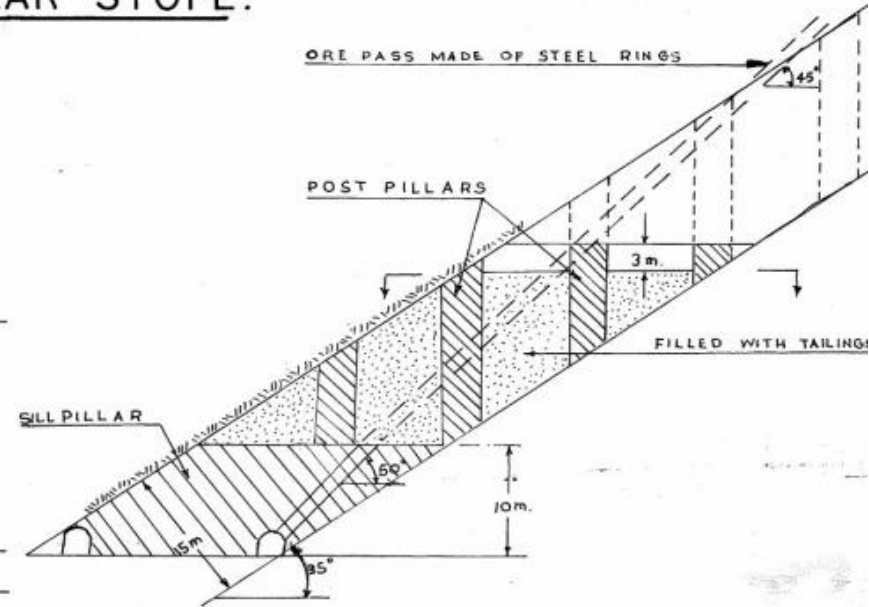
SECTION CC

LAY- OUT OF A POST PILAR STOPE.

SCALE :- 1cm. = 5m.



PLAN OF THE STOPE FLOOR.



SECTION

Sequence of operations

1. Ore passes at 30m. interval inclined at 50°
2. One service Raise in the centre of the stope along F/W. contact, two ventilation Raises at two extremities of the stope along H/W. contact.
3. Development of sill horizon through ventilation, services & ore pass raises.
4. Stripping of sill level to a height of 5m. leaving 4m. x 4m. pillars at spacing of 13m. along strike and 9m. across, back to be rock bolted 1.5m. x 1.5m. pattern.
5. Fill with Tailing leaving an open height of 2.5m.
6. Then 2nd cut 2.5m.
7. For operational convenience a service raise is made at the bottom of the stope.

Location map of HCL mines – Malanjkhand Copper Mine



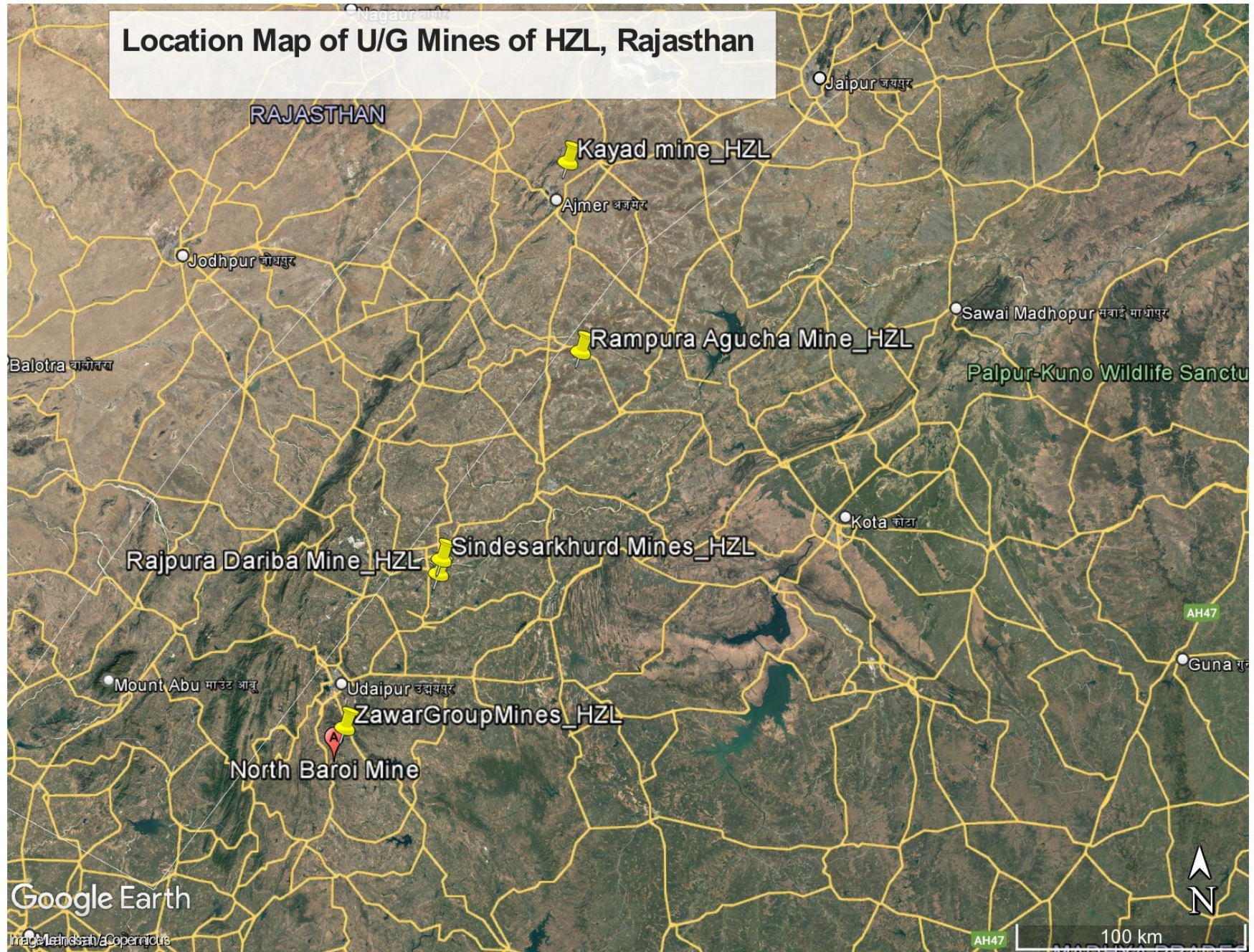
Malanjkhand Copper Project – Salient Features

- Malanjkhand Copper Project was established in 1982.
- Initial project has been set up by Hindustan Copper Ltd to exploit the copper ore through an open pit mine.
- Geological Survey of India took systematic geological exploration at this deposit during 1969.
- Mining lease of the ore was granted to HCL during 1973. With advancement of time this project was enhanced with viable operational developments.
- At present, open pit mining is discontinued due to ultimate pit depth
- Same deposit is to be mined with Underground mining method. Underground mine is on the verge of its commissioning.
- Malanjkhand copper deposit, located 20 km away from the Kanha National Park is the single largest copper deposit of India with nearly 70% of the country's reserve.
- Malanjkhand Mine (Reserve including Resources) = 221.00 million tonnes 1.31% copper
- Other facilities at MCP are concentrator plant Tailings Disposal and Water Reclamation System, Repair Shops, Maintenance Garage, Water Treatment Plant, Warehouses, Fuelling Stations, 132 KV Power Sub-station, Waste Treatment Facilities and Township with modern amenities.

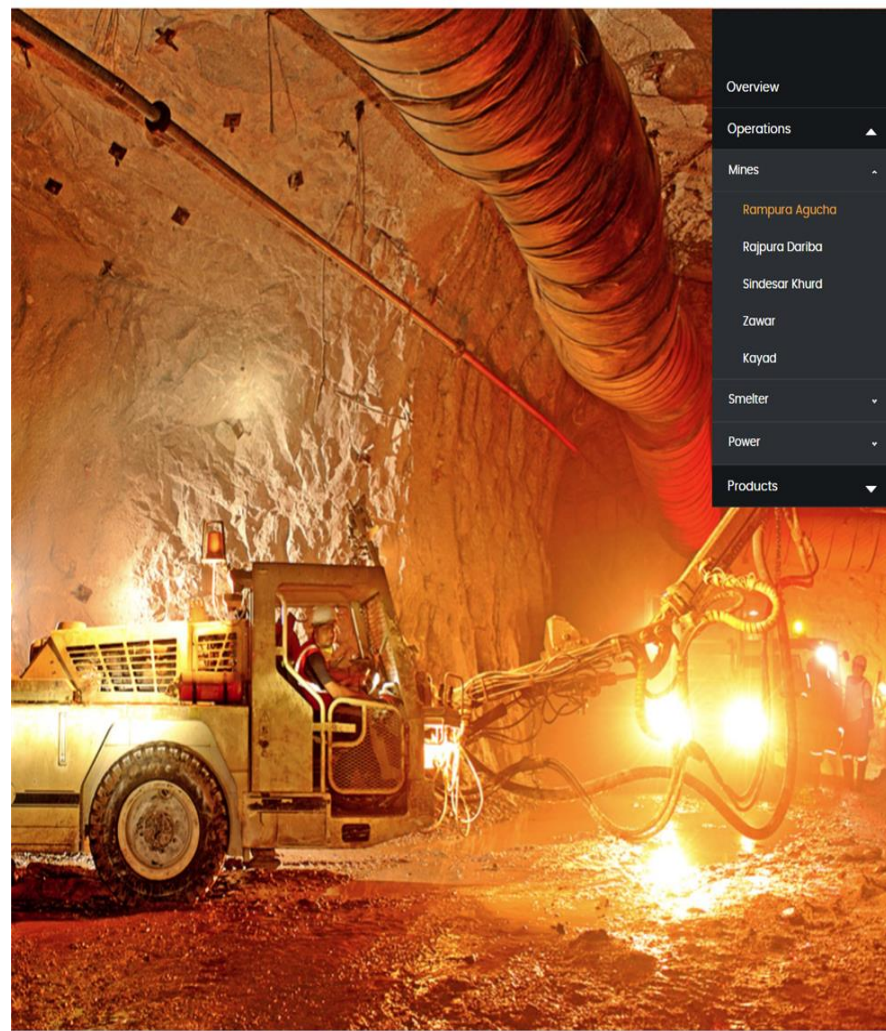
About HZL - Overview

- Hindustan Zinc is India's largest and world's second largest zinc–lead miner.
- More than 50 Years of operational experience, give highest Priority to safety of people and conservation of scarce natural resources through technology and innovation
- It total R&R base of 447.9 Mt and an average zinc-lead grade of 7.8%, mine life is over 25 years
- Fully integrated zinc operation currently hold 80 percent zinc market share in India's Primary zinc Industry .
- It is the 6th largest silver Producers globally with an annual capacity of 800 MT.
- It is a subsidiary of Vedanta Limited Which owns 64.9% stake in the Company While the Government of India retains a 29.5 stake . It is listed on the NSE and BSE.
- Hindustan Zinc's operations comprise lead-zinc mines, hydrometallurgical zinc smelters, lead smelters, Pyro metallurgical lead-zinc smelter as well as sulphuric acid and captive power plants in northwest India
- Total metal production capacity is 1.123 Mt.
- It has facilities located in 5 districts of Rajasthan which includes Udaipur, Chittorgarh, Bhilwara, Rajsamand and Ajmer along with 1 district of Uttarakhand.
- The facilities include Zawar group of Mines , Rajpura dariba Mine, Sindesar Khurd Mine, Rampura Agucha Mine and kayad Mine along with zinc –lead processing facilities which include a smelter at Debari, Chanderiya and Dariba and a silver refinery at panthnagar in uttarakhand .

Location Map of U/G Mines of HZL, Rajasthan



Rampura Agucha Mine – Salient Features



- The Rampura Agucha Mine (RAM) is an underground zinc and lead mine located in the Bhilwara district of Rajasthan State, India. It is located 230 km north of Udaipur
- Rampura Agucha was commissioned in 1991 as an opencast mine.
- Rampura Agucha is stratiform, sediment-hosted, high grade zinc & lead deposit. The ore body is massive and lens shaped with consistent ore grade.
- RAM is the largest and richest, lead-zinc deposit in the country and among the world's largest with 75.0 Mt of ore resources (R&R) with in situ average grade of 10.8% zinc and 2.0% lead on an exclusive basis as on end of FY 2021-2022
- it is also one of the lowest cost zinc producing mine globally.
- The mine is equipped with best-in-class infrastructure ,deploys advanced technology for mining activities for better targeting
- RAM is currently producing at 4.5 Mtpa.
- The mine is highly, mechanized with state of the art technology. Underground mining is carried out using long Hole Open stoping with paste Fill method with mined out stopes backfilled with cemented rock & with cemented tailing in paste form.
- Zinc and lead concentrates produced at Rampura Agucha are transferred to the smelters. The tailing generated due to beneficiation of ore are used by paste filling plants

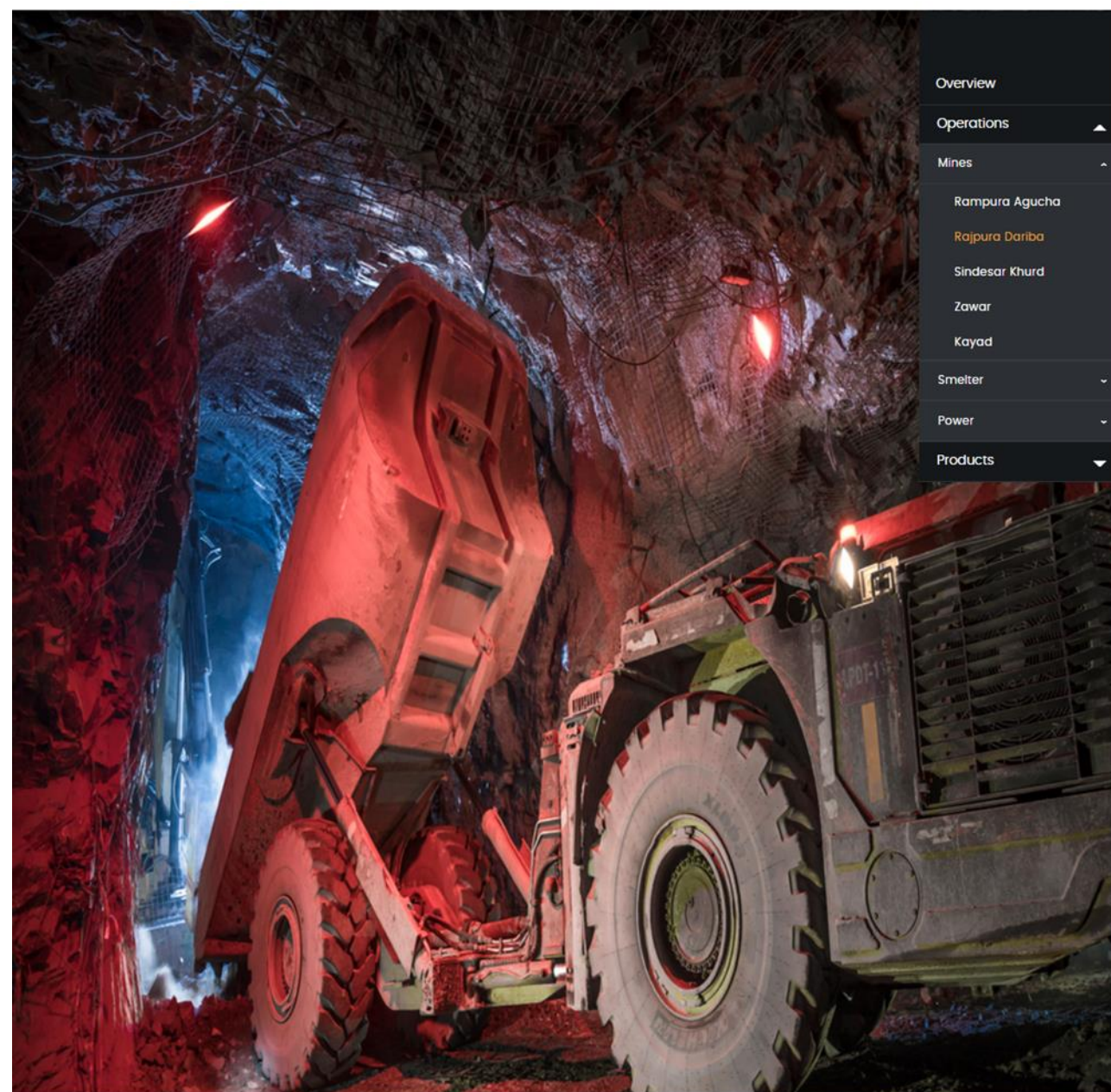
Rampura Agucha Mines of HZL



Google Earth

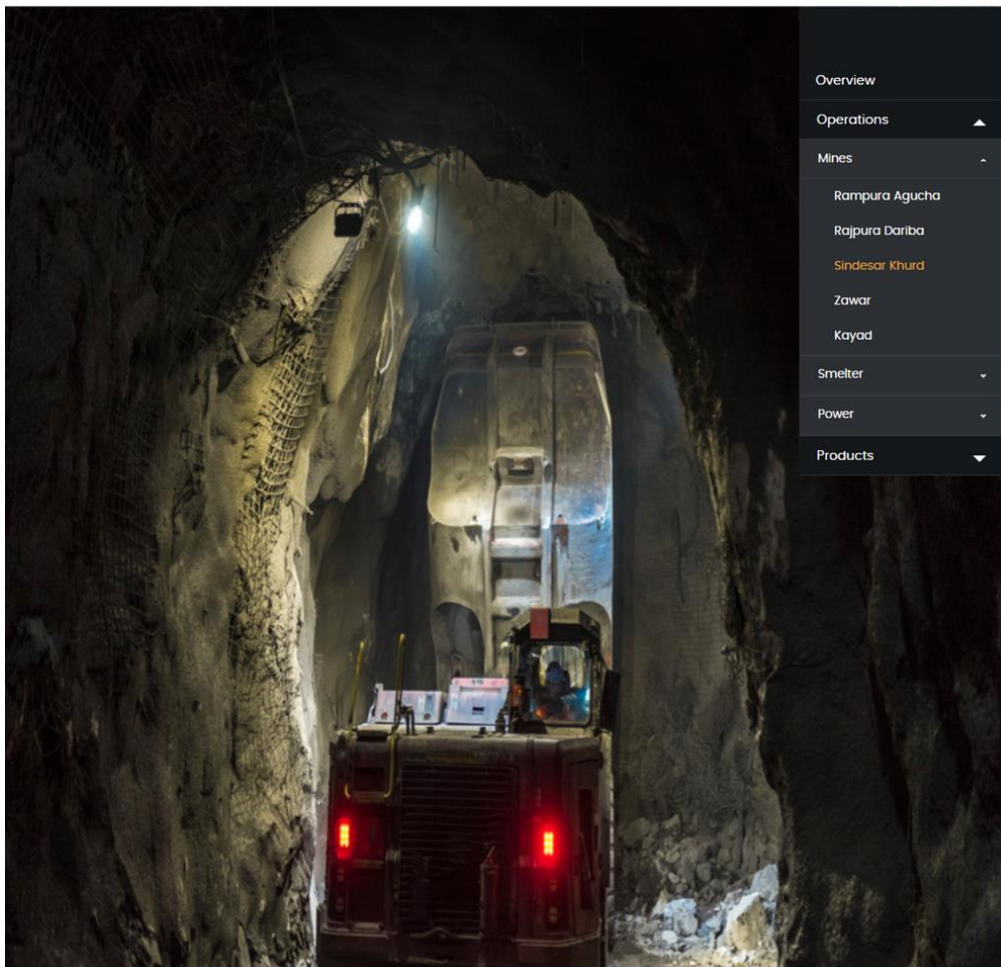
Image © 2022 CNES / Airbus

Rajpura Dariba (RD) Mine – Salient Features



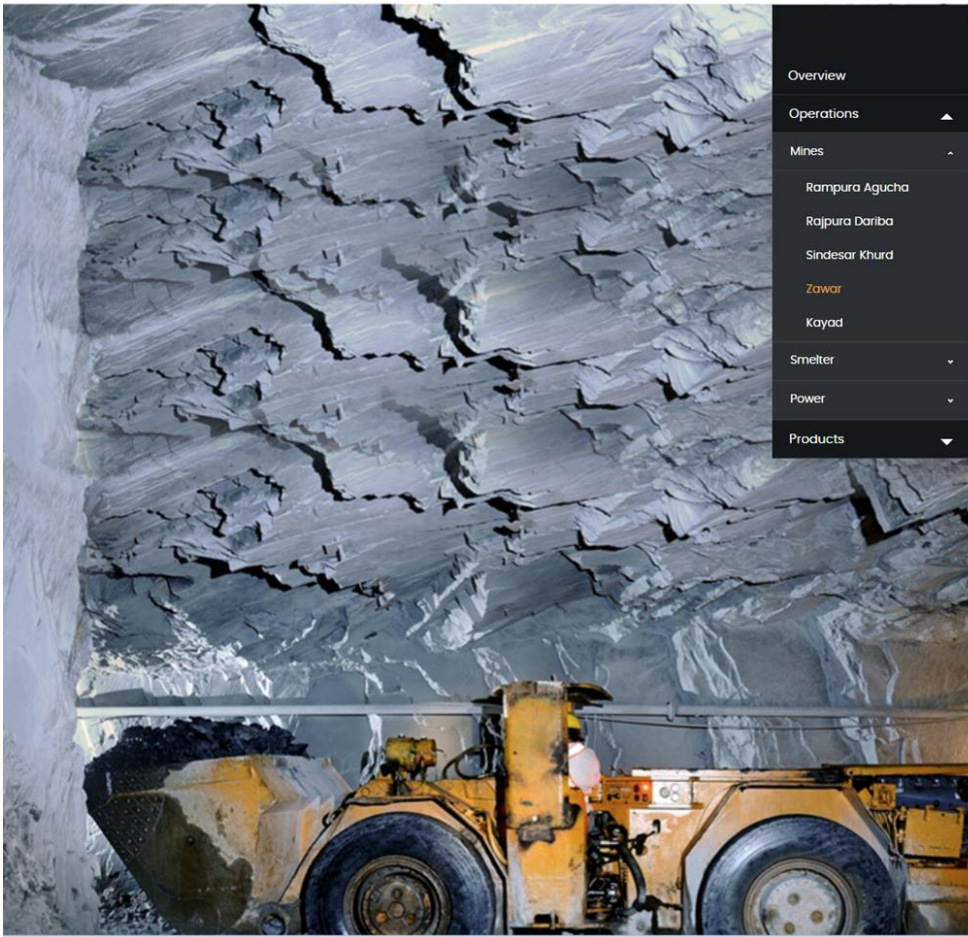
- The underground Rajpura Dariba (RD) mine is one of the oldest mines of HZL, which commenced operation in 1983. It is located about 75 Kilometres north-east of Rajasthan in India and is well connected with roads.
- The mine is accessed via decline, main shaft, and auxiliary shaft.
- HZL is in the process of Augmenting the production capacity from 2.0 Mtpa to 4.0 Mtpa.
- The capacity enhancement will be achieved through the introduction of two new portals for upper east and north lode, through further mechanization and automation .
- Mining is carried out using the Blast Hole Stopping method with mined out stopes backfilled with cemented tailings .
- The ore is crushed underground before being hoisted to the surface. It is then crushed again and milled before undergoing a flotation to produce the concentrate.
- Zinc and lead concentrates produced are transferred to the smelters.

Sindesar Khurd (SK) Mine – Salient Features



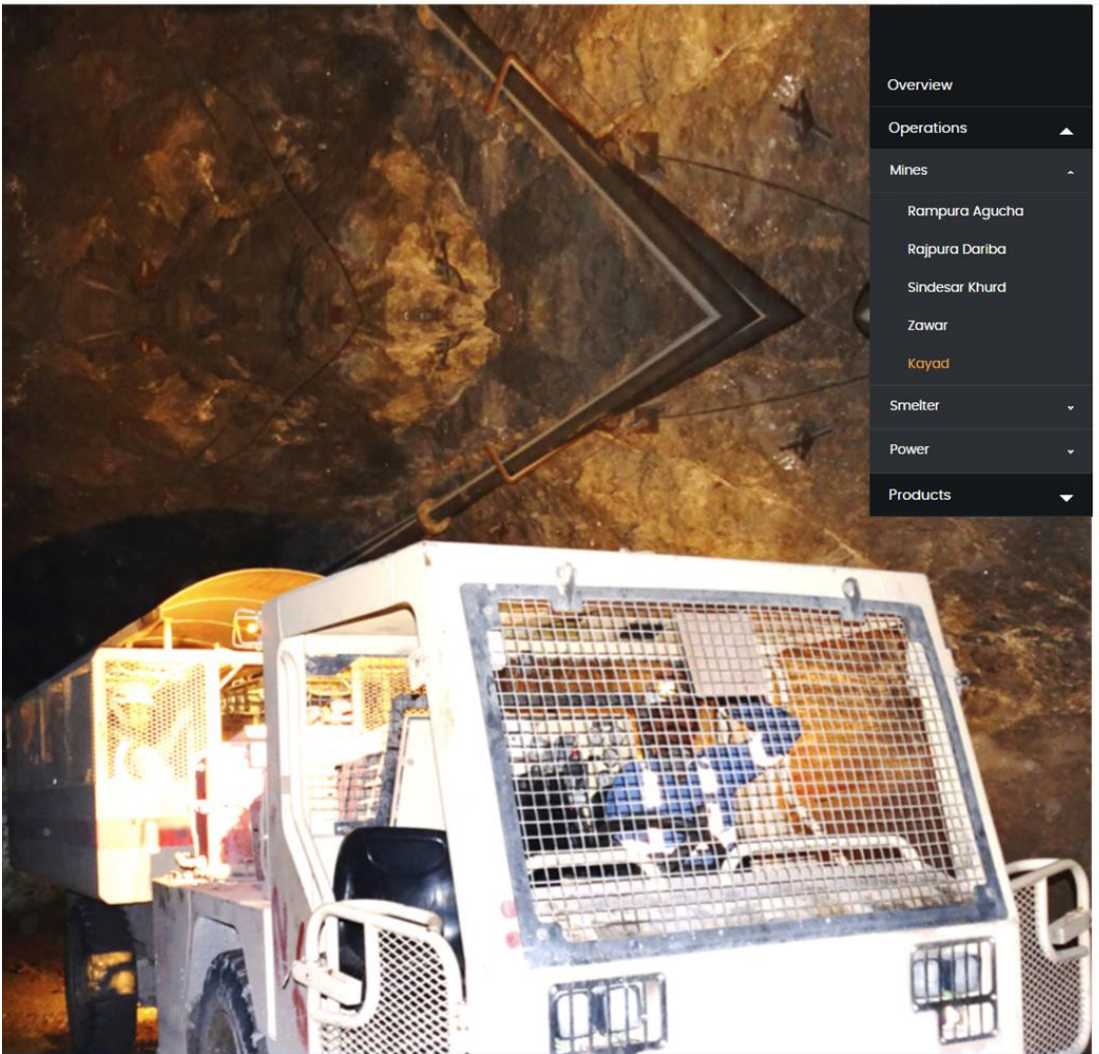
- Sindesar Khurd is an underground mine, commissioned in 2006.
- The mine was Initially opened as a satellite of Rajpura Dariba Mine.
- It is located 82 kms north-east of Udaipur, Rajasthan , India.
- Sindesar Khurd (SK) Mine is a world –class silver –rich mine with state-of-the-art infrastructure facilities and best-in-class mechanization.
- It has a uniform ore body with good zinc and lead grade.
- Since it commenced operation in 2006, the mine has undergone several phases of expansion-from 0.3 Mt to its current capacity of 6.0 Mt.
- SK mine today stands out as one of the largest underground mine and one lowest cost silver producer in India.
- The deposit at SK mine has a main lens and multiple standalone auxillary lenses, creating multiple independent production centers.
- Sindesar Khurd has access to the mine through Decline and shaft.
- Mining method is blast hole open stoping with paste filling.
- The ore is crushed at the surface and transported to the beneficiation plant for further crushing; milling; and flotation process.
- Zinc and lead concentrates produced are transferred to the smelters . The tailing generated due to beneficiation of ore used by paste fill plants.

Zawar Group of Mines – Salient Features



- Zawar group of mines (Mochia, Balaria, Zawarmala, and Baroi) are located about 40 kilometres south-east of Udaipur , in the state of Rajasthan in India. Zawar is very well connected with roads.
- Zawar Mines (ZM) crossed the major Milestone of 4Mt ore production mark in FY 2021-22.
- This group of four heritage mines, with mining history spanning over 2,000 years
- The four underground mines also scaled new levels of Mechanization and digitization.
- Lead-zinc ore is divided into stope blocks which are drilled and blasted using long hole open stoping mining method
- ore is either crushed to underground or at surface to feed mill.
- A backfill and a paste fill plants were recently completed to use tailings from ore beneficiation plants.
- The mine also has a dry stock talling plant, the first of its kind in india.
- The ore is further crushed and milled before undergoing a flotation process, to produce the concentrate.
- Zinc and lead concentrates produced are transferred to the smelters.

Kayad Mines – Salient Features



- It is Located in Kayad, Ajmer (Rajasthan), India
- This underground mine has been delivering a consistent 1.2 Mt of ore with approximately 60 Kt of metal in concentrate over the past five years.
- It is the youngest mine, which started operation in 2011, has produced more than 6 lac tonnes of metal content in the past eight years at the back of high level of mechanization.
- Operations are vertically integrated and supported by continuous and integrated surface and underground exploration.
- In FY 2021-22, Kayad Mine (KM) produced 934 kt of ore, with 44.3 kt of metal in concentrate.
- 7.4 Kilometers of development in existing and upcoming blocks, it is poised to maintain current production rate in the coming years.
- Kayad has access to the mine through decline.
- Mining method is blast hole open stoping with mined out stopes backfilled with cemented rock and with cemented tailing in paste form.
- the ore from Kayad mine is treated in mills at Rampura Agucha Mine.

- Hutti Mine is the main underground Gold Mine of the Hutti Gold Mines Company Limited.
- There are nine parallel gold bearing reefs exposed on the surface .
- Active mining is being carried out in six lodes over a strike of around 1.3 Km .
- The parallel lodes have a general strike of NNW-SSE & dips ranging from 60 to 70 degree due West .
- width of the reefs varies from 2.5 m to as wide as 50 m at some places, the average width being 5 to 8 m.

Location	District Raichur, Karnataka.		
Distance	80 km West of Raichur. 480 km North of Bengaluru. 300 km Southwest of Hyderabad.		
Railway Station	Raichur, South Central Railway - 80 kms		
Mining History	(a) Ancient Mining 2000 years old (Pre Ashokan Times). (b) Modern Mining - Main Mine during Nizams period. (1902-1918). (c) Modern Mining - Rediscovery of Parallel Lodes and mining from 1947 onwards.		
Present Mine Depth	Below 28th level, 933.70 meters.		
Nature of the Deposit	Archaean lode gold deposit. Shear zones controlled Gold-Quartz-Sulphide mineralisation.		
Ore Mined and Gold Produced so far from Hutti Mines.	Ore (Million Tonnes)	Grade (gm/Tonne)	Total Gold (Tonnes)
1. Ore already Mined in the Main Mine(During the period 1902 to 1918)	0.38	19.49	7.41
2. Ore Mines so far from 1948 Onwards in the present mined on the other parallel lodes up to 871 m depth on 31.03.2019.	16.55	4.97	82.18
Total ore mined and gold produced as on 31.03.2019	16.93	5.29	89.59
Ore reserves estimated as on 01.04.2019 (Hutti)			
Proved & Probable Category up to depth of 842 m @ 2.0 g/t cut off		16.30	4.4171.88
Present scale of production			
Total for Hutti Gold Mines Company Limited	8.0 Lakhs Tonnes per Annum		

Modes of Entry to the Mines:

- Hutti Gold Mine is serviced by three shafts namely Mallappa Shaft, Central Shaft, and Village Shaft;
- Both Mallappa & Central are Vertical shafts, with Rectangular cross Section, while village shaft is an incline with elliptical cross section



Mallappa Shaft

- Mallappa shaft with a cross section of 6.1 m x 2.13 m at the surface has been established up to a depth of 910 m.
- this is the main production shaft and currently handling about 80% of the total production from Hutti underground.
- This shaft is equipped with two winders, one Double Drum Winder for transport of men, material and ore/ waste rock hoisting and the other being exclusively used for ore hoisting.



Central Shaft

- Central Shaft has a surface Cross Section 3.76 m x 1.83 m with depth of 871.90 m from the surface.
- At present this shaft is serving as the second outlet from the underground.



Village Shaft

- Village Shaft is an inclined Shaft extending up to 15th level.
- With the completion of the top-level workings the utility of this shaft will be restricted to only pumping and lowering and hoisting of workmen

Evolution of Mining Technology at Hutti Gold Mines:

- **HGML's production has steadily increased over the years. In this process HGML has been able to develop much needed technology expertise in the field and is itself to take new challenges of growth and development.**
- **Over the years, the company has implemented a number of technology updating schemes in mining and processing leading to increased safety and competitiveness of its operation**





Evolution of Mining Technology at Hutti Gold Mines:

- **Hutti Gold Mines started its operation with Shrinkage method of stoping and it continued up to the 15th Level.**
- **This method was abandoned because of safety reasons and below 15th level, Cut and Fill Stoping was adopted.**
- **Since the year 1995, Hutti Gold Mine has completely switched over to bulk mining method of Sub-level Stoping with Large Diameter Blastholes.**
- **High capacity pneumatics Drills were introduced to keep pace with the rate of extraction of ore.**
- **Over the years small loaders have been phased out and the mine is fully equipped with trackless loaders like Load Haul Dump Units**



Evolution of Mining Technology at Hutti Gold Mines:

- To match with higher capacity loaders, Rock Breakers and Crushers are installed in underground to reduce the size of boulders before feeding to the skips for hoisting to surface.
- Major quantity of ore is hoisted by a winder which can be operated in automatic as well as manual mode.
- This led Hutti Gold Mine Gold Mine to phenomenal productivity and increased safety
- considering the need of tomorrow, Hutti Gold Mine has commission the facility for filling of stope voids with classified mill tailings produced in the mill.
- In selective mining areas where ground condition are not favourable in long term aspect, stopes are filled with mill tailings immediately after extraction.

MINE CONSTRUCTION AND EXPLORATORY MINE DEVELOPMENT OF HUTTI SOUTH BLOCK:

- **The Hutti gold deposit has been divided into three parts by intrusion of two major dykes.**
- **The central part, bound by two dykes in north and south, is present operating area of Hutti Gold Mine.**
- **Part of the deposit lying south of the southern dyke is called Hutti south block .**
- **extensive surface exploration by way of trenching, followed by surface diamond drilling has been carried out with encouraging result, in terms of reserve , as well as grade of the deposit .**
- **with a view of detailed exploration, underground mine development was taken up in the year 1983-85.**
- **The activity was continued with further exploration by sinking South Shaft up to a depth of 169.90 m i.e. 5th level and occurrence of all the reefs was confirmed by exploratory underground development through South Shaft .**
- **to continue the process, the proposal for deepening of South Shaft followed by opening of more levels is underway.**

REOPENING OF MAIN MINE:

- History of gold Mining in the Hutti Main Reef dates back to 2000 years.
- Mining in this reef was commenced through two shafts, i.e., Main Shaft and North Shaft.
- Also , a series of old workings along the strike length of this reef testify to this mining activity .
- systematics mining in this reef was started in 1880 and Hutti Mine was commissioned.
- The working in this mine continued up to 1920 and was closed thereafter.
- There had been sporadic attempts to open the Main Hutti Mine (Main Reef) .
- in the year 1951-52, the main Hutti Mine was reclaimed up to a depth of 450 ft. and resampled to assess its grade.
- Again, during the period 1955-57, the mine was reclaimed up to a depth of 640 ft. (190 m) and additional reserve proved.
- However at this point of time, further work in the mine was suspended.
- Afterwards, it was decided to explore the Main Reef through an existing old shaft , Grey's Shaft, located south of the Main Shaft.
- With this objective , the work of Shaft deepening and exploratory development of Main Reef was carried out during the period from January' 2006 to June' 2011 .
- in continuation with the above, proposal for further exploration of Main Reef through underground mine development is in the process

New Project – HGM Co. Ltd.

Uti Gold Mine

Uti Gold Mine

This mine is situated 22 kms North East of Hutti Gold Mine. This deposit is explored by GSI, MECL & H G M L. It has got Mining Lease (ML No: 2126) for 47.96 Ha. Mine development activities are in progress. This mine is working as a satellite mine to Hutti Gold Mine

The Service Shaft & Incline Shaft sinking & associated mine construction works is under progress on lode no. 3 & 4 and will be continued on lode 2, 3 & 4. The south open pit on lode no: 4 will be taken up.



New Project – HGM Co. Ltd.

Hira Buddini Mine

Hira - Buddini Gold Mines

This mine is situated 20 kms East of Hutti Gold Mine. This deposit is explored by GSI. Based on the encouraging results of the preliminary exploration by GSI, considering proximity and viability HGML has started detailed exploration by trenching, diamond drilling & exploratory mining. HGML has got Mining Lease (ML No: 2578, Period) for 22.18 Ha. Exploratory mine development activities are progressing in full swing and experimental stoping operations have started in 2009-2010. This mine is working as a satellite mine to HGM.

Sub level developments & stope drilling will continue. It is planned to continue the detailed exploration by exploratory mining by shaft sinking below the 6th level. Deep Diamond Drilling (2323.80 m) work is completed (explore the depth persistence of Mother lode) and to explore the parallel lodes identified by surface mapping & trenching.



New Project – HGM Co. Ltd.

Wandali Mine

Wandali Gold Mine

This mine is situated 10 kms east of world class Hutti Gold Mine. This deposit explored by GSI & MECL. Based on the encouraging results of the exploration, considering proximity and viability HGML has planned to start detailed exploration by exploratory mining. HGML has got Mining Lease (ML No: 2031, Period: 31.01.1987 to 30.01.2007) for 388.77 Ha. Applied for the renewal of the mining lease and notification for grant of ML renewal is in progress. This mine will be a satellite mine to Hutti Gold Mines. It is planned to explore & exploit the deposit by underground mining.

Mangalore Mine

Mangalore Gold Mines

The mine is situated about 80 kms, north of Hutti Gold Mine. The area was explored by GSI & HGML. HGML has got Mining Lease (ML No: 1305, Period: 12.02.1976 to 11.02.1996) for 55.75 ha. Mining activity held at Mangalur Gold Mine by HGML between 12.02.1976 to 11.02.1996. MGM was closed on 26.04.1996 due to non-viability and due to heavy (52,000 lts/hr) seepage of water in the mine. Applied for the renewal of the mining lease and notification for grant of ML renewal is in progress. It is planned to explore the deposit by detailed exploration by exploratory mining and deep diamond drilling.

Abandoned Mines: Re-opening – HGM Co. Ltd. (Chitradurga Gold Mines)

Ajjanahalli Gold Mine

Ajjanahalli Gold Mine

Ajjanahalli Gold Mine is in Sira taluk, Tumkur Dist. and is situated 75 kms from Chitradurga Gold Unit. The area is explored by GSI & HGML has got Mining Lease (ML No: 2243, period: 22.10.1998 to 22.10.2018). HGML has carried out open cast mining during 1995 to 2002. Subsequently, it was suspended due to non-viability during 2002. i.e., encountering of un-oxidized ore. It is planned to carry out further detailed exploration by diamond drilling and mining. Feasibility studies & associated statutory works are in progress for taking up the exploration & mining.

Ingaldal Copper Mine

Ingaldal Copper Mine (Central Block)

This mine is situated south of Chitradurga. HGML has got Mining Lease (ML No: 2490, Period: 06.03.1993 to 05.03.2013). Mining activities at Ingaldal Copper Mine were carried out from 1965 to 1994. So, Mining work was suspended in 1994 due to world globalization and cause decline in Cu price as well M/s. Hindustan Copper Limited had stopped buying of copper concentrate. It is planned to carry out further detailed exploration by exploratory mining, reclamation. For this statutory works are in progress.

Ingaldal Copper Mine (North Block)

Ingaldal Copper Mine (North Block)

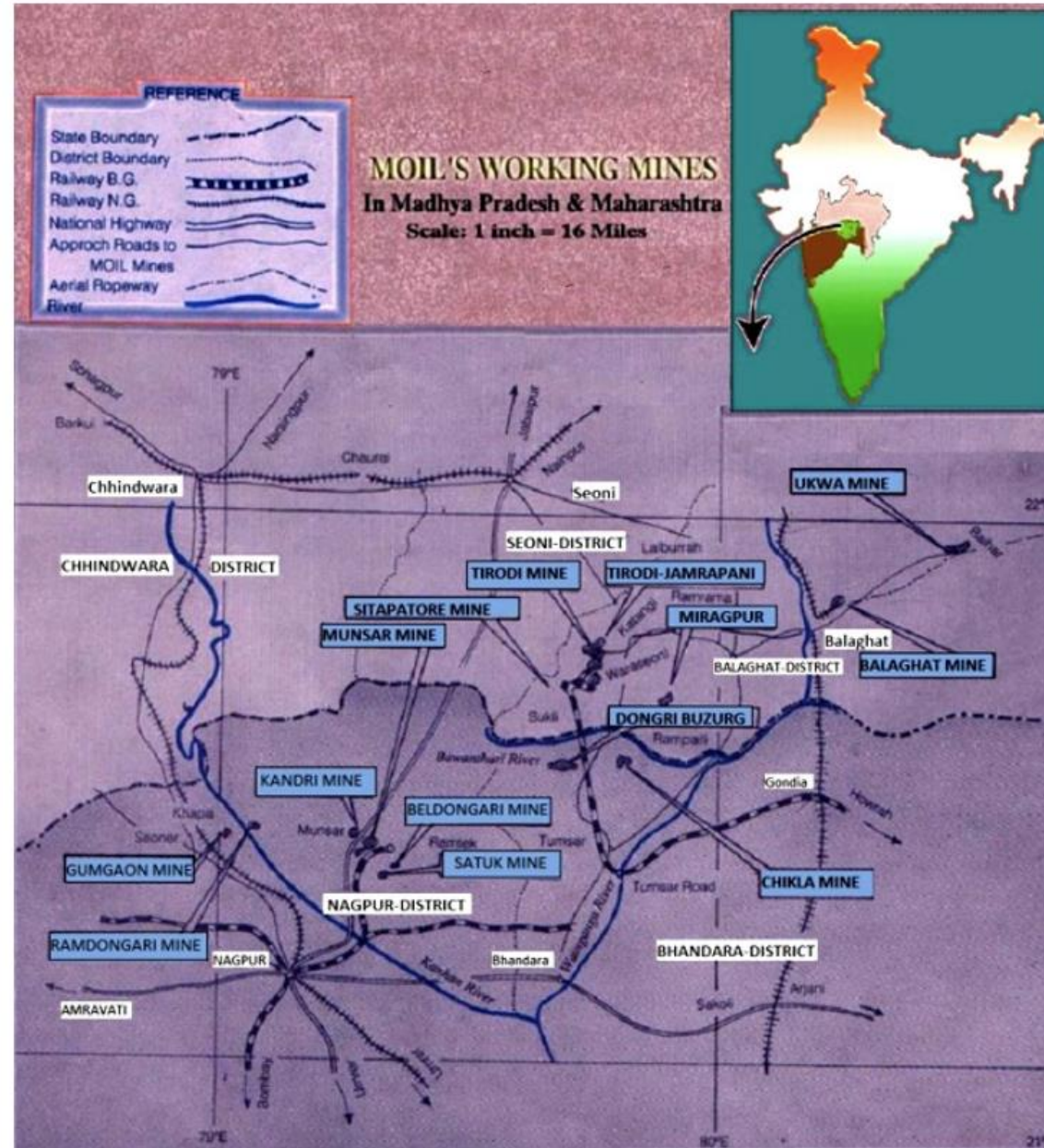
This block is situated 2.50 kms. North of Ingaldal Copper Mine & 6 kms south of Chitradurga. Govt. of Karnataka sanctioned Mining Lease (No: 1125). It is planned to carry out further detailed exploration by exploratory diamond drilling followed by reclamation and mining, reclamation. For this statutory works are in progress.

About MOIL Limited

- MOIL is a Schedule “A” Miniratna Category-I Company.
- It was originally incorporated as Manganese Ore (India) Limited in the year 1962. Subsequently, name of the Company was changed from Manganese Ore (India) Limited to MOIL Limited during the financial year 2010-11.
- MOIL was originally set up in the year 1896 as Central Province Prospecting Syndicate which was later renamed as Central Provinces Manganese Ore Company Limited (CPMO), a British Company incorporated in the UK.
- In 1962, as a result of an agreement between the Government of India and CPMO, the assets of the latter were taken over by the Government and MOIL was formed with 51% capital held between the Govt. of India and the State Governments of Maharashtra and Madhya Pradesh and the balance 49% by CPMO.
- It was in 1977, the balance 49% shareholding was acquired from CPMO and MOIL became a 100% Government Company under the administrative control of the Ministry of Steel.
- During the Financial year 2010-11, MOIL got listed on 15th December, 2010 on National Stock Exchange and Bombay Stock Exchange. After the listing, the shareholding in the company was Govt. of India (71.57%), Govt. of Maharashtra (4.62%) and Govt. of Madhya Pradesh (3.81%) and Public (20%).
- At present the shareholding pattern of the company is Govt. of India (53.35%), Govt. of Maharashtra (5.96%) and Govt. of Madhya Pradesh (5.38%) and Public (35.31%).
- At present, MOIL operates 11 mines, seven located in the Nagpur and Bhandara districts of Maharashtra and four in the Balaghat district of Madhya Pradesh.
- All these mines are about a century old. Except 4, rest of the mines are worked through underground method.
- The Balaghat Mine is the largest mine of the Company. The mine has now reached a mining depth of about 383 meters from the surface.

- Dongri Buzurg Mine located in the Bhandara district of Maharashtra is an opencast mine that produces manganese dioxide ore used by dry battery industry.
- This ore in the form of manganous oxide is used as micro-nutrient for cattle feed and fertilizers.
- MOIL fulfils about 50% of the total requirement of dioxide ore in India.
- At present, the annual production is around 1.1 million tonne which is expected to grow in the coming years.
- MOIL produces and sells different grades of Manganese Ore. They are:
 - High Grade Ores for production of Ferro manganese
 - Medium grade ore for production of Silico manganese
 - Blast furnace grade ore required for production of hot metal and
 - Dioxide for dry battery cells and chemical industries.
- MOIL has set up a plant based on indigenous technology to manufacture 1,500 MT per annum capacity of Electrolytic Manganese Dioxide (EMD).
- This product is used for the manufacture of dry battery cells.
- EMD produced by the Company is of good quality and well accepted by the market.
- MOIL is having a Ferro manganese plant with a capacity of 12,000 MT per annum for value addition.
- In order to promote non-conventional energy resources, MOIL has installed 4.8 MW Wind Energy Farm at Nagda Hills and 15.2 MW Wind Farm at Ratedi Hills, Dist. Dewas in Madhya Pradesh.

Mines Locations





INDIAN METALS & FERRO ALLOYS LTD



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Established in 1961 in the Eastern State of Odisha known for its rich natural resources, IMFA is India's leading fully integrated producer of value added ferro chrome with 190 MVA installed furnace capacity backed up by 204.55 MW captive power generation capacity (including 4.55 MW solar) and extensive chrome ore mining tracts. We are unique from the quality perspective due to comprehensive ISO 9001 Quality Management certification.

We are recognised globally as a reliable and cost-competitive producer with a particular focus on customer satisfaction. Over the years, we have developed long standing relationships with multi-nationals like POSCO of South Korea, Marubeni Corporation and Nisshin Steel of Japan and leading stainless steel producers in China and Taiwan. Within India, Jindal Stainless and Shah Alloys are our leading customers.

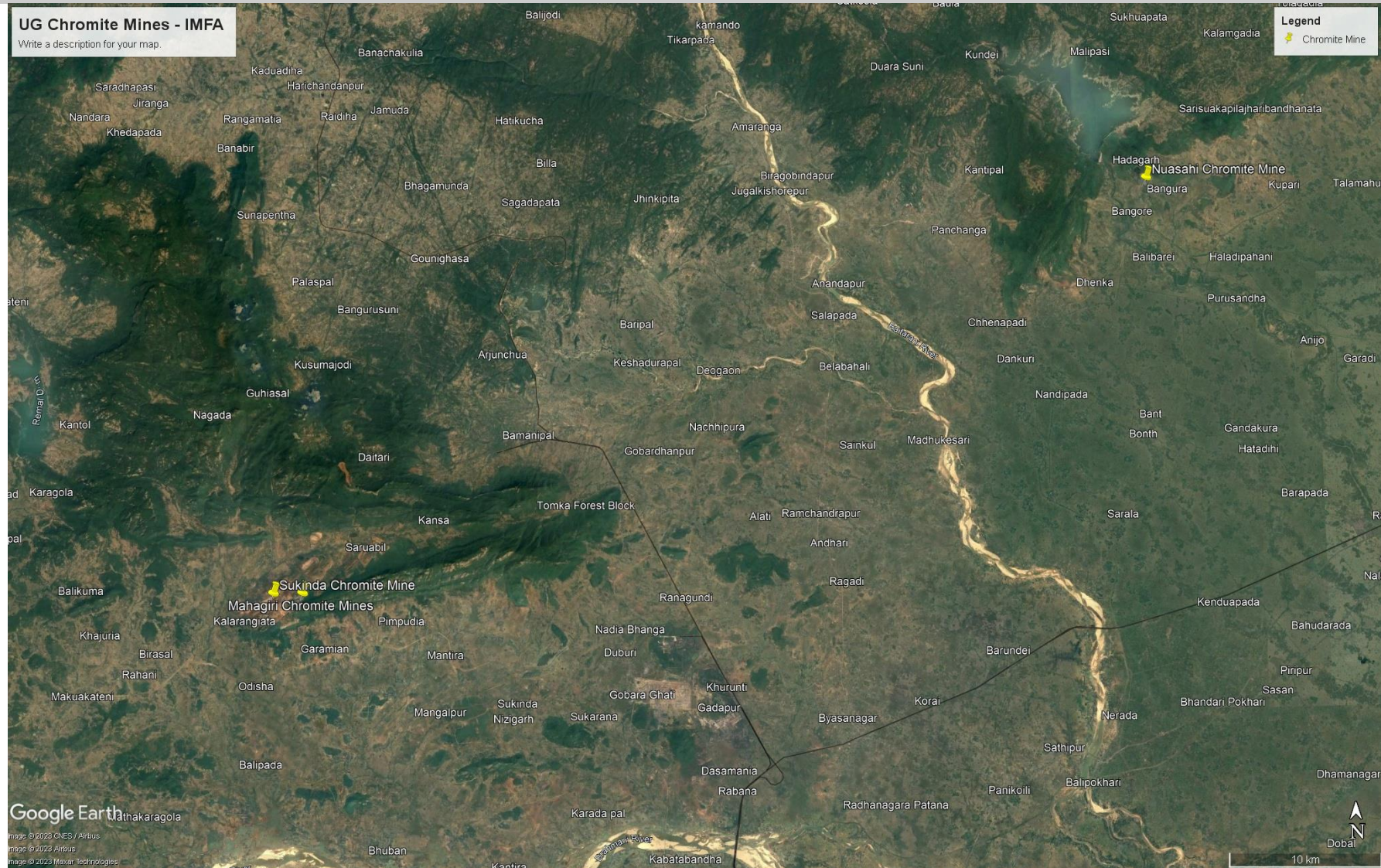
The 50th Foundation Day celebrations in November 2011 has rejuvenated us and we have now embarked upon 'Project ILA' to *Innovate, Lead, Accelerate* the businesses we operate.

Location map of Imfa u/g chromite mines

Sukinda Mines
(Chromite)

Mahagiri Mines
(Chromite)

Nuasahi Chromite Mines



Ever since its inception in 1956, OMC has not only simply focused on propelling Odisha's economic growth to new heights but also worked steadfastly to bring about a change in the lives of the people of the peripheral communities.

OMC's major mines include **Iron, Chrome, Bauxite and Manganese** which cater to the requirement of mineral based industries such as steel, sponge iron, pig iron, ferro-manganese, ferrochrome, and other similar industries.

Presently, Daitari, Gandhamardan – A & B, Kurmitar, Guali, Jilling, Khandbandh, Banspani, Unchabali and Tiringpahar are OMC's major Iron Ore mines, while South Kaliapani and Sukrangi are the major Chrome ore mines. **The Bangur Chrome Ore mine is the first and only underground mine of OMC.** Kodingamali is the only Bauxite mine of OMC.

Further OMC ventured into Khondalite mining for supply of Khondalite stone to Sri Jagannath Temple, Puri & other projects carried out under “**ABADHA**” scheme.

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Location map of OMC u/g chromite Mines



U/G metal Mines: outside India

<https://www.mining-technology.com/projects/kiruna/>

https://en.wikipedia.org/wiki/Kiruna_mine